

The Effect of a Conditional Cash Transfer on Early Marriage: Evidence from Mexico

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Abstract

I study the effect of a conditional cash transfer program on child marriage in rural Mexico and show that it led to a significant increase in early marriage. Using the program's random assignment and staggered implementation, I find that the program more than doubled marriage rates among girls without affecting boys. To isolate the mechanism behind this finding, I study individuals who were ineligible for the transfer but lived in beneficiary households, and thus were exposed only to the additional income. Their marriage rates also rise, evidence of an income effect of the monetary transfer. In a setting where marriage payments and arranged marriages are absent, extra income can increase marriage, an unintended consequence relevant for the design of cash transfer programs.

JEL: I25, J12, J13, I21, I38

1 Introduction

Child and early marriage are widely recognized as a violation of human rights, particularly prevalent in developing countries. They are both consequences and causes of poverty, linked to educational abandonment and reduced participation in formal labor markets.¹ This practice disproportionately affects girls and exposes them to increased risks of early childbearing, violence, abuse, and limited autonomy.² According to UNICEF, around 20% of women worldwide aged 20 to 24 in 2021 married or entered into a union before turning 18 (UNICEF, 2023a). Early marriage is more prevalent in societies characterized by gender inequality, conservative social norms, conflict, insecurity, and acute poverty. Given the belief that education and poverty alleviation can attenuate child marriage, cash transfers conditional on school attendance or payment of school fees have been identified as one of the most promising strategies to decrease early marriage (Kalamar et al., 2016). However, little is known about how these programs affect early marriage decisions in marriage markets with no arranged marriages or marriage payments. Parental involvement in spouse selection is a key determinant of marriage outcomes, especially when parental and child preferences diverge (Vogl, 2013; Banerjee et al., 2013). Additionally, the presence and type of marriage transfers affect how income shocks influence families' marriage decisions for girls (Corno et al., 2020). Given the important role of these institutional features, it is crucial to understand how income transfers impact child marriage in settings where parental interference and marriage payments are absent.

In this paper, I address this question and study one of the world's largest conditional cash transfer programs, Progresa/Oportunidades, implemented in rural Mexico. I show its unintended consequences as it *increased* early marriage and provide evidence of a mechanism behind this unexpected finding. The program provides monetary transfers to poor households, primarily conditional on children's school attendance. Therefore, the program can affect marriage through two main channels: (i) education and (ii) income. I discuss these channels in

¹See Thomson (2003) and Sperling and Winthrop (2015).

²On education and labor market, see Adebawale et al. (2012) and Kalamar et al. (2016). On violence and decision-making power, see Kırdar et al. (2018), Jejeebhoy et al. (1995), and Amin et al. (2017). On fertility choices and children outcomes, see Dahl (2010), Duflo et al. (2015), and Behrman (2015).

turn.

First, education has been shown to affect marriage decisions. Empirically, most evidence points to a negative effect of education on early marriage (Angrist et al., 2002; Skirbekk et al., 2004; Ferré, 2009; Hallfors et al., 2015; Ashraf et al., 2020; Kırdar et al., 2018; Giacobino et al., forthcoming). In the presence of returns to education in the labor market, increasing schooling increases the opportunity cost of marriage, thus leading to decreases or delays in marriage. There is extensive evidence showing that Progres/Oportunidades increased children's years of education significantly (Behrman et al., 2005, 2009; Dubois et al., 2012), and, most recently, that its rollback led to decreases in school enrollment (Marquez-Padilla et al., 2025). Therefore, we could expect Progres/Oportunidades to decrease early marriage through the documented increases in education. A particularity of rural Mexico, however, is the co-existence between schooling and marriage. Rivero and Palma (2017) reports that 17.1% of formally married girls attend school. In this paper's sample, I observe that, across all years, 20% of married girls attend school (see Table A1 for a finer breakdown). Additionally, Attanasio et al. (2012) find little to no returns from education in rural Mexican localities' labor markets. It might be the case, then, that if returns to education are low in the labor market, there is no change in the opportunity cost of marriage. These two observations suggest that increases in education might have a limited effect on marriage decisions.

Second, Progres/Oportunidades might affect marriage decisions through an income effect. This effect is ambiguous ex-ante. On the one hand, increased income may reduce households' reliance on marriage as an insurance mechanism (Amin et al., 2016). On the other hand, it could increase the marriage-market value of beneficiaries or make marriage-related expenses more affordable, thereby facilitating the formation of new households.

I study the overall effect of Progres/Oportunidades on early marriage and isolate the program's income effect. Initially introduced in 1998 in randomly selected localities, Progres/Oportunidades was renamed Oportunidades in 2000 when the initial control-group localities were also incorporated into the program. In 2003, a new set of localities was selected through propensity score matching to serve as the pure control group. The staggered implementation of the program and the available comprehensive panel data allow for a dynamic analysis of the program's causal

effect by comparing the three groups of localities over six years using a staggered differences-in-differences estimator.

I find that exposure to the program increased the probability of marrying before age 18. One year after the start of the program, the effect is 1.2 p.p., statistically different from zero. Five years after the program started, beneficiaries were 3.7 p.p. more likely to be married than the control group. The program more than doubles the marriage probability for treated individuals, from 2% in the absence of the program to around 4.4%. These effects were driven by girls, whose probability of marriage increased by 6 p.p. in 2003. For reference, in that year, the unweighted proportion of married girls under 18 in the control group was about 3%. The program's effect on boys' probability of marriage is indistinguishable from zero.

The program had a greater effect on older girls. However, being exposed longer to Progres a/Oportunidades did not change the magnitude of the program's impact on marriage. I also observe that the program's effect on marriage is larger in magnitude from 2001 onward for both treated groups. This observation suggests that the changes in the program around that time, which included expanding the benefit to high school years, might be relevant to explain the overall effect. I provide supporting evidence for this hypothesis, as I show that receiving the benefit at an age with high marriage risk, which coincides with high school, facilitates marriage for girls in treated localities.

The program's design does not allow for the perfect disentangling of income and education effects. However, to understand the mechanisms behind this unexpected result, I conduct two exercises to isolate the causal effect of income on marriage. I examine two distinct populations that were likely influenced only by the program's income component. First, I analyze children and young adults who were not direct beneficiaries but lived in households eligible for the program. Second, I explore the program's heterogeneous effects on children based on their likelihood of dropping out, focusing on girls whose schooling decisions were unlikely to be influenced by the program - the 'always-takers', or those who would have gone to school even in the absence of the program - thereby isolating its income effect. In both cases, the program increased the probability of marriage, providing evidence that the program's income effect contributes to the increase in child marriage observed in the paper's main results.

The findings in this paper contribute to a vast literature on Progres/Oportunidades' impact on its beneficiaries. Most of the literature documents positive effects of the program on education, labor market participation, network formation, consumption, and poverty, both in the short and medium run (summarized by Parker and Todd, 2017) and in the long run (Araujo and Macours, 2021; Parker and Vogl, 2023). I contribute to this literature by providing causal evidence of the program's effect on child marriage, showing that it led to increased early marriage among girls. This finding is particularly important for any comprehensive cost-benefit analysis of the program, as it highlights an unintended and previously undocumented consequence alongside its well-established positive impacts.

The findings in this paper provide novel insights into the causes and determinants of early marriage. I provide evidence that income effects can explain the increases in marriage due to the program. Additionally, I study this topic in an understudied context regarding early marriage. In this setting, there are no widespread marriage arrangements or payments, such as dowries or bride prices. The results in this paper challenge our conventional wisdom on the relationship between conditional cash transfers and early marriage. These results are important for the design of large-scale programs. They highlight how context-specific marriage-market features may determine the intensities of potentially opposing mechanisms, such as income and education effects, and how these can generate unintended consequences if not accounted for.

This paper adds to the research on how marriage markets and income fluctuations interact. Unlike Baird et al. (2011), who found that higher disposable income delays marriage in Malawi, this study presents the opposite results for rural Mexico. It also contrasts with Handa et al. (2015), who show no impact of unconditional cash transfers on child marriage in Kenya. Understanding marriage market responses to income fluctuations is crucial to understanding the impact of cash transfers. Corno et al. (2020) highlight how income shocks affect early marriage differently across dowry and bride-price systems, suggesting that transfer programs reduce early marriages in bride-price contexts like Malawi and Kenya. This paper's findings offer new insights into the effects of income on early marriage in settings without marriage payments. These are consistent with Bobonis (2011), who looked at the effect of 1.5 more years of Progres/Oportunidades

benefits and found an income effect on marriage among young adult mothers of beneficiary children.

This paper also contributes to the literature on the effect of educational programs on early marriage. My findings are consistent with and complement those of Behrman et al. (2005) and Araujo and Macours (2021). The authors compare the two treatment groups (those who received the program in 1998 and 2000, respectively) and document no significant effect of the program on marriage. By adding a pure control group to the analysis (which did not receive the program until the end of my analysis), I show that this null result conceals the fact that both groups increased their marriage rates compared to the control group. My results are also consistent with Parker and Vogl (2023), who use administrative data to study the long-run impact of the program. They find that the marriage rates of individuals aged 22 to 31 do not differ due to exposure to the program during childhood. Although a direct comparison between the sets of results should be done cautiously —due to the differences in samples and inclusion of marriages after 18 —my results do not conflict with theirs, if we believe that the program changed the intensive margin (when to marry) rather than the extensive margin (marry or not). An income effect can explain this. Therefore, I show evidence that a program like Progresa/Oportunidades can increase marriage if the income effect offsets the education effect. This can be the case, for example, if increases in education do not increase the opportunity cost of marriage. Consistent with this prediction, Gulemetova-Swan (2009) shows that Oportunidades' urban beneficiaries, for whom returns to education are positive, married 1-4 months later than non-beneficiaries. However, a caveat of this analysis is that, contrary to rural areas, where enrollment was close to universal within the randomly selected localities, in urban areas, there was a large extent of household self-selection into the program, hindering the ability to estimate a causal effect.

The rest of this article is organized as follows. Section 2 presents the Mexican context, and Section 3 introduces Progresa/Oportunidades, the data used in this project, and some relevant summary statistics. Section 4 explains and motivates the empirical strategy, and in Section 5, I present the primary analysis results, including a discussion on the relationship between these results and the literature. In Section 6, I provide evidence of the mechanisms in place. Finally,

Section 7 discusses the implications of these findings in terms of welfare and places them in the broad literature on Progresa.

2 Context

In recent decades, Mexico has experienced rapid and prosperous socioeconomic change, and the economic development has been significant for women. In Mexico, between 1970 and 2020, the percentages of women with secondary and tertiary education increased from 5% and 1%, respectively, to 38% and 22%, and female labor-market participation increased from 13% to 47% during the same period (Bhalotra and Fernández, 2021).

Usually, age at marriage correlates positively with economic progress, and early marriage is more prevalent in poorer societies. In Mexico, however, the average marriage age has only increased slightly, from 21.2 to 23 years, despite the strong economic growth in recent decades. For 1990 and 2010, one would predict a lower per-capita GDP for Mexico, given the country's average age at first marriage (see Figure A1). The percentage of women aged 20-24 in consensual unions has decreased from 51% (1993) to 39% (2018) (United Nations, 2024), but early marriage rates have remained relatively constant, around 21%. In 1993, 21% of women aged 20-24 had married before 18; there was a steady, slight increase until 2008, followed by a decline to 21% in 2018 (UNICEF, 2023b).³ Fertility, in Mexico, has been consistently decreasing over the past decades.

In Mexico, there are no widespread dowry practices, and bride prices and arranged marriages are rare. Children are the decision-makers. The main reasons offered to explain the high rates of early marriage in Mexico are: first, women in rural areas are mostly valued by their ability to create and sustain a family rather than their occupation; second, girls marry to initiate their sexual lives without the risk of the stigma attached to out-of-wedlock pregnancy; and third, to escape violent households and protect themselves from exploitative groups in areas

³Data on child marriage among boys are rare. But in Central and Latin America, child marriage rates for boys are estimated to be around 8%. Low rates in Mexico can be expected given two observations: (i) women in general marry older men; and (ii) 'Most girls age 12-17 who are in a union are at least six years younger than their partner(...)', according to Rivero and Palma (2017).

with extreme violence (Girls Not Brides, 2017; Taylor et al., 2019). Finally, marriage also offers economic stability, as formal insurance and labor-market opportunities are limited (UNICEF, 2019; Parrado and Zenteno, 2002).⁴

Most early marriages occur as informal unions. Nowadays, around 75% of girls aged 15 to 17 who were ever married or in a union report being in an informal union (United Nations Children’s Fund, 2019). In 1997, using data from the Mexican National Survey of Demographic Dynamics, I calculate that, in Mexico, about 55% of the girls between 15 and 17 who were in a union were in an informal one. Given this informality, tackling this problem through legislative changes might be inefficient. A change in state laws between 2014 and 2018, which forbade marriages under 18, led to a decrease in legal marriages, offset by an increase in informal unions (Bellés-Obrero and Lombardi, 2020). At the time of Progresa/Oportunidades’ implementation, the minimum legal marriage age varied by state.

Finally, in Mexico, schooling and marriage are not exclusive. According to Rivero and Palma (2017), in 2015, 17.1% and 8.15% of girls who were formally and informally married, respectively, were enrolled in school. School attendance during marriage rarely happens in countries often covered by the literature on early marriage.

3 Progresa/Oportunidades and Data

In 1998, a conditional cash transfer program, Progresa/Oportunidades, was implemented to reduce poverty and its intergenerational cycle in rural areas of Mexico by increasing educational attainment. There were three sets of actions: (i) offering basic health care to all family members; (ii) providing a fixed monetary transfer to be spent on food consumption and nutritional supplements, targeting children under two years old, malnourished children under five years old and pregnant and breast-feeding women; and (iii) monetary transfers to families with children in school, between the third grade of primary school and the third grade of secondary school.

⁴According to Székely Pardo (2003), 70% of the respondents resort to family first when facing problems with lack of money, almost 60% seek family help first to improve housing conditions, around 65% count on family in case of an accident, and 43% when they need a job. Family is a social institution in Mexico; the wider it is, the better insurance it provides.

The benefits scheme for 1998 is in Table 1. Benefits increased with grade and were slightly higher for girls than for boys in middle and secondary school. Transfers, on average, accounted for approximately 14% of eligible households' income (1,400 pesos, equivalent to 173 USD in 1998). To receive these transfers, eligible children had to attend scheduled medical visits and at least 85% of classes/school activities.

Table 1: 1998 Monthly Benefit (pesos)

Primary School			Secondary School		
	Boys	Girls		Boys	Girls
3rd Year	60		1st Year	175	185
4th Year	70		2nd Year	185	205
5th Year	90		3rd Year	195	225
6th Year	120				

Note: This table presents the benefits scheme of ProgresA in its first year of implementation. Children are eligible from the 3rd year of primary school until the third and last year of secondary school. Monetary benefits are increasing with schooling level and are slightly higher for girls than for boys in secondary school.

Eligible households were identified inside each locality through socioeconomic data collected in 1997, assessing their poverty status. On average, 78% of households in the treatment group were eligible for the program, and 97% of these households accepted beneficiary status (Dubois et al., 2012). In the analysis of this paper, I consider only those households within the surveyed localities that were eligible for the benefit (poor households).

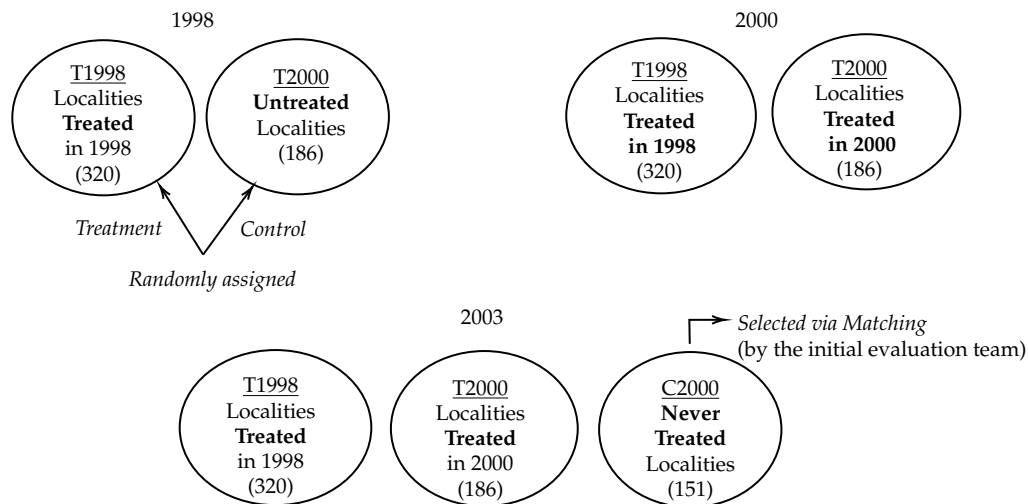
The program was first implemented in 320 randomly chosen rural localities (now referred to as T1998). A further 186 localities were randomly assigned to the control group. All these localities were highly deprived, with access to elementary school, middle school, and a health clinic (Abúndez et al., 2006). In December 1999, all localities in the control group started receiving the program (T2000).⁵ Households in T1998 and T2000 localities were surveyed in November 1997 (ENCASEH97) and March 1998 (before the introduction of the program), in October 1998, and twice in 1999 and 2000 (ENCELs).

In 2000 and 2001, the program underwent some changes, including its geographic expan-

⁵The last survey T2000 answered as a control group was set in November 1999, before the program's introduction. Therefore, for simplicity, I name this group T2000.

sion to T2000, and was renamed Oportunidades. Other significant changes for this analysis are extending benefits to high school (*preparatoria*) students and providing bonuses in case students pass grades.⁶ After the expansion of the program, to evaluate its long-term effects, in 2003, the evaluation team also surveyed a new set of localities that had not received the program until then. These localities were selected to match the treatment localities on aggregated locality characteristics obtained from the 1990 and 1995 censuses. These include housing and demographic characteristics, poverty levels, labor force participation, and ownership of durable goods. Besides, localities had to fulfill the program's eligibility criteria concerning distance to schools and health clinics. I refer to this set of localities as C2000, the pure control group. These 151 localities are also from the same states as the original 506 communities (except for one, for which the neighboring state was used). Figure 1 shows a diagram summarizing the program allocation across localities.

Figure 1: Treatment and Control localities



Note: This figure presents the three groups I will be comparing: T1998, 320 localities that were randomly selected to start treatment in 1998; T2000, 186 localities that were randomly selected to be the control group in 1998, who were then included in the program in 2000; C2000, 151 localities that were selected in 2003, via propensity score matching, to be the pure control group.

In 2003, a new survey (ENCEL2003) included all the households found in T1998, T2000, and C2000. The survey asked the control group, C2000, current and retrospective questions,

⁶Oportunidades introduced *Jóvenes con Oportunidades*, a component of the program that awards a monetary prize to those students who completed high school in less than 4 years and before turning 22.

referring to 1997, 2000, 2001, and 2002.⁷

The design of the program allows for comparisons across the three groups: (i) T1998 is the group of treated localities in 1998; (ii) T2000, the set of localities that started receiving the program only from 2000 onward; and (iii) C2000, the pure control, or the group of localities that did not receive the program until 2003. I use data collected through the ENCASEH and ENCEL to evaluate the program's effect from its start until 2003.

The main outcome of interest for this analysis is *marital status* from 1997 to 2003. I consider an individual married if they report being legally married, living in an informal union, cohabiting, being divorced, or widowed. I choose to do so since I am interested in first marriages and thus do not account for separations. A child is single if she reports her status to be single.⁸

My population of interest is all children aged 6 to 16 in 1997, the baseline year.⁹ Keeping all those whose relevant information is not missing, I have 25 thousand observations, roughly half of which are females. Recall that I am considering only poor households within each locality, meaning those eligible for the program.

Summary Statistics Table 2 presents the proportion of married individuals by group and year for the whole sample (Panel (a)) and those younger than 18 (Panel (b)). In the first three rows of the descriptive tables, I present the unweighted average of married individuals in each group, T1998, T2000, and C2000. Due to the non-randomness of the pure control group, C2000, I also present in the fourth and fifth rows the weighted average of married individuals for the pure control group, weighted by the probability of being first treated in 1998 and 2000, respectively, versus being in the control group.

In the first years of analysis, from 1997 to 1999, there are more married individuals in the

⁷I use information from the 2007 survey to fill in missing data on year and age at marriage.

⁸In ENCEL2003 and ENCEL2007, individuals were asked about age at first marriage or union. This information allows me to retrieve the year and age at marriage for individuals for whom I did not have that information in previous surveys and for those who married after 2003. The data from 2003 allows me to check for consistency in civil status reporting. I use the 2007 data only for descriptive statistics and to complete the marriage status when information is missing from the other surveys.

⁹Only 1.5% of ever-married children declared marrying younger than 12; therefore, I assume that a child becomes at risk of marriage only at that age. I excluded from the sample all children who had not turned 12 by 2003. I do not consider children who were over the age of 16 in 1997 either, given that they were exposed to the program near their 18th birthday.

control group, but the proportions are fairly close across groups - albeit statistically different -, starting to diverge after 1999. After 2000, there were more married children in the treatment groups than in the control group. Note that for the empirical strategy used in this paper, and described in the next section, child marriage rates do not have to be the same at baseline, since I will rely on the parallel trends assumption. The proportion of girls who are married is systematically larger than the proportion of boys who are married across all groups (see Tables A2 to A3 in Appendix A).

Of those reporting age at marriage, 1.5% married before turning 12, 18.6% married between 12 and 15, 30% married at 16 or 17, and almost 50% at 18 or later. From those who married before turning 18, 60% married at 16 and 17 (see Figure A2 in the Appendix). The average age at marriage is 1.1 years lower for girls (17) than for boys (18.1).

Table 2: Proportion of Married by Group and Year (in %)

(a) All							
	1997	1998	1999	2000	2001	2002	2003
T1998	0.79	1.69	3.25	5.82	10.26	13.38	15.68
T2000	0.92	1.85	3.43	6.10	11.00	14.17	16.52
C2000	1.46	1.99	2.99	4.37	6.66	9.74	11.80
C2000(IPW1998)	1.64	2.43	3.77	5.12	7.42	10.14	12.17
C2000(IPW2000)	1.69	2.57	3.63	5.04	7.29	10.07	12.01

(b) Under 18 years old							
	1997	1998	1999	2000	2001	2002	2003
T1998	0.79	1.60	2.58	3.49	4.56	5.29	5.50
T2000	0.92	1.77	2.65	3.41	5.05	5.68	5.89
C2000	1.46	1.99	1.66	1.86	2.22	2.90	3.29
C2000(IPW1998)	1.64	2.43	1.64	1.66	2.05	2.38	2.91
C2000(IPW2000)	1.69	2.57	1.79	1.71	2.07	2.40	2.64

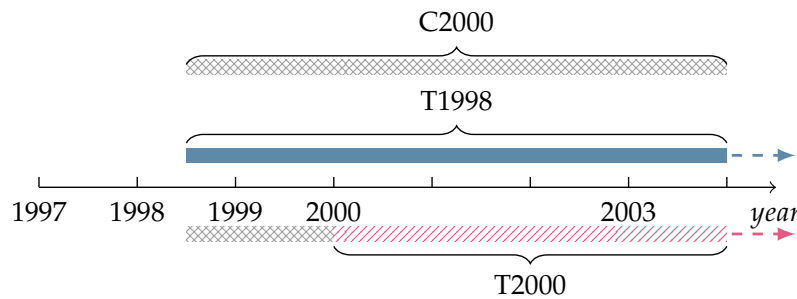
Note: This table presents the proportion of married individuals by group and year. Panel (a) refers to all individuals between 6 and 16 years old in 1997, and Panel (b) refers to the same individuals until they turn 18. T1998 are those individuals who started receiving the program in 1998. T2000 is the set of individuals who first received the program in 2000. C2000 is the control group. C2000(IPW1998) and C2000(IPW2000) are the control group weighted by the probability of being first treated in 1998 and 2000, respectively, versus being in the control group.

4 Empirical Strategy

To estimate the program's causal effect on early marriage, I exploit the random and quasi-random allocation of the program across localities and the variation in the timing of implementation. Recall that I have information on three groups: (i) T1998, the group of localities (318) receiving the treatment in 1998 and beyond; (ii) T2000, the group that first received treatment in 2000 (182 localities); and (iii) a pure control group, C2000, which was never treated in the analysis period until 2003 (143 localities). In total, the analysis covers 643 localities. I observe these groups from 1997 until 2003.¹⁰

As described before, the program was randomly allocated at the start of the implementation. Localities in T1998 were randomly selected to receive the treatment in 1998, and localities in T2000 were randomly selected as the control group. In 2000, the control group was incorporated into the program and started receiving the benefits. C2000 is the group of localities non-randomly selected in 2003, to be the pure comparison group. Figure 2 illustrates the program allocation across groups, the years of the analysis, and the role they represent in the empirical strategy.

Figure 2: Treatment and Control Groups Across Years of Analysis



Note: This figure presents the three groups I will be comparing: T1998, in full and blue, the first treated group; T2000, in both crosshatched gray and pink slanted stripes, to emphasize that the same group of localities is a control group until 2000 (crosshatched gray) and joins the treated group from that year onward (pink slanted stripes); and C2000, the control group selected through propensity score matching which was never treated, crosshatched and gray.

The staggered implementation of the program and the rich panel structure of the data allow

¹⁰The analysis covers 14 fewer localities than were surveyed. This is due to the processing of the data, exclusion of observations with important missing information, etc. For more details on the data processing, see Appendix B.3.

the estimation of dynamic causal treatment effects by comparing the three groups over 6 years. Therefore, I will use a differences-in-differences estimator that accounts for multiple periods and staggered treatment allocation. In particular, I use the doubly-robust estimator proposed by Callaway and Sant’Anna (2021) for three reasons. First, it has been shown that, in staggered designs, two-way fixed-effect models with staggered treatment cannot be interpreted causally when treatment effects are heterogeneous. The intuition behind this is that the estimate for the causal effect at a certain period might be contaminated by the treatment effects from other periods, even if the parallel trends and no anticipation assumptions hold.¹¹

Second, this estimator allows us to use individual pre-treatment characteristics for more credible parallel trend assumptions. Controlling for these characteristics allows us to compare more similar individuals across the groups of localities. Improving this comparison is particularly important in this setting. The treatment allocation was done at the locality level, and the empirical analysis is at the individual level, thus the randomization does not guarantee balancing. Additionally, the pure control group (C2000) is not an experimental group. On average, unobserved characteristics of localities in T1998 and T2000 are uncorrelated with treatment allocation due to the random assignment to the program across these localities. However, the selection of localities in C2000 assumes that, given the observed characteristics, the treatment allocation was as good as random. Including individual attributes strengthens the plausibility of the assumption since the comparison is then across similar individuals in similar municipalities. The estimator proposed in Callaway and Sant’Anna (2021), henceforth the CS estimator, allows for the use of pre-treatment characteristics by combining outcome regression and inverse probability weighting. Outcome regression adjustment allows for covariate-specific trends in potential outcomes across groups. For example, if the potential outcome (marriage) evolution in the case of non-treatment depends on covariates (e.g., gender and age), conditional parallel trends are less restrictive. The causal treatment effect is identified as long as the remaining unobserved characteristics affecting the outcome are time-invariant. Inverse probability-weighting allows re-weighting the observations by the estimated treatment assignment probability to improve

¹¹See, for example, Goodman-Bacon (2018), Athey and Imbens (2022), Borusyak and Jaravel (2018), de Chaisemartin and D’Haultfœuille (2020), Callaway and Sant’Anna (2021), Sun and Abraham (2021).

comparability across groups. The identifying assumption is that, conditional on these characteristics, the treatment assignment was as good as random. Third, this doubly robust estimator identifies the average treatment effect for each group at a given point in time, even if either the propensity score model *or* the outcome regression models are misspecified, but not both.

The CS estimator identifies a group-time causal effect if the following assumptions hold. First, I need to assume that the overlapping condition is satisfied. Meaning that at least a small fraction of the population is treated at each ‘starting’ period (when treatment starts for each group) and that, for all periods, the propensity score is uniformly bounded away from one.¹² Second, treatment must be irreversible, meaning that, if a group is treated at time t , then it is treated at $t + 1$ for any t , which this design satisfies. The third assumption requires limited treatment anticipation: individuals could not anticipate that they would be beneficiaries of the program prior to its implementation. Attanasio et al. (2012) find no evidence of anticipatory behavior by any of the cohorts. The fourth and final assumption is the conditional parallel trends assumption: in the absence of treatment, the average conditional outcome of the group first treated at a given year and the groups not yet treated would have evolved in parallel.

A common practice used to provide evidence on the plausibility of the parallel trends assumption is to test whether there are different pre-treatment trends for treated and control groups. The idea is that, conditional on observed characteristics, the change in the outcome that the treated group would have if they had not participated in the treatment is the same as the change observed for the untreated group. Conditional on the observed characteristics, the groups’ evolution differs only by treatment status.

The estimand of interest is the average treatment effect at time t for the group that was first treated in period g , using the groups that were not yet treated for comparison. It is defined as

$$ATT_{dr}^{ny}(g, t) = \mathbb{E} \left[\left(\frac{G_g}{\mathbb{E}[G_g]} - \frac{\frac{p_{g,t}(X)(1-D_t)(1-G_g)}{1-p_{g,t}(X)}}{\mathbb{E} \left[\frac{p_{g,t}(X)(1-D_t)(1-G_g)}{1-p_{g,t}(X)} \right]} \right) \left(Y_t - Y_{g-1} - m_{g,t}^{ny}(X) \right) \right],$$

¹²In practice, I exclude from my total sample 15 observations that have an estimated propensity score higher than 0.999. See Figures A3 to A8 for the distribution of the propensity scores. The main results of this paper are robust to restricting the analysis to observations with a propensity score less than 0.8 and between 0.2 and 0.8.

where $g \in \mathcal{G}$ is the first treatment year for a given cohort, or group, $p_{g,t}(X)$ is the propensity score, or the probability of being first treated in period g conditional on covariates X and conditional on either being treated the first time at g , ($G_g = 1$), or ‘not yet treated’, $((1 - D_s)(1 - G_g) = 1)$. Y_t is the outcome of interest at time t , and Y_{g-1} is the outcome at baseline before the unit is treated. Finally, $m_{g,t}^{ny}(X)$ is the expected outcome evolution from baseline to time t , conditional on covariates X for the ‘not yet treated’, $m_{g,t}^{ny}(X) = \mathbb{E} [Y_t - Y_{g-1} \mid X, D_t = 0, G_g = 0]$.

The estimation follows a two-step strategy. The first step estimates the propensity score and outcome regression, $p_{g,t}(X)$ and $m_{g,t}^{ny}(X)$. In the second step, the fitted values of these estimands are plugged into the sample analog of the ATT to obtain its estimate. I cluster the standard errors at the locality level, since this was the unit of treatment randomization. Standard errors are estimated using 10 thousand bootstrap iterations. I present two types of confidence intervals. In the figures showing the treatment effects, because the object of interest is the dynamic path of the effect, I present 95% simultaneous (uniform) confidence bands, which cover the entire path of dynamic effects jointly. Such a band is constructed so that, with 95% probability, the true $ATT(t)$ lies within it at every event time simultaneously - in contrast to the pointwise 95% intervals reported in the tables, which have that coverage one event time at a time.

Threats to Identification The pure control group, C2000, was selected by the program’s evaluation team using a matching model to select localities that were the closest possible to the treatment groups. However, we cannot guarantee that, on average, these groups are equal in observed and unobserved characteristics. Since we cannot control for potential unobserved differences across individuals in the treated and control localities, it is important to ensure that we compare individuals for whom, conditional on a set of characteristics, treatment was equally likely and/or for whom outcomes would have evolved similarly. Using a differences-in-differences design allows us to relax the assumption that the groups were identical at baseline. It only requires assuming parallel trends, as explained before. Nevertheless, to increase the likelihood that this is the case, I use a doubly robust estimator, which requires specifying the propensity score model and the outcome regression models. I use two sets of characteristics: (i) those that are important to determine outcome progression — motivated by the literature on

the determinants of marriage; (ii) and those that are determinants of treatment status — stated and used by the program authorities. Despite the different motivations for including the variables (either their relevance to the evolution of the outcome or to the treatment status), both models include all variables. Table A4 presents the summary statistics of all variables used in the models, as well as the outcome variable, for all groups. Tables A5, A6, A7, A8, A9, and A10 show the balance test of these characteristics across groups and gender.

The final specification of both outcome regression and propensity score models is very close to the one used by Diaz and Handa (2006), who show that propensity-score matching performs well in the evaluation of Progresa, replicating the RCT results.¹³ They show that, for outcomes that are measured comparably across survey instruments, which is the case of marriage, matching estimates on a non-experimental sample are not statistically different from the experimental estimates. They also show that the larger the set of (relevant) covariates, the larger the reduction in the bias. Assuming that a specification that yields unbiased estimates in sample also does so out of sample, we can interpret this as evidence that the propensity score model is well specified, which supports the assumption that the used doubly-robust differences-in-differences estimator is identifying the ATT.¹⁴

Another caveat I need to address is that, for the pure control group (C2000), the pre-treatment information on the used covariates is recall data collected in 2003 regarding 1997. Therefore, there could be recall bias regarding the baseline characteristics, which could then lead to biased estimates. Since the recall data was only collected for C2000 and not T1998 and T2000, it is hard

¹³My specification includes the same variables as those used in Diaz and Handa (2006), except for access to social security. I add more variables that are important determinants of wealth, treatment heterogeneity, and marriage. My specification includes: **Housing characteristics**: dummy variables for dirt floor, inferior-quality wall, inferior-quality roof, number of bedrooms, piped water, electricity, ownership of animals, land, blender, refrigerator, gas stove, gas heater, radio, TV, dishwasher, car or truck; **Household composition**: the number of members in the household and dummy variables for having at least one child between 0 and 5, at least one teenager between 16 and 19, at least one woman between 20 and 39, 40 and 59, and 60+, respectively, and at least one man between 20 and 39, 40 and 59, and 60+, respectively; **Marriage predictors**: gender, age, education level at baseline, indigenous background, and household head and spouse characteristics.

¹⁴My specification is also similar to the one used by Behrman et al. (2011), which estimates the program's effect on education. The most significant difference is that I am not using missing variable flags; instead, I am losing the observations for which there is no information on baseline characteristics (see Appendix B.3). Despite these differences, **I can replicate the results reported in the paper mentioned above regarding the program's effect on educational achievement.**

to judge the accuracy of this data. One way of sensing how problematic this might be is to remove from the propensity score and outcome regressions the variables that are more likely to be subject to recall bias, such as asset possession. I kept those unlikely to be affected by that issue, such as household composition and parental education. The results are robust to this specification (see Tables A11, A12 and A13). Additionally, I looked at control localities in the 1990 census, and I observe that the summary statistics of the baseline characteristics are very close to C2000, for similar populations (for example, children between 6 and 16 - as used in this paper - and children between 0 and 9 in 1990 - those who would be between 6 and 16 in 1997). That the means of the variables of interest are the same as in the census is reassuring regarding the absence of recall bias. For a more detailed discussion on this, see Section B.2.

One might worry that marriage status may be subject to misreporting in the ENCASEH and ENCEL data, particularly among adolescents, due to social stigma or misremembering civil status. This misreporting can also differ by treatment status. To assess the reliability of marriage reporting, I conducted a series of empirical exercises: (i) comparing the self-reported data I use in this paper's analysis with Census data and administrative records, (ii) examining potential inconsistencies in marriage-status reports, and (iii) testing the role of treatment on misreporting. The results, presented in detail in Section B.1, suggest that marriage reporting is reliable, with similar proportions of individuals reporting marriage status across datasets, very low rates of inconsistencies across the years, and no significant correlation between treatment intensity and the probability of misreporting child marriage. Overall, I conclude that marriage misreporting is unlikely to be a significant problem in my analysis. Note that early marriage was a common and widely accepted practice in the late 1990s and early 2000s, particularly in rural areas, and therefore unlikely to be stigmatized (Wiedemann, 2021; Taylor et al., 2019). Marriage, whether in a formal or informal union, is also an extremely important decision girls make and are unlikely to forget. I have also tested whether treated and control localities differed in their marriage rates prior to the program being implemented. I used the 1990 Census and 1995 *Encuesta*/Survey data to show that there was no significant difference in child marriage rates in these years. This analysis provides additional support for the plausibility of the parallel trends assumption. More details can be read in Section B.2.

Additionally, an in-depth analysis of attrition and missing data is available in Appendix B.3. Since attrition increases with years (it is higher for T1998 than T2000, statistically significant from November 1999 onward), and the treatment groups are more likely to have missing information regarding marriage, I perform a series of robustness checks where I show that attrition is mostly driven by household attrition, rather than individual. I also show that observed characteristics do not simultaneously predict marriage and attrition, and that despite attrition and marriage being positively correlated, this does not differ by treatment group. These exercises are detailed in section B.3. Additionally, using Lee bounds with inverse probability weights and tight bounds, and treating the data as a repeated cross-section, I estimate a lower bound for the aggregate effect for girls of 2 p.p., statistically different from zero at 1%, CI=[0.016, 0.025]. Besides attrition, some individuals' ages do not progress as expected, or their gender changes. These might indicate a mismatch in the IDs or misreporting of gender or age. For the main analysis, I exclude all those observations in which gender is inconsistent and age decreases. If I am stricter and drop those observations that show any inconsistency in age (either decreasing or unreasonably increasing), I obtain qualitatively similar results with larger magnitudes. A third problem concerns missing data regarding baseline characteristics, mainly in the control group. I exclude all observations for which I do not have complete information on these characteristics. Imputing missing values would introduce bias in the propensity score estimates due to the non-zero covariance across the predictors. Finally, 34% of the control group lacks baseline education information. Since the literature suggests that education is an extremely important predictor of marriage decisions, I further exclude those observations with missing education, rather than excluding the variable from the outcome and propensity score regressions.

5 Results

5.1 Probability of Early Marriage

I start by analyzing whether the program has affected the probability of early marriage. In this set of results, unless otherwise stated, I consider only individuals until they turn 18. Table 3 shows that the program increased, on average, the probability of early marriage by 2.4 percentage points (p.p.) (with the lower bound of the 95% confidence interval at 0.008 and the upper bound at 0.039; hereafter, CI=[0.008, 0.039]), significant at 1%. This effect corresponds to more than doubling the marriage rate compared to the control group (the average marriage rate for C2000 is 2%).

Table 3: Progresa/Oportunidades Average Treatment Effect on Early Marriage

	All		
	All	T1998	T2000
ATT	0.0237*** (0.008) [0.0081 , 0.0393]	0.0221** (0.0103) [0.0018 , 0.0423]	0.0267*** (0.0092) [0.0087 , 0.0447]
Control Mean		0.02	0.021
Effect in %		107.82	127.87
N		25645	

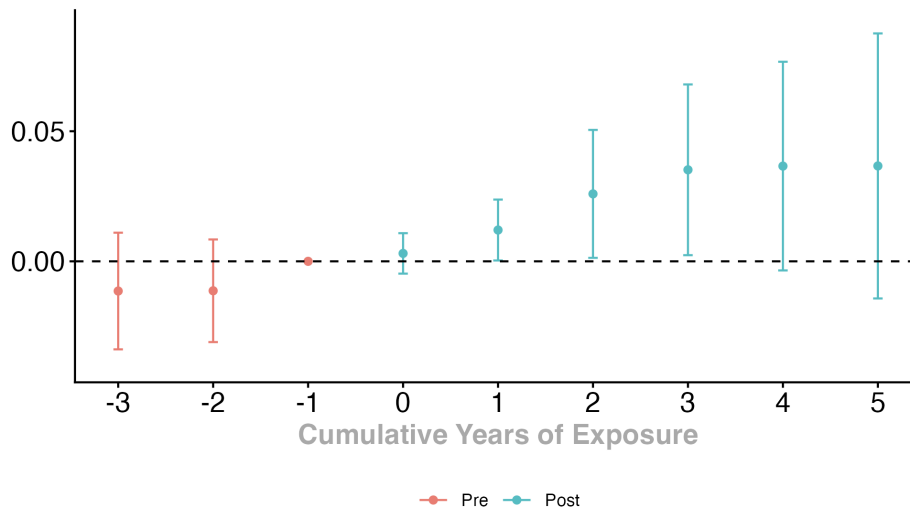
Note: This table presents the aggregated average treatment effect on the treated. 'All' represents the estimate using treatment groups T1998 and T2000. The second and third columns separately present the average treatment effect over time for treatment groups T1998 (who first received the treatment in 1998) and T2000 (who first received the treatment in 2000). Standard errors are clustered at the randomization level (locality). The average marriage rate for C2000(IPW1998) is 2% and for C2000(IPW2000) 2.1%. N represents the number of observations used for the analysis. Asterisks denote pointwise statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Then, I explore how this effect varied with the length of exposure to Progresa/Oportunidades. Figure 3 shows the effect of the program on the probability of being married by the number of years exposed to the benefit (these results are also in Table A14 in the Appendix). For instance, time -1 represents one period before treatment, so for group T1998, $t = -1$ corresponds to 1997, and to 1999 for group T2000. Similarly, time 2 represents two years after treatment. Note that

the effects in times 4 and 5 are only estimated using T1998, the only group treated for more than 3 years in the studied period. It is similar for period -2, which is only observed for T2000. In the pre-treatment periods, I do not reject the null hypothesis of no effect of the program at any conventional significance level, supporting the plausibility of the parallel trends assumption.

Then, I observe that the program did not affect early marriage in its first year of implementation ($t = 0$). However, it began to lead to increased marriage rates after one year of exposure. One year after receiving the benefit ($t = 1$), treatment groups were 1.2 p.p. (CI=[0.003, 0.021], statistically significant at 1%) more likely to be married than the control group. This effect increased to 3.5 p.p. (CI=[0.010, 0.061], significant at 1%) in the third year and 3.7 p.p. (CI=[-0.003, 0.077], significant at 10%) after five years. For reference, the unconditional and unweighted proportion of married individuals in the control group was about 3% in 2003, so the effect corresponds to more than doubling marriage incidence.

Figure 3: Progresa/Oportunidades' Effect on Early Marriage, by Length of Exposure



Note: This figure presents the average treatment effect on the treated by the length of exposure to treatment. Time -1 represents one period prior to treatment. For T1998 (the group that first received the treatment in 1998), time -1 corresponds to 1997, and for T2000 (the group that first received the treatment in 2000) to 1999. Period 2 represents two years after treatment, and so on. In red are the estimates before treatment started, and in blue after treatment started. Standard errors are clustered at the randomization level (locality). Confidence intervals shown are 95% uniform (simultaneous) confidence bands from the multiplier bootstrap of Callaway and Sant'Anna (2021).

Across treatment groups, I observe a positive trend in the estimated coefficients one year

after the program started for T1998. Though noisy, the point estimates for T1998 are relevant from 2000 onward. In 2000, T1998 beneficiaries were 2.3 p.p. (CI=[-0.004,0.045], significant at 10%) more likely to be married before 18 than non-beneficiaries. The estimate increases to 3.7 p.p. in 2002 (CI=[0.005, 0.068], significant at 5%) and remains constant in 2003 (CI=[-0.003, 0.076], significant at 10%), over twice as likely as the control group (C2000(IPW1998)'s marriage rate is 2.91% in 2003). Figure A9 and Table A15 in the Appendix show these results.

For the second treatment group, T2000, the program increased marriage after the first year of implementation. In 2001, the effect is 2.8 p.p. (CI=[0.004, 0.051], significant at 5%), 3.2 p.p. (CI=[0.01, 0.054], significant at 1%) in 2002 and 3.8 p.p. (CI=[0.009, 0.067], significant at 5%) in 2003. These results hint that the changes made in the program around 2001 (from Progres a to Oportunidades) were important in explaining the program's positive effect on marriage.

Since the analysis is restricted to children aged 6-16 in 1997, nearly 80% of individuals in the sample are under 14 in 1997, and approximately 60% are under 14 by 1999. A potential concern is that marriage rates may be uniformly low across all treatment and control groups in the pre-treatment periods, thus the pre-trend comparison in the main analysis might be uninformative. To address this, I test for pre-treatment differences between treatment and control groups in three subsamples: (i) those aged 10-17 in 1997; (ii) those aged 12-17 in 1997; and (iii) those aged 14-17 in 1997.¹⁵ These groups are at greater risk of early marriage during the pre-treatment periods. Tables A16, A17, and A18 show that children in these age ranges in treatment and control groups also do not differ regarding marriage behavior before the program was implemented.

5.2 Heterogeneous Effects

5.2.1 Gender

Around the world, early and child marriage is a more prevalent phenomenon among girls than boys. Also, the socioeconomic consequences associated with early marriage are known to be more damaging for females than males due to early childbearing, exposure to violence, and a

¹⁵These age brackets take into consideration that before 12 years old girls were essentially never married, so I split the samples such that there were mostly 'prone-to-marry' children at pre-treatment years (1997 and 1999).

higher likelihood of formal labor-market exclusion (Fan and Koski, 2022; Parsons et al., 2015; Edmeades et al., 2022). Hence, in this section, I examine the program's heterogeneous effects by gender. Table 4 shows that the large treatment effects on girls drive the overall effects. On average, the program increased the probability of early marriage for girls by 3.8 p.p. (CI=[0.014, 0.062], significant at 1%). This effect is not significant for boys, 0.8 p.p. (CI=[-0.008, 0.024]).

Table 4: Progresa/Oportunidades' Average Treatment Effect on Early Marriage, by Gender

Girls			
	All	T1998	T2000
ATT	0.0377*** (0.0122) [0.0138 , 0.0615]	0.034** (0.014) [0.0065 , 0.0614]	0.0442*** (0.0165) [0.0118 , 0.0766]
Control Mean		0.03	0.032
Effect in %		112.25	139.38
N		12357	
Boys			
	All	T1998	T2000
ATT	0.0078 (0.0083) [-0.0085 , 0.024]	0.0077 (0.0082) [-0.0082 , 0.0237]	0.0078 (0.0137) [-0.0191 , 0.0347]
Control Mean		0.01	0.01
Effect in %		77.10	77.16
N		13288	

Note: This table presents the aggregated average treatment effect on the treated by gender. In the first column of each gender, 'All' represents the estimate using as treatment groups both T1998 and T2000. The second and third columns present the average treatment effect over time for treatment groups T1998 and T2000, respectively. Standard errors are clustered at the randomization level (locality). Asterisks denote pointwise statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 4 shows the dynamic effects of the program for both girls' and boys' marriage decisions. After 1 year of exposure to Progresa/Oportunidades, girls living in a beneficiary locality were, on average, 1.9 p.p. (CI=[0.004, 0.034], significant at 5%) more likely to be married (see Table A19). After 5 years, the probability of early marriage increased by 6 p.p. (CI = [0.015,

0.106], significant at 1%) due to the program. In 2003, in the weighted control groups, around 3.5% of the girls were married (see Table A2); thus, the program more than doubled the likelihood of marriage for girls. The point estimates are positive and increasing for girls in both treatment groups across the years. However, it is after 2001 that they start being meaningful (see Figure A10 and Table A20 in the Appendix for the estimates for each treatment cohort separately). Looking at the pre-treatment coefficients, there was no difference in marriage behavior before the program was implemented, providing evidence that the effect stems from the program rather than from pre-existing differences. Similarly to the exercise for the whole sample, I have extended the test of pre-treatment differences for age groups at highest risk of marriage before the program was implemented, and I reach the same conclusion (see Tables A21, A22, and A23).

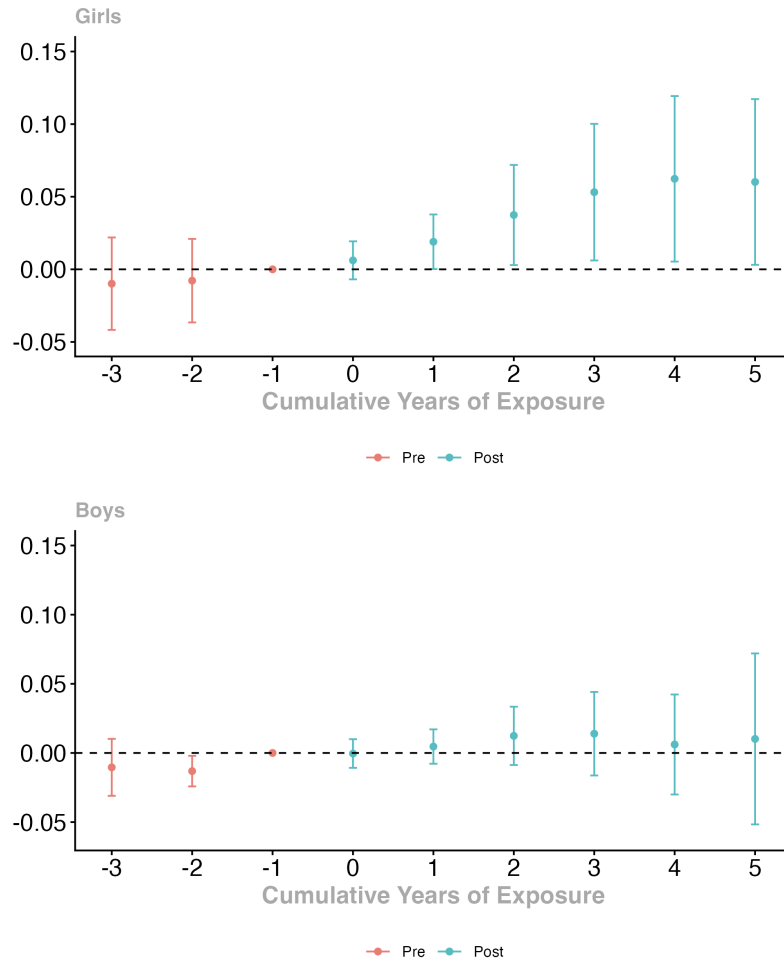
Results for boys, presented in the bottom panel of Figure 4 (and in Figure A11 in the Appendix), should be interpreted with caution, as I reject the null hypothesis of no pre-trends. Before the program started for boys in T2000, there was a negative trend, which hints at a different pre-treatment behavior. Thus, the post-treatment results may not be due to the program, but a product of those pre-existing differences. Despite overall positive point estimates, most are not statistically different from zero and have low magnitudes. For the disaggregated results by treatment group, see Tables A24 and A25, in the Appendix.

Those girls for whom I have information on their partners' ages were, on average, 3.5 years younger than their partners. 60% of these girls married older men, so I look at the program's effect on young men up to 30 years old at baseline. For this population, I find that older men in eligible and non-eligible households in treated localities were more likely to marry than those in control localities.¹⁶

In summary, after Progresa/Oportunidades was introduced, girls in beneficiary localities eligible for the program were more likely to be married before age 18 than similar girls in localities that did not receive the conditional cash transfer program. The same does not happen for boys under 18.

¹⁶See Table A26. As in Bobonis (2011), I also find a positive effect among older women who were single at baseline.

Figure 4: Progresa/Oportunidades' Effect on Early Marriage, by Year and Gender



Note: This figure presents the average treatment effect on the treated by length of exposure for girls and boys separately. In red are the estimates before treatment started, and in blue after treatment started. The top panel restricts the analysis to girls, and the bottom panel to boys. Standard errors are clustered at the randomization level (locality). Confidence intervals shown are 95% uniform (simultaneous) confidence bands from the multiplier bootstrap of Callaway and Sant'Anna (2021).

5.2.2 Age

Since marriage is positively associated with age, I investigate whether the program had heterogeneous effects across this dimension. Given the results in the previous section, I restrict this analysis to girls.¹⁷ I split the sample into three age groups, defined at baseline: (i) girls aged between 6 and 8 in 1997, (ii) girls from 9 to 11 years of age, and (iii) girls from 12 to 14 years old. Recall that I stop considering individuals once they turn 18. Therefore, the last year I observe the oldest group is 2002, since in 2003 all of these children would have turned 18. For the same reason, I do not consider girls 15 and 16 years old at baseline, since I would not have post-treatment periods for those in T2000.

Figure A12 shows the effect of Progresa/Oportunidades on early marriage separately for girls in T1998 (left panel) and T2000 (right panel). The light blue circles represent the estimates for the age group 6-8, the dark blue triangles for ages 9-11, and the red squares for the 12-14 cohort. Note that girls in T1998 are being compared to those in T2000 until 1999 (including) and those in the pure control group, C2000 across all years. Those in T2000 are being compared exclusively to the pure control group. The fewer observations in each age group make the estimates noisier, but the point estimates are consistent with the aggregate results. I find positive point estimates across all ages and treatment groups, with estimates larger for the two older groups, despite the oldest group being underpowered. Those girls aged 13 to 17 between 2001 and 2003 are the ones for whom the program has the largest and statistically significant effect. The magnitude of the effect increases with age, but conditional on age, there is no difference in the effect across treatment groups. This suggests that the length of exposure to the program does not affect marriage decisions. What appears relevant is exposure to the program and the age at which it occurs.

Age and censoring. Because the sample includes children as young as 6 in 1997, the youngest cohorts are right-censored: I do not observe them reaching 18 within the analysis window, so I cannot see whether they marry between their last observed age and 18. The full-sample

¹⁷Analyzing just boys, results suggest positive but small effects at younger ages and no significant effect for the oldest age group.

estimates could therefore be a lower bound on the program's effect on marriage before 18. To have a sense of the magnitude for a fully-observed cohort, I restrict the analysis to girls aged 12-16 in 1997, all of whom turn 18 by 2003. Unlike the age breakdown above, which caps the cohorts at 14 so that both treatment groups have post-treatment periods, this cut prioritizes observing the complete marriage-risk window to 18, at the cost of including girls (the oldest, in T2000) who are observed for only a short period. As a consequence, the estimates are noisier. Nevertheless, the point estimates are large: after one year of exposure, treated girls are roughly 3 p.p. (CI=[-0.017, 0.082]) more likely to be married, rising to about 11 p.p. (CI=[-0.014, 0.231], significant at 10%) after 4 years (Figure A13 and Table A27).

5.3 Discussion

The experimental component of Progresa compares T1998 to T2000. In 1998 and 1999, since T2000 had not yet received the program, I identify the effect of being a beneficiary versus not. From 2000 onward, both groups received the program, and the comparison identifies the effect of T1998's 1.5 additional years of exposure. As presented in Tables A28, A29, and A30, and in Figure A14, under this design I do not find any impact on child marriage. This null is consistent with two features of my main results: the program's effect on marriage takes more than 1.5 years to materialize, so the 1998-1999 window is too short to detect it, and once both groups are treated, the residual difference is small. These findings are consistent with and complement those of Behrman et al. (2005), Araujo and Macours (2021), and Parker and Vogl (2023). The authors use the experimental component of the program and estimate the effect of receiving 1.5 additional years of the program by comparing the outcomes of the two treatment groups. Behrman et al. (2005) examines marriage rates in 2003 among individuals aged 9 to 15 at baseline (1997). They find no statistically significant difference in marriage rates between T1998 and T2000. Their results are not directly comparable to the ones presented in this paper since they do not focus on child marriage (they include marriages after 18). Araujo and Macours (2021) also compares T1998 and T2000 in 2017, 20 years after the start of the program. Similarly, they find no significant effect of the program on the probability of marriage. They find a half-year

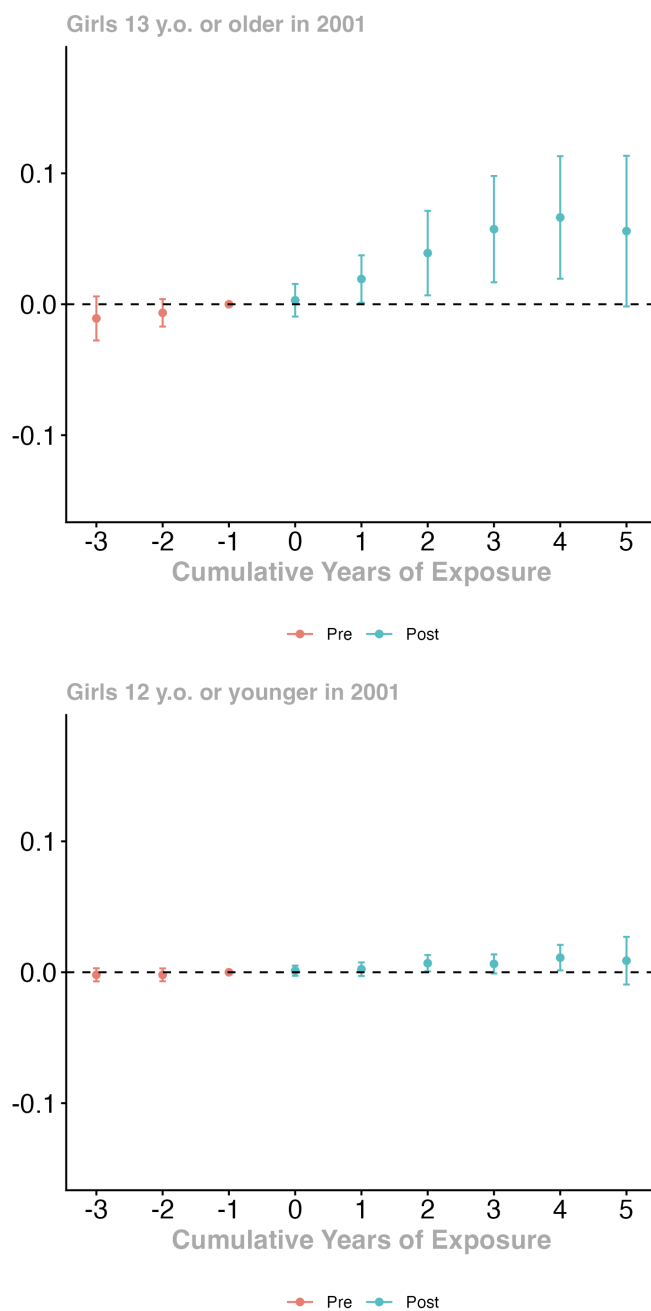
delay in marriage age, but the delay is not statistically significant for women. Like Behrman et al. (2005), they do not restrict this analysis to marriages below 18. Finally, Parker and Vogl (2023) use 10% of census data and a differences-in-differences strategy to compare individuals who were fully exposed to the program (younger than 14 in 1997) to those who were 15 and older and for whom the program did not affect schooling. They interact the age cohort dummies with a measure of the program's incidence during the first roll-out phase. They find that the program did not affect the likelihood of marriage. By 2010, when individuals are between 22 and 31, treated individuals are as likely to be married as control individuals.

The caveat in Progresas's experimental analysis is that we can only test the program's effect for 1.5 years. From 2000 onward, the comparison is always regarding the length of exposure to the program. This does not allow us to test the effect of receiving versus not receiving the program after this year, nor to take into account the changes that occurred in that same year. The differences-in-differences analysis overcomes these drawbacks since it allows us to analyze the effect of the program for up to 6 years, by calendar year and exposure length. Using this strategy, I find both treatment groups increased their marriage rates from 2000 onward, whereas the pure control group did not (see Figure A10). I also find that the program began to significantly affect marriage decisions after the year 2000, which might be explained by the changes implemented by the program, particularly the extension of benefits to high school.

Students are supposed to reach high school at around 14-15 years old if they do not repeat any year. Since most early marriages in Mexico happen between 15 and 17 years old, it might be the case that receiving the benefit at this age facilitates marriage decisions for girls in treated localities. In an exercise similar to the one executed by Parker and Vogl (2023), I compare the program's effect across two groups: (i) all the girls 13 or older in 2001, who were likely to be in high school between then and 2003 - highly exposed, and (ii) all girls 12 or younger in 2001, who were unlikely to be in high school during this period - not exposed. Figure 5 shows that if we look at all girls who were likely to be in high school in the analysis period, those who could receive the benefit during high school (treatment groups) were more likely to marry than those who could not receive the benefit (control group). If, in turn, we look at the effect of the program on younger girls, we observe positive and statistically significant effects, but in a much

smaller magnitude. The effect is significantly larger for the older cohort than the younger one (see Figure A15 in the Appendix). Note that these results do not conflict with Parker and Vogl (2023) if we believe the program did not change the extensive margin on whether to marry, but it affected the intensive margin and anticipated the decision for those individuals who would marry in the following years. Meaning that those individuals who were already likely to marry had their decision facilitated and anticipated by the program. An income effect could explain this, as I discuss in the next section of the paper.

Figure 5: Progres/Oportunidades' Effect on Early Marriage: High School Benefits



Note: This figure presents the average treatment effect on the treated by length of exposure for two different groups: (i) girls 12 or younger in 2001, and (ii) girls 13 or older in 2001. In red are the estimates before treatment started, and in blue after. Standard errors are clustered at the randomization level (locality). Confidence intervals shown are 95% uniform (simultaneous) confidence bands from the multiplier bootstrap of Callaway and Sant'Anna (2021); they hold jointly across all event times and are therefore wider than the pointwise 95% intervals reported in the tables.

6 Mechanism: Income Effect

In the previous analysis, I found that the conditional cash transfer program Progresa/Oportunidades increased the probability of marriage among girls under 18. Although the program was not targeted at reducing early marriage, this result might be surprising since the program led to increases in education, which is often an important mechanism for decreasing child and early marriage.

Given the nature of Progresa/Oportunidades, we cannot disentangle the effect of education from the overall program's effect. This is because if beneficiary children are exposed to more education due to the program, then their household should be receiving the benefit. However, we can hypothesize if and how education is changing marriage decisions in rural Mexico. If there are labor-market returns to education, increases in education should lead to an increase in the opportunity cost of marriage. In Mexico, however, this might not be the case. First, there is evidence that education may be an imperfect measure of human capital accumulation. Behrman et al. (2005) find no evidence that the program led to better grades, and Dubois et al. (2012) find that the program harmed grade progression for secondary school students. Second, Attanasio et al. (2012) show that the relationship between wages and education is flat in rural Mexican localities. If this is the case, education may not directly affect marriage decisions. To test the plausibility of this argument, I compare Progresa/Oportunidades' effect on early marriage between localities with returns to education above the median and localities below the median.¹⁸ I find that, on average, the program's effect is larger in localities where returns to education are below the median compared to localities above the median (see Figure A16). However, this difference is not statistically significant in any analysis year.

We cannot disregard the hypothesis that education actually increases early marriage. Agarwal et al. (2023), for example, show that in India — where dowries are common practice —

¹⁸I estimate a Mincerian regression using municipal-level data from the 1995 census. Of the 657 surveyed localities in the Progresa/Oportunidades sample, I could match returns to education for 261. The availability of this information is not statistically different between treatment and control localities, and early marriage rates do not statistically differ between localities for which there is information on returns to education and those for which the information is unavailable. The overall effect of the program on localities for which I observe returns to education is similar to the effect in the entire sample.

education and youth are valuable in the marriage market, leading to young educated girls marrying earlier than less-educated ones. These findings are consistent with Andrew and Adams (2022), who show that parents believe education is valuable in the marriage market, but not age. Thus, early school abandonment might push parents to marry off their daughters earlier, leading to the positive effect of education on marriage. In the Mexican case, despite the absence of arranged marriages, it can also be that education is valuable in the marriage market, or a way for girls to meet their partners. In fact, Attanasio and Kaufmann (2017) find that girls believe college has positive returns in both labor and marriage markets. Further investigation is necessary to determine whether the returns materialize and whether this replicates for lower levels of education.

An alternative channel through which the program might affect marriage decisions is its income effect. The monetary transfer received by eligible households that complied with the conditionality requirements might increase marriage rates. Ex ante, however, the direction of the income effect is unclear. The transfers may reduce reliance on marriage as a safety net by relaxing budget constraints. On the other hand, the household's extra income may increase boys' and girls' desirability on the marriage market, change their network, and/or it may facilitate marriage by making marriage-related expenditures - such as setting up a new household, having more people to support within a household, or losing workforce - more affordable.¹⁹

To investigate further the hypothesis that the income effect of the program is driving increases in marriage, I perform two empirical exercises. In both exercises, the aim is to restrict the analysis to a population that is only affected by income and not schooling. I exploit two different populations for this purpose.

¹⁹In a Working Paper, Berge et al. (2022) show that an entrepreneurship program increased young women's fertility, in Tanzania. They explain this through an income effect, for which they provide suggestive evidence looking at the association between income and the probability of being a teen mom. Interestingly, they find this positive correlation only for self-employed women. They argue, in line with the literature, that self-employment is more compatible with motherhood due to more flexible labor arrangements.

6.1 Income Effect on Marriage (I)

In this first exercise, I test whether positive income transfers causally affect marriage decisions by exploiting variation in household composition to separate the program's income effect from its overall impact. I focus on the sub-sample of individuals between 6 and 16 years of age at baseline who were exposed to the income effect only. I restrict the analysis to those individuals who are not eligible for the benefit themselves since they completed, in 1997, the final grade of middle school or higher, but live in the same household as an eligible child. For example, these could be older siblings who have completed middle school and whose younger sibling(s) is(are) eligible for the program.²⁰ The sample consists of 3,115 individuals, 46% of them female and, on average, 15.51 years old at baseline. In this exercise, I consider marriage both before and after 18 years old, otherwise I would not have observations to run the exercise. Since the goal is to test whether marriage is a normal good in Mexico, I believe it is sufficient to examine whether a positive income shock leads to an increase in marriage in rural Mexico in general, without focusing on the age at which the union occurred.

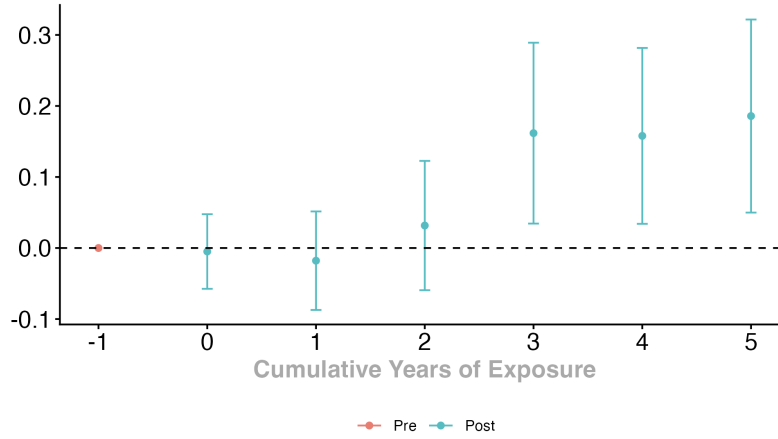
If the program incentivized older siblings to pursue more years of education, then I could not disentangle the two effects. Since the benefit was calculated to cover about two-thirds of a child's wage, it is likely insufficient to cover both the beneficiary child's wage and the older sibling's. Empirically, I do not observe different levels of education between treated and control groups in 1997, 2000, and 2003, which is suggestive evidence of no education 'spillover' effects on non-eligible members within the household.

Figure 6 shows the effect of a positive income shock on the probability of marriage. Initially, the program had no significant effect on marriage. However, from 2000 onward, I observe positive and substantial effects: the income component of the program led to an increase in the probability of marriage of between 16 p.p. (CI=[0.061, 0.263], significant at 1%) and 19 p.p. (CI=[0.078, 0.294], significant at 1%).

These results suggest that, in this population, receiving a positive income transfer enables individuals to marry.

²⁰Given changes in eligibility rules in 2000, I compare T1998 and C2000 to avoid eligibility misclassification.

Figure 6: Causal Effect of an Income Shock on the Probability of Marriage



Note: This figure presents the average treatment effect on the treated, by length of exposure, for the sample of individuals who would not be eligible for the program, but share the household with an eligible individual. It includes marriages after 18. Standard errors are clustered at the randomization level (locality). Confidence intervals shown are 95% uniform (simultaneous) confidence bands from the multiplier bootstrap of Callaway and Sant'Anna (2021); they hold jointly across all event times and are therefore wider than the pointwise 95% intervals reported in the tables.

6.2 Income Effect on Marriage (II)

In this second exercise, I exploit a different population to estimate the effect of the program on girls for whom it did not change schooling decisions. These would be the equivalent of the always-takers - girls who would reach the level of education they reached independently of the program. The assumption is that the program only has an impact on their decisions through the income effect since their education choices were not affected.

To do this, I start by predicting the dropout probability for every girl in the sample. A girl is classified as likely to have dropped out if, by 2003, her completed years of education fall at least two years below the level expected for her age. For example, a girl who was 12 years old in 2003 would be expected to have 6 years of education; if she had 4 or fewer, she is classified as a likely dropout. If her 2003 education level meets or exceeds this threshold, or if she is still enrolled in school, she is classified as unlikely to have dropped out. Dropout probabilities are predicted by age at baseline using the control group's baseline characteristics described in Section 4.²¹ The

²¹Since dropout probability varied substantially with age, which is highly correlated with both marriage and educational attainment, predicted dropout probabilities are estimated using a probit model run separately for each age group on 1997 baseline characteristics, and then pooled into a single global ranking.

sample is then divided into deciles of predicted dropout probability, and the program's effect on child marriage is estimated separately for each decile using the same empirical strategy as in the main analysis.

Deciles 1 and 2 correspond to girls with very low predicted dropout probability, that is, the always-takers.²² For these girls, the program's effect on marriage can be attributed primarily to the income channel. Girls in decile 10, by contrast, have very high predicted dropout probabilities, meaning the program had scope to affect both their schooling and marriage decisions. A decreasing pattern of program effects across deciles would therefore be suggestive evidence of an income effect on marriage.

Figure 7 presents the results. Consistent with this prediction, the estimated effect of the program on marriage tends to be larger for girls with lower dropout probabilities, and smaller or negligible for those with higher dropout probabilities. For the latter group, the income and schooling effects likely work in opposite directions, attenuating the net impact on marriage. For girls in the lowest deciles, where the effect can be interpreted as reflecting the income channel primarily, the estimates are consistent with the earlier finding of a positive income effect on marriage. These patterns are broadly consistent with a marriage market in which education delays marriage while income accelerates it.

The decreasing pattern across deciles provides supporting evidence for the role of the income effect in driving child marriage increases. However, due to the noisy estimates, these findings should be interpreted as suggestive rather than conclusive, complementing the previous section's exercise.²³

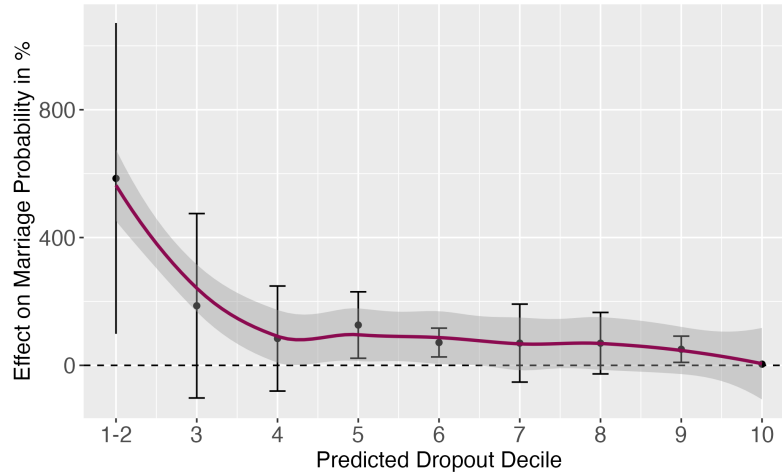
Discussion By looking at two populations that were likely exposed to the program's income effect alone, I find supporting evidence that the increases in child marriage driven by the program were a result of income increases.

Income impacts marriage decisions through several channels. The first is through the costs

²²Deciles 1 and 2 are combined because baseline marriage rates in the control group are effectively zero in decile 1 alone, making the normalization uninformative. Tables A31 and A32 show the average predicted probability and marriage rates in each decile for all girls and for the control group, respectively.

²³Results are robust to using 8 quantiles of the dropout probability instead of deciles, as shown in Figure A17.

Figure 7: Effect of Progresa/Oportunidades on the Probability of Marriage by Predicted Dropout Decile



Note: This figure shows the overall ATT on marriage probability for girls in each predicted dropout decile, expressed as a percentage of the control group's mean marriage rate across the observation period. Deciles 1 and 2 are combined because baseline marriage rates in decile 1 are effectively zero, making the normalization uninformative. Error bars show 95% confidence intervals from the CS estimator, clustered at the locality level. The curve is a LOESS smooth fitted through the 9 decile estimates; the gray band is the 95% confidence interval of the LOESS estimated trend.

of marriage. Even if there is no formal wedding, as most unions at this age are informal, there are still expenses associated with marriage. First, income availability increases the affordability of marriage, such as by allowing for the costs of a new in-law in the household or helping to support the creation of new households. Mier y Terán (2004) argues that higher earnings and opportunities for better jobs in rural Mexico enable young people to form independent households and marry earlier. Rubalcava and Teruel (2005), who studies the effect of Progresa (T1998 vs T2000) on living arrangements, finds that the transfer led to young adults (children of the head of the household) both leaving the household and constituting their own family, and to an inflow of new members, some of whom were sons-in-law and daughters-in-law. Importantly, this could be either because the girls' household can now afford to support a new family, or because the husband's household can, provided they are also beneficiaries.²⁴

Second, the program might be changing marriage markets. Beneficiary girls' and young

²⁴It could also be the case that girls in treatment localities marry to have children themselves and continue receiving the program through their children. However, this is unlikely to be the case, as Parker and Ryu (2023) find no evidence suggesting that any age group of women increased fertility in response to the Progresa program.

men's value in the marriage market might increase due to better economic conditions after the program. Mier y Terán (2004), for example, argues that early marriage is more common in land-holding households, suggesting resources are an asset in the marriage market.²⁵

Third, the income effect might change girls' bargaining power in the parents' household. Since the benefit is received only if girls regularly attend school, they might use it to improve their household situation. Higher bargaining power could entail reducing their home production, thus decreasing the cost of leaving the household. Further, higher bargaining power might also lead to more independent decision-making, such as marrying and exiting the household. Usually, higher bargaining power is associated with lower marriage rates. This is true in settings where marriages are arranged, and girls are not the decision-makers. In Mexico, more decision-making power might lead to earlier marriages if the girl chooses so. If the girl wants to leave the household and marry, she can do so as long as she continues attending school and the family continues to receive the benefit.²⁶ Evidence that supports this is that married girls in treatment localities attend school at a higher rate than married girls in control localities. In 2003, beneficiary-married girls were 6 p.p. more likely to attend school than non-beneficiary-married girls. Girls might stay in school, even after marriage, to continue receiving the benefits. A caveat of this theory is that it can be the case that all married girls want to go to school (despite their arrangement within the household), but only beneficiaries can afford it, which also explains the observed pattern.

These are all plausible explanations that support the presented evidence on the causal effect of income on marriage. Despite being unable to pinpoint which of the previously discussed channels is prevailing, these are important results for future research and policy design. Given that marriage is a normal good in Mexico and there are no countervailing forces in society and the economy, monetary transfers to young people lead them to marry more.

²⁵Unrelated to income but due to schooling, girls might also have a broader marriage market. They might be meeting new potential partners at a higher rate, which increases the likelihood of marriage. However, girls are unlikely to meet their partners in school since, on average, their partners are 3.5 years older than they are. Schooling could also act as a deterrent, thus decreasing child marriage.

²⁶Note that, according to the program's regulations, girls older than 16 can receive the benefit themselves, with the mother's approval. Unfortunately, there is no data available to determine who benefited from this rule.

7 Conclusion

I study the effect of a conditional cash transfer program implemented in rural Mexico, Progresas/Oportunidades, on child and early marriage. Leveraging the random assignment of the program at the locality level and its subsequent expansion, I show that the program led to an increase in early marriage rates for girls. I empirically test the program's income channel and show that positive income inflows lead to higher marriage rates in rural Mexico. Therefore, the program's unintended consequences regarding early marriage can be explained by this income mechanism.

Given that Mexico is a context where children are the decision-makers, it is reasonable to question whether this practice in rural Mexico is as harmful as in contexts with arranged marriages and marriage payments. If children decide to marry, they must receive some utility from it. Is it, then, detrimental to their future? Due to self-selection into marriage and age at marriage, it is extremely challenging to understand the causal effect of early marriage on girls' education, well-being, and labor-market outcomes. However, we can analyze the association between early marriage and female well-being, which are the indicators governments and institutions use to call for the end of this practice. As in other regions in the world, early marriage in rural Mexico is associated with several adverse outcomes: girls who marry before turning 18 years old are, on average, less educated, participate less in the labor market, have more children, and are subject to more violence.

It is important to highlight that, despite the unintended effect I present in this paper, Progresas/Oportunidades was a successful program with respect to many other social and economic outcomes.²⁷ The program improved beneficiary children's physical development, increased their schooling years, reduced child labor, and increased the probability of working and working for a wage as adults. It also increased the likelihood that beneficiaries would have a micro-enterprise. Descriptive statistics also suggest that, even though the program has increased early marriage, it might have attenuated the negative consequences of this practice, given that those girls in treated localities are more educated and have better labor-market out-

²⁷See Parker et al. (2007) for a comprehensive summary and discussion of research on Progresas/ Oportunidades.

comes, independently of their marital status. In the long run, Araujo and Macours (2021) and Parker and Vogl (2023) documented positive effects of the program on education, labor market outcomes, housing, and durable goods ownership, particularly for women. The program has also reduced household poverty and increased consumption and investment in children and livestock. It also had positive externalities on non-beneficiaries, as documented by Lalive and Cattaneo (2009); Bobonis and Finan (2009) and Angelucci et al. (2010). An important caveat, as mentioned before, is that there is no evidence of improvement in cognitive development or achievement tests due to the program. These might be explained by the low investment in the supply side of education, whether within the household or the educational system.

It is worth noting that the data used in this paper does not allow for a serious and in-depth analysis of the effect of the program on marriage quality or on the consequences of child marriage. Therefore, further research should assess early marriage's economic and social consequences and conduct a cost-benefit analysis, accounting for the documented positive economic consequences of similar programs, to determine the overall program's effect. Future research should also focus on understanding the causal consequences of early marriage in this context and how education and social norms motivate marriage decisions.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process During the preparation of this work, I used Claude Code for language and code revision. I have since reviewed and edited the output as needed and take full responsibility for the content of the published article.

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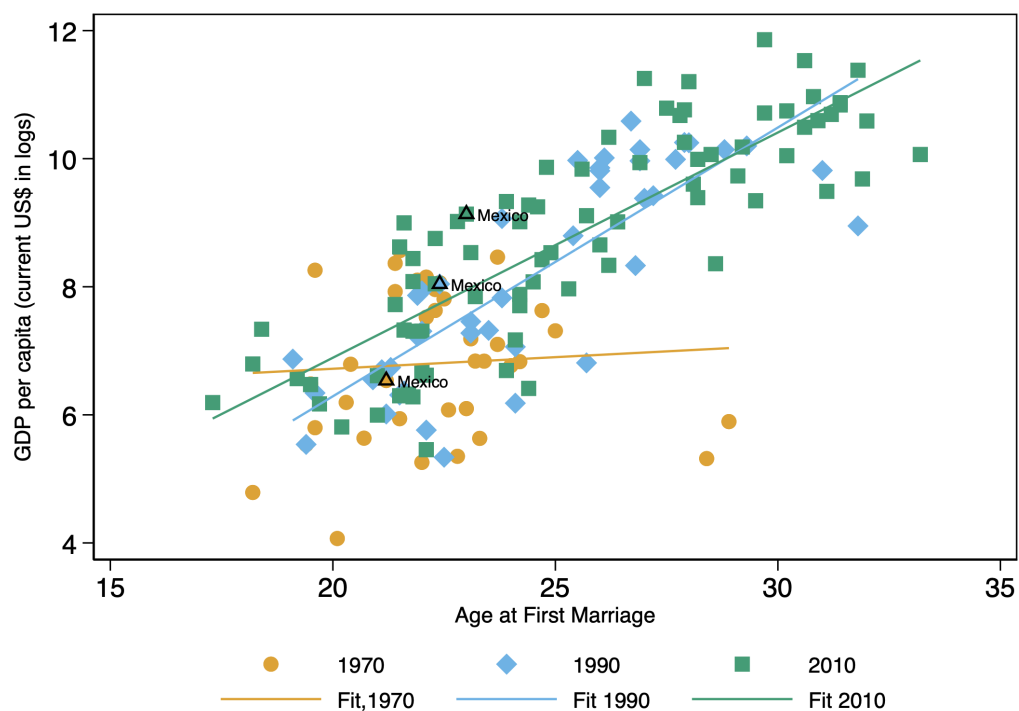
A Appendix

Table A1: Proportion of Children Attending School Conditional on Being Married

Year of Marriage	Attends School				
	1997	1998	1999	2000	2003
1997	51.32	38.03	26.67	7.74	3.2
1998		51.67	33.77	21.74	13.04
1999			50	40.99	31.91
2000				46.45	34.96
2001					8.43
2002					8
2003					20.24

Note: This table shows the proportion of children attending school in the year of or after declaring marriage.

Figure A1: Correlation between Age at First Marriage and GDP per capita



Note: This graph shows a correlation between a country's GDP and age at marriage across several countries over three years. Each data point corresponds to a country in a given year. The lines represent the prediction for GDP from a linear regression of GDP on age at marriage. Data is from the World Bank Data Gender Portal. Data displayed in yellow circles corresponds to 1970, blue diamonds to 1990, and green squares to 2010.

Table A2: Proportion of Married by Group and Year (in %): Girls

(a) All							
	1997	1998	1999	2000	2001	2002	2003
T1998	1.38	2.56	4.57	7.95	14.27	18.73	21.48
T2000	1.46	2.64	4.62	8.41	15.38	19.58	22.13
C2000	2.73	3.21	4.69	6.72	10.04	13.81	16.17
C2000(IPW1998)	2.75	3.34	4.79	6.65	9.88	12.66	15.32
C2000(IPW2000)	2.77	3.68	4.91	6.77	10.12	13.05	15.42
(b) Under 18 years old							
	1997	1998	1999	2000	2001	2002	2003
T1998	1.38	2.46	3.57	4.77	6.78	8.07	7.88
T2000	1.46	2.54	3.48	4.74	7.46	8.49	8.06
C2000	2.73	3.21	2.83	3.13	3.66	4.60	4.94
C2000(IPW1998)	2.75	3.34	2.54	2.64	3.24	3.28	3.69
C2000(IPW2000)	2.77	3.68	2.82	2.65	3.42	3.60	3.45
(c) Between 12 and 16 years old							
	1997	1998	1999	2000	2001	2002	2003
T1998	3.38	6.27	10.63	17.72	31.12	38.66	42.40
T2000	3.74	6.60	11.29	19.87	33.87	40.76	44.94
C2000	6.80	7.90	11.58	16.08	23.25	30.15	32.90
C2000(IPW1998)	6.73	8.11	11.67	15.87	23.03	28.45	32.18
C2000(IPW2000)	6.76	8.93	11.93	16.14	23.35	28.94	32.65

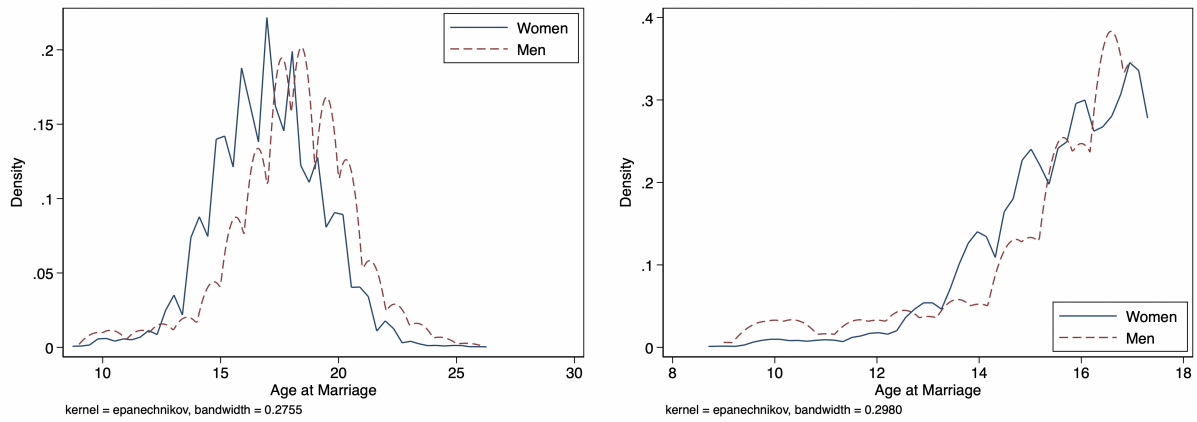
Note: This table presents the proportion of married individuals by group and year, restricting the analysis to girls. Panel (a) refers to all girls between 6 and 16 years old in 1997, Panel (b) refers to the same girls until they turn 18, and Panel (c) refers to the girls who are between 12 and 16 at baseline. T1998 are those individuals who started receiving the program in 1998. T2000 is the set of individuals who first received the program in 2000. C2000 is the control group. C2000(IPW1998) and C2000(IPW2000) are the control group weighted by the probability of being first treated in 1998 and 2000, respectively, versus being in the control group.

Table A3: Proportion of Married by Group and Year (in %): Boys

(a) All							
	1997	1998	1999	2000	2001	2002	2003
T1998	0.25	0.88	2.04	3.86	6.57	8.45	10.34
T2000	0.41	1.09	2.30	3.91	6.83	9.02	11.17
C2000	0.27	0.86	1.40	2.19	3.53	5.96	7.74
C2000(IPW1998)	0.33	1.36	2.61	3.40	4.80	7.48	8.83
C2000(IPW2000)	0.39	1.32	2.28	3.24	4.54	7.27	8.56
(b) Under 18 years old							
	1997	1998	1999	2000	2001	2002	2003
T1998	0.25	0.81	1.67	2.30	2.45	2.64	3.25
T2000	0.41	1.04	1.85	2.12	2.72	2.96	3.77
C2000	0.27	0.86	0.59	0.71	0.91	1.35	1.80
C2000(IPW1998)	0.33	1.36	0.70	0.67	0.88	1.51	2.15
C2000(IPW2000)	0.39	1.32	0.77	0.85	0.84	1.38	1.98
(c) Between 12 and 16 years old							
	1997	1998	1999	2000	2001	2002	2003
T1998	0.61	2.02	4.17	7.27	13.57	17.49	20.47
T2000	1.00	2.66	4.92	7.72	14.44	19.03	22.29
C2000	0.70	2.20	3.61	5.54	8.80	14.34	17.06
C2000(IPW1998)	0.73	3.01	5.79	7.54	10.34	15.43	16.96
C2000(IPW2000)	0.89	3.02	5.22	7.40	10.22	15.50	17.02

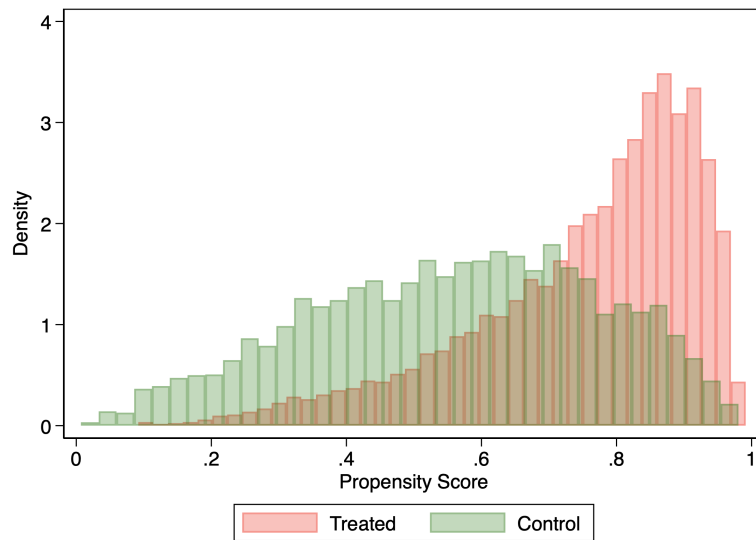
Note: This table presents the proportion of married individuals by group and year, restricting the analysis to boys. Panel (a) refers to all boys between 6 and 16 years old in 1997, Panel (b) refers to the same boys until they turn 18, and Panel (c) refers to the boys who are between 12 and 16 at baseline. T1998 are those individuals who started receiving the program in 1998. T2000 is the set of individuals who first received the program in 2000. C2000 is the control group. C2000(IPW1998) and C2000(IPW2000) are the control group weighted by the probability of being first treated in 1998 and 2000, respectively, versus being in the control group.

Figure A2: Distribution of Age at Marriage



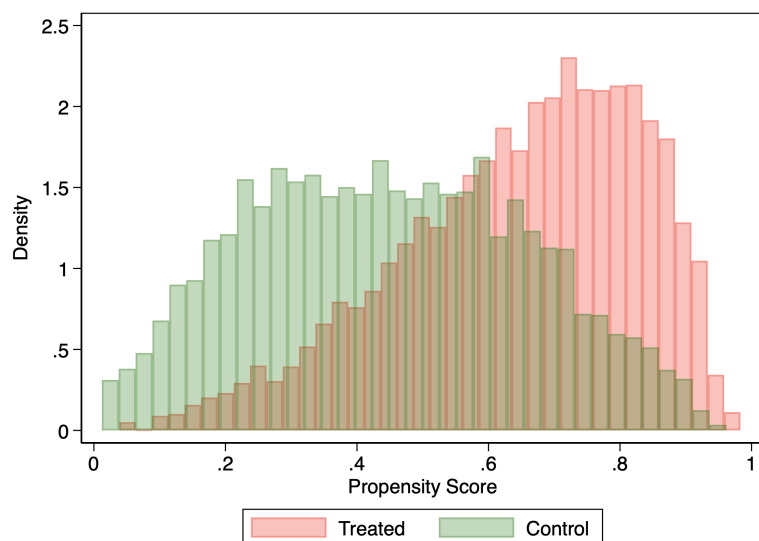
Note: The left panel presents the age distribution at marriage for the entire sample, separately for men and women. The right panel presents the same distribution but only considers individuals who married before 18 years old.

Figure A3: Distribution of the Propensity Score by Group: T1998 VS C2000



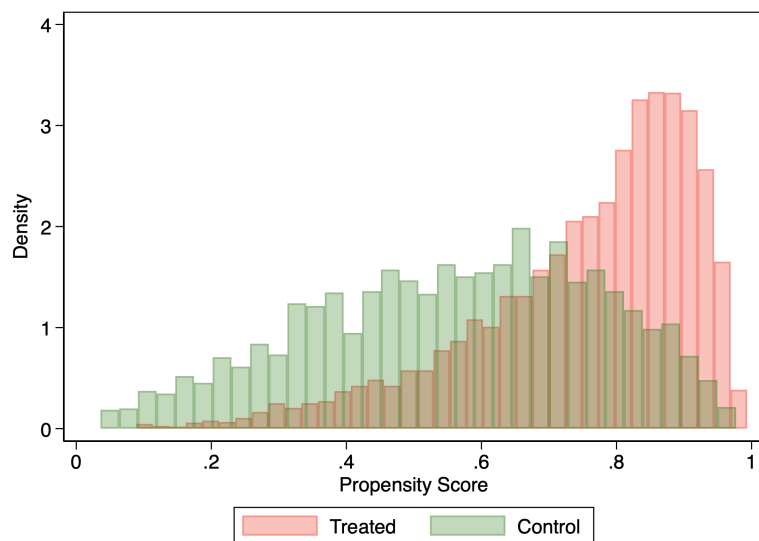
Note: This figure separately presents the histogram of the propensity score for treated (T1998) and control (C2000) groups.

Figure A4: Distribution of the Propensity Score by Group: T2000 VS C2000



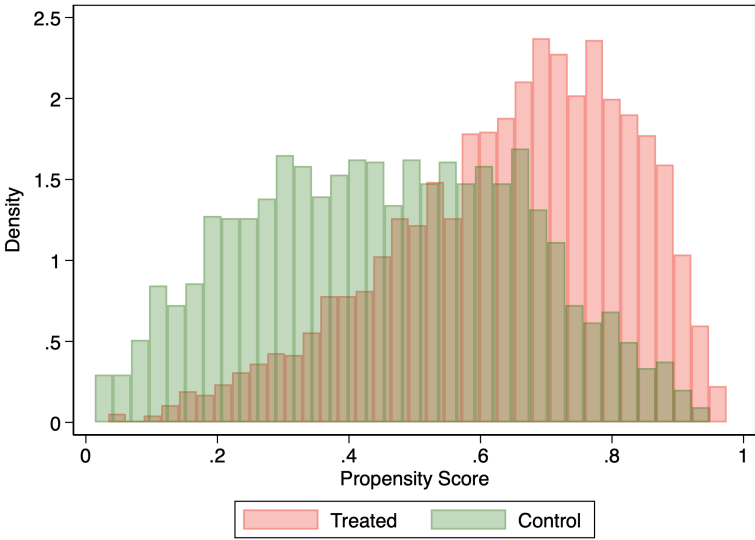
Note: This figure separately presents the histogram of the propensity score for treated (T2000) and control (C2000) groups.

Figure A5: Distribution of the Propensity Score by Group: Girls T1998 VS C2000



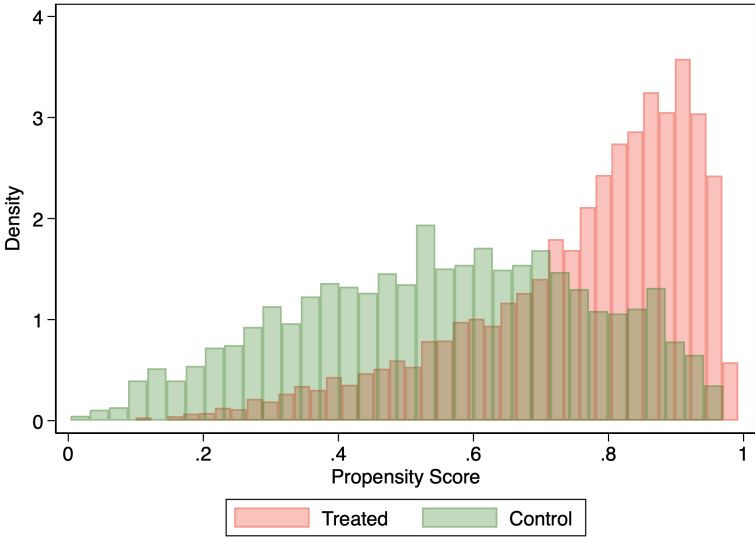
Note: This figure separately presents the histogram of the propensity score for girls in treated (T1998) and control (C2000) groups.

Figure A6: Distribution of the Propensity Score by Group: Girls T2000 VS C2000



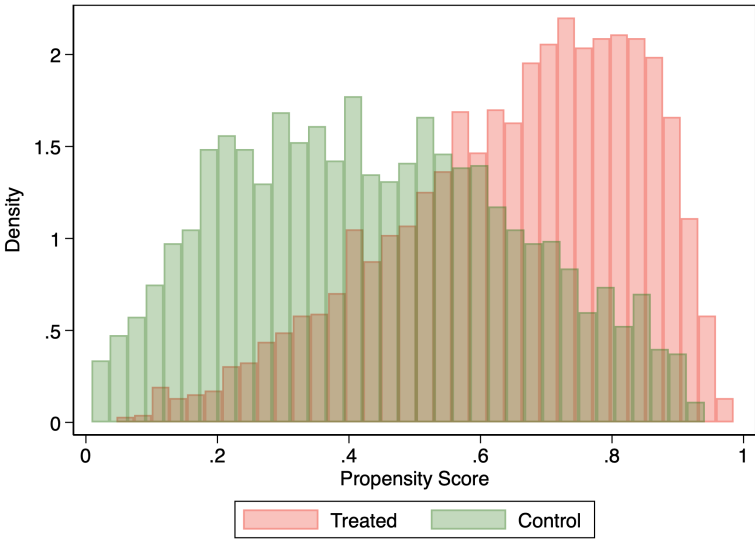
Note: This figure separately presents the histogram of the propensity score for girls in treated (T2000) and control (C2000) groups.

Figure A7: Distribution of the Propensity Score by Group: Boys T1998 VS C2000



Note: This figure separately presents the histogram of the propensity score for boys in treated (T1998) and control (C2000) groups.

Figure A8: Distribution of the Propensity Score by Group: Boys T2000 VS C2000



Note: This figure separately presents the histogram of the propensity score for boys in treated (T2000) and control (C2000) groups.

Table A4: Summary Statistics Baseline Characteristics

Variable	T1998	T2000	C2000	All
Married	0.008	0.009	0.015	0.010
Education Level	3.520	3.439	3.587	3.512
Age in 1997	10.687	10.637	10.522	10.637
Dirt Floor	0.683	0.715	0.691	0.694
Inferior Quality Wall	0.661	0.694	0.607	0.658
Inferior Quality Roof	0.188	0.213	0.259	0.210
No. of Bedrooms	1.756	1.707	1.674	1.725
Piped Water	0.360	0.277	0.427	0.352
Electricity	0.681	0.696	0.699	0.689
Animals	0.429	0.404	0.290	0.391
Land	0.679	0.644	0.485	0.627
Blender	0.227	0.253	0.235	0.236
Refrigerator	0.074	0.052	0.095	0.073
Gas Stove	0.193	0.192	0.283	0.212
Gas Heater	0.020	0.016	0.024	0.020
Radio	0.610	0.615	0.555	0.599
TV	0.393	0.421	0.368	0.395
Video Player	0.022	0.014	0.016	0.018
Dish Washer	0.024	0.017	0.018	0.021
Car	0.007	0.003	0.018	0.008
Truck	0.023	0.026	0.034	0.026
HH Chief or Spouse Have Gone to School	0.720	0.713	0.641	0.701
HH Chief or Spouse Worked the Week Before	0.927	0.911	0.930	0.923
Number of Individuals in the HH	7.397	7.464	7.508	7.440
Age of Household Head	43.138	43.220	42.059	42.924
Female Household Head	0.065	0.062	0.116	0.075
Anyone in the HH Speaks an Indigenous Language	0.415	0.431	0.284	0.391
At least one child between 0 and 5 y.o	0.679	0.694	0.673	0.682
At least one teen between 16 and 19 y.o	0.443	0.424	0.384	0.424
At least one woman between 20 and 39 y.o	0.713	0.740	0.689	0.715
At least one woman between 40 and 59 y.o	0.374	0.355	0.385	0.371
At least one woman over 60 y.o	0.094	0.100	0.135	0.105
At least one man between 20 and 39 y.o	0.554	0.572	0.528	0.553
At least one man between 40 and 59 y.o	0.481	0.461	0.480	0.475
At least one man over 60 y.o	0.100	0.097	0.141	0.108
Guerrero	0.104	0.058	0.061	0.082
Hidalgo	0.180	0.117	0.103	0.145
Michoacan	0.139	0.134	0.126	0.135
Puebla	0.151	0.165	0.094	0.142
Queretaro	0.044	0.042	0.108	0.058
San Luis Potosi	0.143	0.126	0.118	0.133
Observations	12874	7144	5627	25645

Note: This table reports the mean for all variables used for the analysis of the paper, for all individuals, by treatment group.

Table A5: Balance Test on Baseline Characteristics: All (1)

	T1998	T2000	T1998(IPW)	T2000(IPW)	Control
Married	-0.007 (-3.066)	-0.01 (-2.25)	-0.008 (-2.526)	-0.01 (-2.43)	0.01
Education Level	-0.067 (-0.928)	-0.15 (-1.88)	-0.026 (-0.289)	0.04 (0.47)	3.59
Age in 1997	0.165 (2.652)	0.12 (1.76)	-0.087 (-0.938)	-0.05 (-0.67)	10.52
Dirt Floor	-0.007 (-0.193)	0.02 (0.60)	-0.029 (-0.858)	0.00 (0.11)	0.69
Inferior Quality Wall	0.054 (1.356)	0.09 (2.03)	0.012 (0.260)	0.02 (0.46)	0.61
Inferior Quality Roof	-0.072 (-1.984)	-0.05 (-1.19)	-0.005 (-0.156)	-0.01 (-0.29)	0.26
No. of Bedrooms	0.083 (1.671)	0.03 (0.60)	-0.044 (-0.571)	-0.00 (-0.07)	1.67
Piped Water	-0.067 (-1.327)	-0.15 (-2.81)	0.023 (0.463)	0.00 (0.09)	0.43
Electricity	-0.018 (-0.417)	-0.00 (-0.07)	0.017 (0.365)	0.02 (0.36)	0.70
Animals	0.139 (4.785)	0.11 (3.59)	-0.002 (-0.048)	-0.02 (-0.45)	0.29
Land	0.194 (4.700)	0.16 (3.59)	-0.000 (-0.005)	-0.01 (-0.18)	0.48
Blender	-0.008 (-0.321)	0.02 (0.62)	0.021 (0.785)	0.03 (0.86)	0.24
Refrigerator	-0.021 (-1.320)	-0.04 (-2.68)	0.016 (1.335)	0.01 (0.55)	0.10
Gas Stove	-0.089 (-2.099)	-0.09 (-2.01)	0.002 (0.057)	-0.00 (-0.06)	0.28
Gas Heater	-0.004 (-0.760)	-0.01 (-1.73)	-0.002 (-0.254)	-0.00 (-0.56)	0.02
Radio	0.055 (2.187)	0.06 (2.21)	-0.013 (-0.467)	-0.01 (-0.23)	0.56
TV	0.025 (0.667)	0.05 (1.33)	0.009 (0.259)	0.02 (0.52)	0.37
Video Player	0.006 (1.436)	-0.00 (-0.40)	0.004 (0.681)	0.00 (0.13)	0.02
Dish Washer	0.006 (1.031)	-0.00 (-0.12)	0.000 (0.058)	0.00 (0.59)	0.02
Car	-0.011 (-2.699)	-0.01 (-4.02)	0.000 (0.147)	-0.00 (-0.19)	0.02
Truck	-0.011 (-1.822)	-0.01 (-1.00)	0.003 (0.447)	-0.00 (-0.20)	0.03

Note: This table reports parameter estimates and t-statistics (in parentheses) for regressions of baseline characteristics on a treatment indicator. T1998 (T2000) equals 1 if the individual belongs to the set of localities treated in 1998 (2000) and 0 if it otherwise. The first two columns report the parameters without any re-weighting. The third and fourth columns report the estimates obtained by re-weighting the control group, as described. In the four regressions, standard errors were clustered at the locality level. The last column presents the unconditional, unweighted mean of each characteristic for the control group C2000.

Table A6: Balance Test on Baseline Characteristics: All (2)

	T1998	T2000	T1998(IPW)	T2000(IPW)	Control
HH Chief or Spouse Have Gone to School	0.079 (2.348)	0.07 (2.10)	0.006 (0.179)	-0.01 (-0.32)	0.64
HH Chief or Spouse Worked the Week Before	-0.003 (-0.393)	-0.02 (-1.95)	-0.011 (-1.127)	-0.02 (-1.41)	0.93
Number of Individuals in the HH	-0.110 (-0.863)	-0.04 (-0.33)	0.092 (0.675)	0.10 (0.56)	7.51
Age of Household Head	1.079 (3.363)	1.16 (2.87)	-0.644 (-1.394)	0.11 (0.23)	42.06
Female Household Head	-0.051 (-4.910)	-0.05 (-4.99)	0.007 (0.925)	0.01 (0.80)	0.12
Anyone in the HH Speaks an Indigenous Language	0.131 (2.180)	0.15 (2.17)	-0.026 (-0.442)	-0.02 (-0.22)	0.28
At least one child between 0 and 5 y.o	0.006 (0.359)	0.02 (1.13)	-0.002 (-0.104)	0.01 (0.25)	0.67
At least one teen between 16 and 19 y.o	0.059 (3.695)	0.04 (2.24)	0.004 (0.159)	0.01 (0.49)	0.38
At least one woman between 20 and 39 y.o	0.023 (1.847)	0.05 (3.33)	0.011 (0.682)	0.00 (0.25)	0.69
At least one woman between 40 and 59 y.o	-0.010 (-0.719)	-0.03 (-1.74)	-0.017 (-0.802)	-0.00 (-0.15)	0.38
At least one woman over 60 y.o	-0.041 (-3.011)	-0.03 (-2.29)	0.012 (1.250)	0.01 (0.99)	0.13
At least one man between 20 and 39 y.o	0.026 (1.585)	0.04 (2.15)	0.018 (0.790)	0.01 (0.29)	0.53
At least one man between 40 and 59 y.o	0.001 (0.078)	-0.02 (-1.06)	0.009 (0.402)	0.01 (0.28)	0.48
At least one man over 60 y.o	-0.041 (-2.857)	-0.04 (-2.86)	-0.004 (-0.341)	0.01 (0.67)	0.14
Guerrero	0.043 (1.139)	-0.00 (-0.10)	0.040 (1.060)	0.02 (0.89)	0.06
Hidalgo	0.077 (1.746)	0.01 (0.33)	-0.006 (-0.111)	0.00 (0.06)	0.10
Michoacan	0.012 (0.289)	0.01 (0.16)	-0.016 (-0.314)	0.00 (0.06)	0.13
Puebla	0.057 (1.411)	0.07 (1.47)	-0.032 (-0.543)	-0.04 (-0.63)	0.09
Queretaro	-0.064 (-1.023)	-0.07 (-1.04)	0.001 (0.020)	-0.00 (-0.07)	0.11
San Luis Potosi	0.025 (0.469)	0.01 (0.14)	0.016 (0.370)	0.02 (0.37)	0.12

Note: This table reports parameter estimates and t-statistics (in parentheses) for regressions of baseline characteristics on a treatment indicator. T1998 (T2000) equals 1 if the individual belongs to the set of localities treated in 1998 (2000) and 0 if otherwise. The first two columns report the parameters without any re-weighting. The third and fourth columns report the estimates obtained by re-weighting the control group, as described. In the four regressions, standard errors were clustered at the locality level. The last column presents the unconditional, unweighted mean of each characteristic for the control group C2000.

Table A7: Balance Test on Baseline Characteristics: Girls (1)

	T1998	T2000	T1998(IPW)	T2000(IPW)	Control
Married	-0.014 (-3.255)	-0.01 (-2.88)	-0.014 (-2.274)	-0.01 (-2.46)	0.03
Education Level	-0.111 (-1.163)	-0.17 (-1.67)	0.104 (0.882)	0.07 (0.61)	3.64
Age in 1997	0.071 (0.824)	0.01 (0.12)	0.018 (0.139)	-0.04 (-0.33)	10.58
Dirt Floor	0.004 (0.097)	0.03 (0.65)	-0.024 (-0.710)	0.01 (0.17)	0.68
Inferior Quality Wall	0.044 (1.086)	0.08 (1.83)	0.000 (0.011)	0.02 (0.44)	0.61
Inferior Quality Roof	-0.077 (-2.157)	-0.05 (-1.25)	-0.002 (-0.069)	-0.01 (-0.20)	0.26
No. of Bedrooms	0.090 (1.816)	0.06 (1.00)	-0.070 (-0.822)	-0.02 (-0.26)	1.67
Piped Water	-0.074 (-1.478)	-0.15 (-2.76)	0.021 (0.427)	0.00 (0.05)	0.44
Electricity	-0.032 (-0.757)	-0.02 (-0.44)	0.015 (0.316)	0.02 (0.35)	0.72
Animals	0.137 (4.793)	0.11 (3.36)	0.003 (0.084)	-0.01 (-0.27)	0.29
Land	0.186 (4.410)	0.14 (3.13)	0.005 (0.135)	-0.01 (-0.22)	0.49
Blender	-0.013 (-0.513)	0.02 (0.78)	0.022 (0.765)	0.02 (0.60)	0.24
Refrigerator	-0.026 (-1.475)	-0.05 (-2.61)	0.016 (1.238)	0.00 (0.30)	0.10
Gas Stove	-0.090 (-2.069)	-0.09 (-1.95)	0.006 (0.180)	0.00 (0.01)	0.29
Gas Heater	-0.001 (-0.113)	-0.01 (-1.89)	-0.001 (-0.201)	-0.00 (-0.61)	0.02
Radio	0.053 (1.893)	0.06 (1.99)	-0.015 (-0.469)	-0.00 (-0.10)	0.56
TV	0.021 (0.523)	0.05 (1.08)	0.013 (0.345)	0.02 (0.42)	0.38
Video Player	0.005 (1.005)	-0.00 (-0.60)	0.005 (1.007)	0.00 (0.19)	0.02
Dish Washer	0.004 (0.510)	-0.00 (-0.58)	0.000 (0.051)	0.00 (0.51)	0.02
Car	-0.010 (-1.923)	-0.01 (-3.38)	-0.000 (-0.038)	-0.00 (-0.30)	0.02
Truck	-0.008 (-1.263)	-0.01 (-0.58)	0.001 (0.174)	-0.00 (-0.06)	0.03

Note: This table reports parameter estimates and t-statistics (in parentheses) for regressions of baseline characteristics on a treatment indicator for girls. T1998 (T2000) equals 1 if the individual belongs to the set of localities treated in 1998 (2000) and 0 if otherwise. The first two columns report the parameters without any re-weighting. The third and fourth columns report the estimates obtained by re-weighting the control group, as described. In the four regressions, standard errors were clustered at the locality level. The last column presents the unconditional, unweighted mean of each characteristic for the control group C2000.

Table A8: Balance Test on Baseline Characteristics: Girls (2)

	T1998	T2000	T1998(IPW)	T2000(IPW)	Control
HH Chief or Spouse Have Gone to School	0.078 (2.181)	0.06 (1.53)	0.009 (0.241)	-0.01 (-0.38)	0.65
HH Chief or Spouse Worked the Week Before	0.000 (0.057)	-0.03 (-2.31)	-0.010 (-1.044)	-0.02 (-1.40)	0.93
Number of Individuals in the HH	-0.072 (-0.550)	-0.05 (-0.32)	0.078 (0.515)	0.09 (0.50)	7.52
Age of Household Head	0.646 (1.649)	1.22 (2.43)	-0.673 (-1.240)	0.33 (0.59)	42.17
Female Household Head	-0.056 (-4.855)	-0.05 (-4.06)	0.006 (0.755)	0.00 (0.48)	0.12
Anyone in the HH Speaks an Indigenous Language	0.129 (2.101)	0.14 (1.99)	-0.026 (-0.418)	-0.01 (-0.12)	0.29
At least one child between 0 and 5 y.o	0.017 (0.902)	0.02 (0.89)	0.007 (0.312)	0.01 (0.39)	0.68
At least one teen between 16 and 19 y.o	0.038 (2.199)	0.03 (1.50)	-0.002 (-0.075)	0.01 (0.32)	0.40
At least one woman between 20 and 39 y.o	0.029 (2.006)	0.05 (2.69)	0.019 (0.980)	0.01 (0.30)	0.69
At least one woman between 40 and 59 y.o	-0.016 (-0.948)	-0.02 (-1.09)	-0.021 (-0.844)	0.00 (0.08)	0.38
At least one woman over 60 y.o	-0.056 (-3.505)	-0.05 (-2.93)	0.005 (0.412)	0.01 (0.48)	0.14
At least one man between 20 and 39 y.o	0.032 (1.908)	0.05 (2.45)	0.020 (0.860)	0.01 (0.28)	0.52
At least one man between 40 and 59 y.o	-0.003 (-0.182)	-0.03 (-1.47)	0.006 (0.227)	0.00 (0.16)	0.48
At least one man over 60 y.o	-0.053 (-3.215)	-0.05 (-2.71)	0.000 (0.025)	0.01 (1.09)	0.15
Guerrero	0.033 (0.839)	-0.01 (-0.24)	0.034 (0.924)	0.02 (0.81)	0.07
Hidalgo	0.074 (1.668)	0.01 (0.26)	-0.003 (-0.061)	0.00 (0.10)	0.10
Michoacan	0.008 (0.167)	0.01 (0.11)	-0.018 (-0.346)	-0.00 (-0.03)	0.13
Puebla	0.055 (1.306)	0.07 (1.37)	-0.037 (-0.610)	-0.04 (-0.57)	0.10
Queretaro	-0.058 (-0.936)	-0.06 (-0.99)	0.001 (0.042)	-0.00 (-0.09)	0.11
San Luis Potosi	0.034 (0.682)	0.02 (0.31)	0.020 (0.460)	0.02 (0.40)	0.11

Note: This table reports parameter estimates and t-statistics (in parentheses) for regressions of baseline characteristics on a treatment indicator for girls. T1998 (T2000) equals 1 if the individual belongs to the set of localities treated in 1998 (2000) and 0 if otherwise. The first two columns report the parameters without any re-weighting. The third and fourth columns report the estimates obtained by re-weighting the control group, as described. In the four regressions, standard errors were clustered at the locality level. The last column presents the unconditional, unweighted mean of each characteristic for the control group C2000.

Table A9: Balance Test on Baseline Characteristics: Boys (1)

	T1998	T2000	T1998(IPW)	T2000(IPW)	Control
Married	-0.000 (-0.180)	0.00 (0.97)	-0.001 (-0.460)	0.00 (0.10)	0.00
Education Level	-0.026 (-0.333)	-0.13 (-1.50)	-0.126 (-1.118)	0.01 (0.07)	3.54
Age in 1997	0.252 (3.097)	0.21 (2.48)	-0.158 (-1.289)	-0.05 (-0.51)	10.47
Dirt Floor	-0.017 (-0.456)	0.02 (0.55)	-0.031 (-0.836)	0.00 (0.08)	0.70
Inferior Quality Wall	0.064 (1.551)	0.10 (2.14)	0.024 (0.535)	0.03 (0.60)	0.61
Inferior Quality Roof	-0.067 (-1.778)	-0.05 (-1.10)	-0.008 (-0.234)	-0.02 (-0.40)	0.26
No. of Bedrooms	0.076 (1.409)	0.01 (0.19)	-0.025 (-0.321)	-0.00 (-0.01)	1.68
Piped Water	-0.060 (-1.163)	-0.15 (-2.81)	0.023 (0.448)	0.00 (0.07)	0.41
Electricity	-0.004 (-0.091)	0.01 (0.27)	0.019 (0.405)	0.02 (0.35)	0.68
Animals	0.140 (4.480)	0.12 (3.53)	-0.003 (-0.094)	-0.02 (-0.57)	0.29
Land	0.202 (4.840)	0.18 (3.92)	-0.000 (-0.011)	-0.00 (-0.05)	0.48
Blender	-0.003 (-0.123)	0.01 (0.43)	0.018 (0.631)	0.03 (1.00)	0.23
Refrigerator	-0.017 (-1.060)	-0.04 (-2.54)	0.015 (1.182)	0.01 (0.64)	0.09
Gas Stove	-0.089 (-2.084)	-0.09 (-2.04)	-0.004 (-0.136)	-0.01 (-0.20)	0.27
Gas Heater	-0.007 (-1.243)	-0.01 (-1.24)	-0.002 (-0.343)	-0.00 (-0.49)	0.02
Radio	0.058 (2.212)	0.06 (2.17)	-0.008 (-0.288)	-0.01 (-0.32)	0.55
TV	0.029 (0.787)	0.06 (1.52)	0.007 (0.182)	0.02 (0.56)	0.36
Video Player	0.007 (1.629)	-0.00 (-0.12)	0.002 (0.241)	0.00 (0.16)	0.02
Dish Washer	0.009 (1.466)	0.00 (0.36)	0.001 (0.148)	0.00 (0.53)	0.02
Car	-0.013 (-3.010)	-0.01 (-3.72)	0.001 (0.360)	0.00 (0.19)	0.02
Truck	-0.014 (-2.052)	-0.01 (-1.30)	0.004 (0.625)	-0.00 (-0.29)	0.04

Note: This table reports parameter estimates and t-statistics (in parentheses) for regressions of baseline characteristics on a treatment indicator for boys. T1998 (T2000) equals 1 if the individual belongs to the set of localities treated in 1998 (2000), and 0 otherwise. The first two columns report the parameters without any re-weighting. The third and fourth columns report the estimates obtained by re-weighting the control group, as described. In the four regressions, standard errors were clustered at the locality level. The last column presents the unconditional, unweighted mean of each characteristic for the control group C2000.

Table A10: Balance Test on Baseline Characteristics: Boys (2)

	T1998	T2000	T1998(IPW)	T2000(IPW)	Control
HH Chief or Spouse Have Gone to School	0.081 (2.410)	0.09 (2.56)	0.003 (0.078)	-0.00 (-0.15)	0.63
HH Chief or Spouse Worked the Week Before	-0.006 (-0.703)	-0.01 (-1.23)	-0.013 (-1.148)	-0.02 (-1.16)	0.93
Number of Individuals in the HH	-0.145 (-1.079)	-0.04 (-0.31)	0.089 (0.653)	0.11 (0.65)	7.49
Age of Household Head	1.478 (4.299)	1.10 (2.75)	-0.544 (-1.041)	-0.15 (-0.28)	41.96
Female Household Head	-0.047 (-4.120)	-0.06 (-4.90)	0.008 (0.970)	0.01 (1.07)	0.12
Anyone in the HH Speaks an Indigenous Language	0.133 (2.219)	0.16 (2.32)	-0.023 (-0.382)	-0.02 (-0.24)	0.28
At least one child between 0 and 5 y.o	-0.004 (-0.262)	0.02 (1.19)	-0.012 (-0.522)	0.00 (0.17)	0.67
At least one teen between 16 and 19 y.o	0.078 (4.436)	0.05 (2.52)	0.009 (0.335)	0.02 (0.60)	0.37
At least one woman between 20 and 39 y.o	0.019 (1.254)	0.05 (3.22)	-0.000 (-0.005)	0.00 (0.03)	0.69
At least one woman between 40 and 59 y.o	-0.005 (-0.339)	-0.04 (-2.07)	-0.010 (-0.440)	-0.01 (-0.35)	0.39
At least one woman over 60 y.o	-0.027 (-2.058)	-0.02 (-1.32)	0.020 (1.958)	0.02 (1.65)	0.13
At least one man between 20 and 39 y.o	0.020 (1.052)	0.04 (1.60)	0.018 (0.666)	0.01 (0.38)	0.53
At least one man between 40 and 59 y.o	0.005 (0.317)	-0.01 (-0.43)	0.011 (0.428)	0.00 (0.15)	0.48
At least one man over 60 y.o	-0.031 (-2.150)	-0.04 (-2.67)	-0.006 (-0.484)	0.00 (0.07)	0.13
Guerrero	0.053 (1.409)	0.00 (0.05)	0.047 (1.187)	0.02 (0.93)	0.06
Hidalgo	0.080 (1.794)	0.02 (0.40)	-0.009 (-0.164)	-0.00 (-0.03)	0.10
Michoacan	0.017 (0.409)	0.01 (0.21)	-0.014 (-0.280)	0.01 (0.13)	0.12
Puebla	0.060 (1.502)	0.07 (1.55)	-0.027 (-0.464)	-0.05 (-0.66)	0.09
Queretaro	-0.069 (-1.101)	-0.07 (-1.08)	0.000 (0.004)	-0.00 (-0.04)	0.11
San Luis Potosi	0.017 (0.294)	0.00 (0.00)	0.012 (0.277)	0.01 (0.30)	0.13

Note: This table reports parameter estimates and t-statistics (in parentheses) for regressions of baseline characteristics on a treatment indicator for boys. T1998 (T2000) equals 1 if the individual belongs to the set of localities treated in 1998 (2000) and 0 if otherwise. The first two columns report the parameters without any re-weighting. The third and fourth columns report the estimates obtained by re-weighting the control group, as described. In the four regressions, standard errors were clustered at the locality level. The last column presents the unconditional, unweighted mean of each characteristic for the control group C2000.

Table A11: Robustness Test (Recall): Effect of Progresa/Oportunidades on the Probability of Marriage by Length of Exposure

(a) By Exposure				(b) By Group and Year			
Event-Time	ATT(t)	Std. Error	Conf. Interval	Time	ATT(g,t)	Std. Error	Conf. Interval
-3	-0.0125*	0.0074	[-0.027 , 0.0021]	T 1998 1997	0	NA	[NA , NA]
-2	-0.0102	0.0064	[-0.0227 , 0.0022]	T 1998 1998	2e-04	0.0019	[-0.0036 , 0.0039]
-1	0	NA	[NA , NA]	T 1998 1999	0.0045	0.0044	[-0.0042 , 0.0131]
0	0.0025	0.0029	[-0.0033 , 0.0082]	T 1998 2000	0.0209	0.0131	[-0.0047 , 0.0465]
1	0.011**	0.0046	[0.002 , 0.0201]	T 1998 2001	0.0296*	0.0163	[-0.0023 , 0.0615]
2	0.0227**	0.0093	[0.0045 , 0.0409]	T 1998 2002	0.0338**	0.0137	[0.0069 , 0.0607]
3	0.0305**	0.012	[0.0071 , 0.054]	T 1998 2003	0.034*	0.0204	[-0.006 , 0.074]
4	0.0338**	0.0136	[0.0071 , 0.0605]	T 2000 1997	-0.0125*	0.0074	[-0.0269 , 0.002]
5	0.034*	0.0205	[-0.0062 , 0.0743]	T 2000 1998	-0.0102	0.0064	[-0.0227 , 0.0023]
N	25645			T 2000 1999	0	NA	[NA , NA]
				T 2000 2000	0.0066	0.0079	[-0.009 , 0.0221]
				T 2000 2001	0.0228*	0.012	[-6e-04 , 0.0463]
				T 2000 2002	0.0259**	0.0108	[0.0048 , 0.0469]
				T 2000 2003	0.0323**	0.016	[0.001 , 0.0636]
				N	25645		

Note: This table shows average treatment effect on the treated using only variables that are less prone to recall bias in the specification of the outcome regression and the propensity score. Asterisks denote pointwise statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A12: Robustness Test (Recall): Effect of Progresa/Oportunidades on the Probability of Marriage by Length of Exposure: Girls

(a) By Exposure				(b) By Group and Year				
Event-Time	ATT(t)	Std. Error	Conf. Interval	Time	ATT(g,t)	Std. Error	Conf. Interval	
-3	-0.014	0.0125	[-0.0384 , 0.0104]	T 1998	1997	0	NA	[NA , NA]
-2	-0.0083	0.012	[-0.0317 , 0.0152]	T 1998	1998	0.0016	0.0031	[-0.0044 , 0.0077]
-1	0	NA	[NA , NA]	T 1998	1999	0.0071	0.0083	[-0.0091 , 0.0233]
0	0.0047	0.0053	[-0.0058 , 0.0152]	T 1998	2000	0.0268	0.0191	[-0.0105 , 0.0642]
1	0.0173**	0.0074	[0.0028 , 0.0318]	T 1998	2001	0.0435*	0.0234	[-0.0025 , 0.0894]
2	0.0337**	0.0139	[0.0065 , 0.0608]	T 1998	2002	0.0597***	0.0227	[0.0153 , 0.1041]
3	0.046***	0.0176	[0.0114 , 0.0805]	T 1998	2003	0.0563**	0.0231	[0.011 , 0.1017]
4	0.0597***	0.0221	[0.0164 , 0.103]	T 2000	1997	-0.014	0.0128	[-0.0391 , 0.0111]
5	0.0563**	0.0221	[0.0131 , 0.0996]	T 2000	1998	-0.0083	0.0123	[-0.0324 , 0.0159]
N				T 2000	1999	0	NA	[NA , NA]
12357				T 2000	2000	0.0101	0.0141	[-0.0175 , 0.0377]
				T 2000	2001	0.0354*	0.0201	[-0.004 , 0.0749]
				T 2000	2002	0.0458**	0.0214	[0.0039 , 0.0877]
				T 2000	2003	0.0504**	0.0222	[0.0068 , 0.094]
				N				
				12357				

Note: This table shows average treatment effect on the treated girls using only variables that are less prone to recall bias in the specification of the outcome regression and the propensity score. Asterisks denote pointwise statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A13: Robustness Test (Recall): Effect of Progresa/Oportunidades on the Probability of Marriage by Length of Exposure: Boys

(a) By Exposure				(b) By Group and Year				
Event-Time	ATT(t)	Std. Error	Conf. Interval		Time	ATT(g,t)	Std. Error	Conf. Interval
-3	-0.0095	0.0092	[-0.0274 , 0.0085]	T 1998	1997	0	NA	[NA , NA]
-2	-0.0111***	0.0039	[-0.0187 , -0.0035]	T 1998	1998	-0.0012	0.0026	[-0.0062 , 0.0038]
-1	0	NA	[NA , NA]	T 1998	1999	0.0017	0.0038	[-0.0058 , 0.0092]
0	2e-04	0.004	[-0.0077 , 0.008]	T 1998	2000	0.0143*	0.0086	[-0.0025 , 0.031]
1	0.0044	0.0053	[-0.006 , 0.0148]	T 1998	2001	0.0136	0.0158	[-0.0173 , 0.0446]
2	0.0113	0.0074	[-0.0033 , 0.0258]	T 1998	2002	0.0066	0.011	[-0.015 , 0.0282]
3	0.0129	0.012	[-0.0107 , 0.0365]	T 1998	2003	0.011	0.0351	[-0.0578 , 0.0798]
4	0.0066	0.0112	[-0.0154 , 0.0286]	T 2000	1997	-0.0095	0.0091	[-0.0273 , 0.0083]
5	0.011	0.0349	[-0.0574 , 0.0793]	T 2000	1998	-0.0111***	0.0038	[-0.0186 , -0.0036]
N	13288			T 2000	1999	0	NA	[NA , NA]
				T 2000	2000	0.0026	0.0122	[-0.0214 , 0.0266]
				T 2000	2001	0.0092	0.0143	[-0.0188 , 0.0372]
				T 2000	2002	0.0057	0.0149	[-0.0235 , 0.0349]
				T 2000	2003	0.0114	0.025	[-0.0375 , 0.0603]
N				13288				

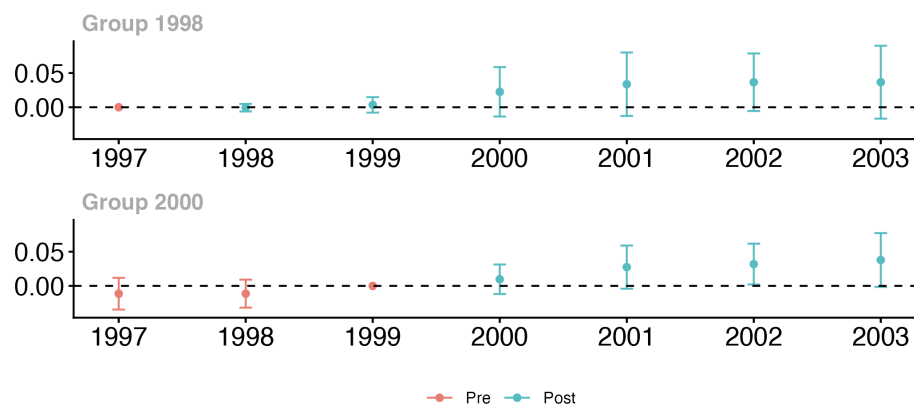
Note: This table shows the average treatment effect on the treated boys using only variables that are less prone to recall bias in the specification of the outcome regression and the propensity score. Asterisks denote pointwise statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A14: Effect of Progresa/Oportunidades on the Probability of Marriage by Length of Exposure

Event-Time	ATT(t)	Std. Error	Conf. Interval
-3	-0.0114	0.0089	[-0.0289 , 0.0061]
-2	-0.0113	0.0079	[-0.0267 , 0.0041]
-1	0	NA	[NA , NA]
0	0.003	0.0031	[-0.003 , 0.0091]
1	0.012***	0.0047	[0.0029 , 0.0212]
2	0.0259***	0.0098	[0.0067 , 0.0451]
3	0.0352***	0.0131	[0.0095 , 0.0608]
4	0.0366**	0.016	[0.0053 , 0.0679]
5	0.0367*	0.0203	[-0.0032 , 0.0765]
N	25645		

Note: This table shows the average treatment effects by length of exposure and the respective standard errors and confidence intervals. N is the number of observations. Event-Time refers to the period relative to the treatment year. Asterisks denote pointwise statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A9: Effect of Progresa/Oportunidades on the Probability of Marriage by Group



Note: This figure presents the average treatment effect on the treated by treatment group and period. Group 1998, or T1998, is the group that first received treatment in 1998, and Group 2000, or T2000, is the group that first received treatment in 2000. In red are the estimates before treatment started, and in blue after treatment started. Standard errors are clustered at the randomization level (locality). Confidence intervals shown are 95% uniform (simultaneous) confidence bands from the multiplier bootstrap of Callaway and Sant'Anna (2021); they hold jointly across all event times and are therefore wider than the pointwise 95% intervals reported in the tables.

Table A15: Effect of Progresa/Oportunidades on the Probability of Marriage by Group and Year

	Time	ATT(g,t)	Std. Error	Conf. Interval
T 1998	1997	0	NA	[NA , NA]
T 1998	1998	-6e-04	0.0021	[-0.0048 , 0.0035]
T 1998	1999	0.0035	0.0043	[-0.005 , 0.0119]
T 1998	2000	0.0226*	0.0137	[-0.0043 , 0.0495]
T 1998	2001	0.0337*	0.0176	[-8e-04 , 0.0682]
T 1998	2002	0.0366**	0.016	[0.0053 , 0.0679]
T 1998	2003	0.0367*	0.0202	[-0.003 , 0.0763]
T 2000	1997	-0.0114	0.0088	[-0.0287 , 0.0059]
T 2000	1998	-0.0113	0.0078	[-0.0266 , 0.0039]
T 2000	1999	0	NA	[NA , NA]
T 2000	2000	0.0097	0.0082	[-0.0064 , 0.0257]
T 2000	2001	0.0275**	0.012	[0.004 , 0.0509]
T 2000	2002	0.0319***	0.0113	[0.0097 , 0.0541]
T 2000	2003	0.0378**	0.0149	[0.0086 , 0.0671]
N			25645	

Note: This table shows the average treatment effects by group and length of exposure, and the respective standard errors and confidence intervals. N is the number of observations. Asterisks denote pointwise statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A16: Pre-trends Test on the Probability of Marriage: Age 10-17 in 1997

Event-Time	ATT(t)	Std. Error	Conf. Interval
-3	-0.016	0.0134	[-0.0424 , 0.0103]
-2	-0.0153	0.0121	[-0.039 , 0.0085]
-1	0	NA	[NA , NA]
0	0.0035	0.0059	[-0.0081 , 0.0151]
N		17000	

Note: This table shows the average treatment effects by length of exposure and the respective standard errors and confidence intervals. N is the number of observations. Event-Time refers to the period relative to the treatment year. This table restricts the analysis to individuals aged between 10 and 17 in 1997. Asterisks denote pointwise statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A17: Pre-trends Test on the Probability of Marriage: Age 12-17 in 1997

Event-Time	ATT(t)	Std. Error	Conf. Interval
-3	-0.0163	0.0154	[-0.0466 , 0.0139]
-2	-0.0153	0.0134	[-0.0417 , 0.011]
-1	0	NA	[NA , NA]
0	0.0027	0.0101	[-0.0171 , 0.0226]
N	11791		

Note: This table shows the average treatment effects by length of exposure and the respective standard errors and confidence intervals. N is the number of observations. Event-Time refers to the period relative to the treatment year. This table restricts the analysis to individuals aged between 12 and 17 in 1997. Asterisks denote pointwise statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A18: Pre-trends Test on the Probability of Marriage: Age 14-17 in 1997

Event-Time	ATT(t)	Std. Error	Conf. Interval
-3	-0.005	0.0565	[-0.1157 , 0.1056]
-2	-0.0049	0.0293	[-0.0623 , 0.0525]
-1	0	NA	[NA , NA]
0	-0.002	0.0092	[-0.02 , 0.016]
N	7084		

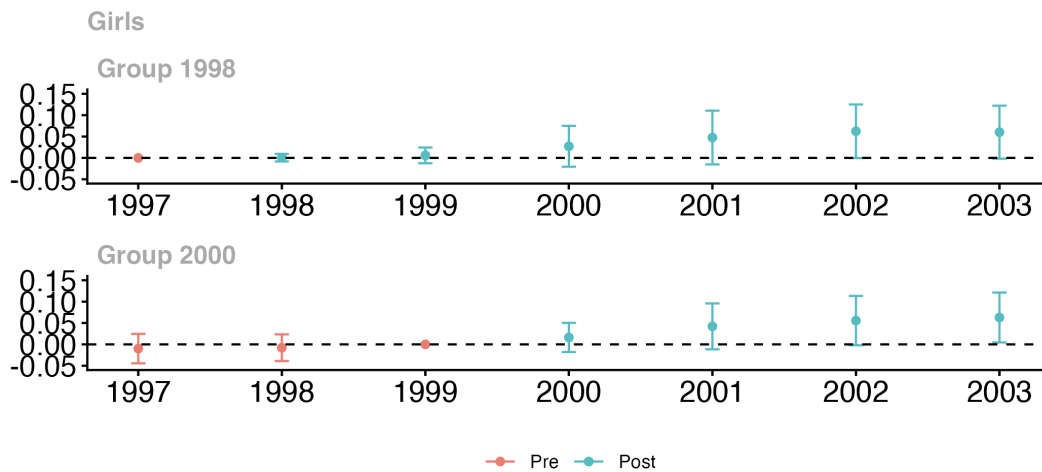
Note: This table shows the average treatment effects by length of exposure and the respective standard errors and confidence intervals. N is the number of observations. Event-Time refers to the period relative to the treatment year. This table restricts the analysis to individuals aged between 14 and 17 in 1997. Asterisks denote pointwise statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A19: Effect of Progresa/Oportunidades on the Probability of Marriage by Length of Exposure: Girls

Event-Time	ATT(t)	Std. Error	Conf. Interval
-3	-0.0099	0.013	[-0.0353 , 0.0156]
-2	-0.0078	0.0117	[-0.0308 , 0.0152]
-1	0	NA	[NA , NA]
0	0.0062	0.0053	[-0.0043 , 0.0167]
1	0.019**	0.0077	[0.004 , 0.034]
2	0.0374***	0.0141	[0.0099 , 0.065]
3	0.0531***	0.0192	[0.0156 , 0.0907]
4	0.0623***	0.0232	[0.0168 , 0.1079]
5	0.0602***	0.0233	[0.0146 , 0.1058]
N	12357		

Note: This table shows the average treatment effects by length of exposure and the respective standard errors and confidence intervals. N is the number of observations. Event-Time refers to the time period relative to the treatment year. Asterisks denote pointwise statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A10: Effect of Progresa/Oportunidades on the Probability of Marriage by Group: Girls



Note: This figure presents the average treatment effect on the treated by treatment group and period. Group 1998, or T1998, is the group that first received treatment in 1998, and Group 2000, or T2000, is the group that first received treatment in 2000. In red are the estimates before treatment started, and in blue after treatment started. Standard errors are clustered at the randomization level (locality). Confidence intervals shown are 95% uniform (simultaneous) confidence bands from the multiplier bootstrap of Callaway and Sant'Anna (2021); they hold jointly across all event times and are therefore wider than the pointwise 95% intervals reported in the tables.

Table A20: Effect of Progresa/Oportunidades on the Probability of Marriage by Group and Year: Girls

	Time	ATT(g,t)	Std. Error	Conf. Interval
T 1998	1997	0	NA	[NA , NA]
T 1998	1998	6e-04	0.0034	[-0.0061 , 0.0072]
T 1998	1999	0.0059	0.0071	[-0.0079 , 0.0197]
T 1998	2000	0.0271	0.0183	[-0.0088 , 0.0631]
T 1998	2001	0.0476**	0.0241	[4e-04 , 0.0948]
T 1998	2002	0.0623***	0.024	[0.0153 , 0.1094]
T 1998	2003	0.0602**	0.0237	[0.0136 , 0.1067]
T 2000	1997	-0.0099	0.0132	[-0.0357 , 0.016]
T 2000	1998	-0.0078	0.0119	[-0.0312 , 0.0156]
T 2000	1999	0	NA	[NA , NA]
T 2000	2000	0.0162	0.0131	[-0.0094 , 0.0418]
T 2000	2001	0.0422**	0.0206	[0.0019 , 0.0825]
T 2000	2002	0.0556**	0.0221	[0.0123 , 0.0989]
T 2000	2003	0.0629***	0.0224	[0.0191 , 0.1067]
N			12357	

Note: This table shows the average treatment effects by group and length of exposure, and the respective standard errors and confidence intervals. N is the number of observations. Asterisks denote pointwise statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A21: Pre-trends Test on the Probability of Marriage: Age 10-17 in 1997 - Girls

Event-Time	ATT(t)	Std. Error	Conf. Interval
-3	-0.01	0.0198	[-0.0487 , 0.0287]
-2	-0.0105	0.0178	[-0.0454 , 0.0245]
-1	0	NA	[NA , NA]
0	0.009	0.0109	[-0.0122 , 0.0303]
N		8199	

Note: This table shows the average treatment effects by length of exposure and the respective standard errors and confidence intervals. N is the number of observations. Event-Time refers to the period relative to the treatment year. This table restricts the analysis to girls aged between 10 and 17 in 1997. Asterisks denote pointwise statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A22: Pre-trends Test on the Probability of Marriage: Age 12-17 in 1997 - Girls

Event-Time	ATT(t)	Std. Error	Conf. Interval
-3	-0.0052	0.0256	[-0.0554 , 0.0449]
-2	-0.0078	0.0243	[-0.0555 , 0.0399]
-1	0	NA	[NA , NA]
0	0.0131	0.0128	[-0.012 , 0.0382]
N	5652		

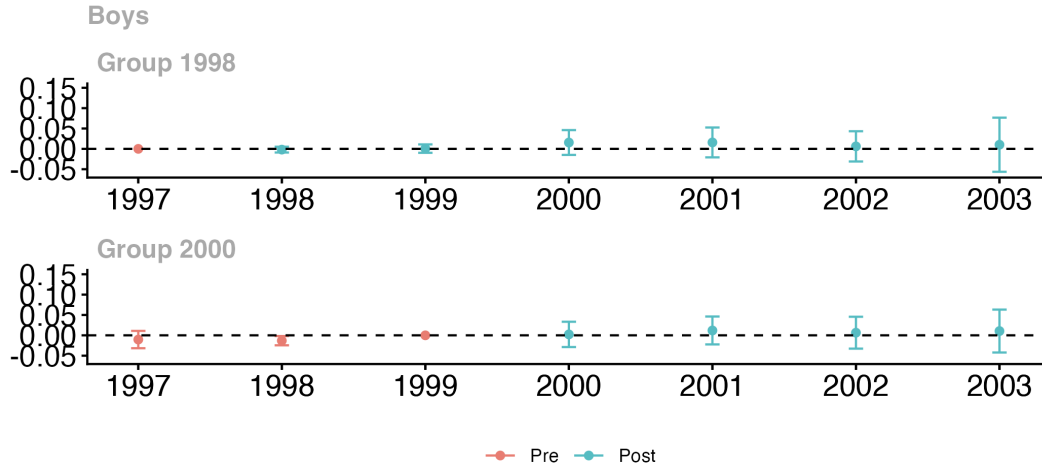
Note: This table shows the average treatment effects by length of exposure and the respective standard errors and confidence intervals, for girls only. N is the number of observations. Event-Time refers to the period relative to the treatment year. This table restricts the analysis to girls aged between 12 and 17 in 1997. Asterisks denote pointwise statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A23: Pre-trends on the Probability of Marriage: Age 14-17 in 1997 - Girls

Event-Time	ATT(t)	Std. Error	Conf. Interval
-3	0.0082	0.0581	[-0.1058 , 0.1221]
-2	0.004	0.0468	[-0.0876 , 0.0957]
-1	0	NA	[NA , NA]
0	7e-04	0.0194	[-0.0373 , 0.0386]
N	3411		

Note: This table shows the average treatment effects by length of exposure and the respective standard errors and confidence intervals. N is the number of observations. Event-Time refers to the period relative to the treatment year. This table restricts the analysis to girls aged between 14 and 17 in 1997. Asterisks denote pointwise statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A11: Effect of Progresa/Oportunidades on the Probability of Marriage by Group: Boys



Note: This figure presents the average treatment effect on the treated by treatment group and period. Group 1998, or T1998, is the group that first received treatment in 1998, and Group 2000, or T2000, is the group that first received treatment in 2000. In red are the estimates before treatment started, and in blue after treatment started. Standard errors are clustered at the randomization level (locality). Confidence intervals shown are 95% uniform (simultaneous) confidence bands from the multiplier bootstrap of Callaway and Sant'Anna (2021); they hold jointly across all event times and are therefore wider than the pointwise 95% intervals reported in the tables.

Table A24: Effect of Progresa/Oportunidades on the Probability of Marriage by Length of Exposure: Boys

Event-Time	ATT(t)	Std. Error	Conf. Interval
-3	-0.0104	0.0088	[-0.0276 , 0.0068]
-2	-0.0131***	0.0047	[-0.0223 , -0.0039]
-1	0	NA	[NA , NA]
0	-4e-04	0.0044	[-0.0091 , 0.0082]
1	0.0046	0.0053	[-0.0057 , 0.015]
2	0.0124	0.009	[-0.0052 , 0.03]
3	0.0139	0.0129	[-0.0113 , 0.0391]
4	0.0061	0.0154	[-0.0241 , 0.0363]
5	0.0102	0.0263	[-0.0414 , 0.0618]
N	13288		

Note: This table shows the average treatment effects by length of exposure and the respective standard errors and confidence intervals. N is the number of observations. Event-Time refers to the time period relative to the treatment year. Asterisks denote pointwise statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A25: Effect of Progresa/Oportunidades on the Probability of Marriage by Group and Year: Boys

	Time	ATT(g,t)	Std. Error	Conf. Interval
T 1998	1997	0	NA	[NA , NA]
T 1998	1998	-0.0019	0.0029	[-0.0075 , 0.0038]
T 1998	1999	6e-04	0.0042	[-0.0076 , 0.0089]
T 1998	2000	0.0156	0.0124	[-0.0087 , 0.0399]
T 1998	2001	0.0158	0.0149	[-0.0135 , 0.045]
T 1998	2002	0.0061	0.0151	[-0.0235 , 0.0358]
T 1998	2003	0.0102	0.0271	[-0.0429 , 0.0632]
T 2000	1997	-0.0104	0.0087	[-0.0274 , 0.0066]
T 2000	1998	-0.0131***	0.0046	[-0.0222 , -0.004]
T 2000	1999	0	NA	[NA , NA]
T 2000	2000	0.0023	0.0127	[-0.0225 , 0.0271]
T 2000	2001	0.012	0.0139	[-0.0154 , 0.0393]
T 2000	2002	0.0065	0.0159	[-0.0248 , 0.0377]
T 2000	2003	0.0105	0.0215	[-0.0315 , 0.0525]
N			13288	

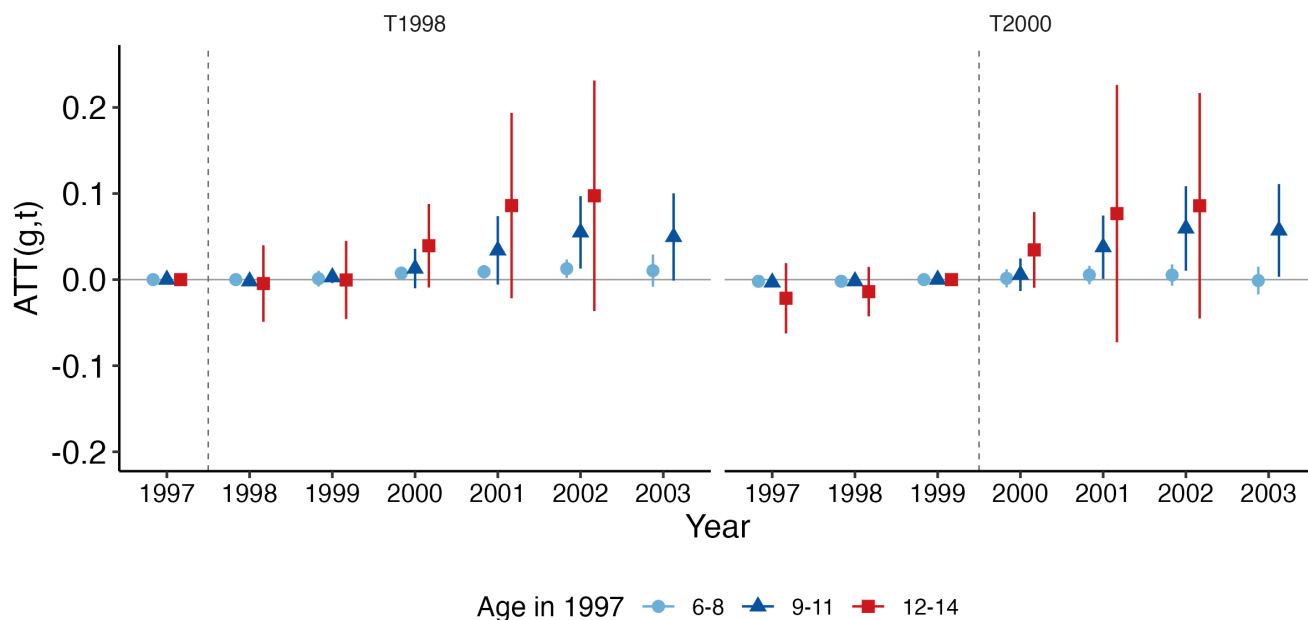
Note: This table shows the average treatment effects by group and length of exposure, and the respective standard errors and confidence intervals. N is the number of observations. Asterisks denote pointwise statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A26: Program's effect on Marriage (adult men)

Rel. time	ATT	SE
-3	-0.048	(0.041)
-2	-0.027	(0.041)
-1	NA	
0	0.011	(0.008)
1	0.021*	(0.011)
2	0.061*	(0.031)
3	0.073**	(0.034)
4	0.084*	(0.050)
5	0.098*	(0.052)

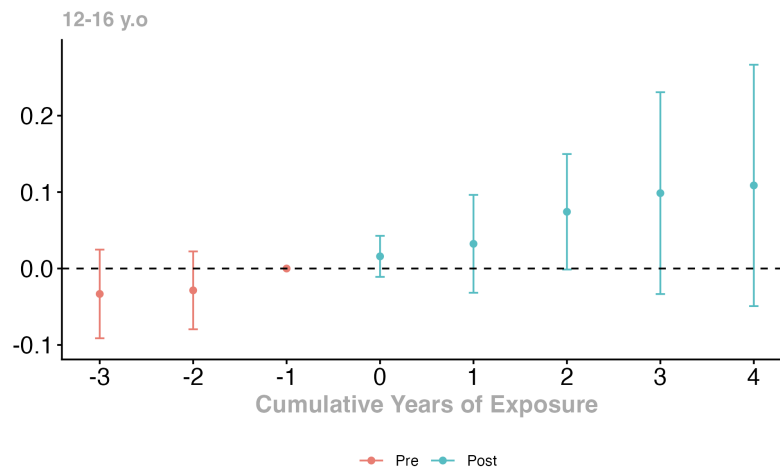
Note: This table presents the average treatment effect on the treated by length of exposure for adult men in the sample. Standard errors, clustered at the randomization level (locality), are in parentheses. Asterisks denote pointwise statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A12: Effect of Progresa/Oportunidades on the Probability of Marriage: Girls, by Age



Note: This figure presents the average treatment effect on the treated girls, $ATT(g,t)$, by treatment cohort (T1998 and T2000) and year, for three baseline age groups: girls aged 6–8, 9–11, and 12–14 in 1997. Colour and marker shape denote the age group. The dashed vertical line marks each cohort's treatment onset; estimates to its left are pre-treatment. The 12–14 group is not observed in 2003, since those girls have all turned 18 by then. Standard errors are clustered at the randomization level (locality). Confidence intervals shown are 95% uniform (simultaneous) confidence bands from the multiplier bootstrap of Callaway and Sant'Anna (2021).

Figure A13: Progresa/Oportunidades' Effect on Early Marriage, by Length of Exposure: Girls aged 12 to 16 in 1997



Note: This figure presents the average treatment effect on the treated girls by the length of exposure to treatment. Time -1 represents one period prior to treatment. For T1998 (the group that first received the treatment in 1998) time -1 corresponds to 1997, and for T2000 (the group that first received the treatment in 2000) to 1999. Period 2 represents two years after treatment, and so on. In red are the estimates before treatment started, and in blue after treatment started. Standard errors are clustered at the randomization level (locality). Confidence intervals shown are 95% uniform (simultaneous) confidence bands from the multiplier bootstrap of Callaway and Sant'Anna (2021); they hold jointly across all event times and are therefore wider than the pointwise 95% intervals reported in the tables.

Table A27: Progresa/Oportunidades' Effect on Early Marriage, by Length of Exposure: Girls aged 12 to 16 in 1997

Event-Time	ATT(t)	Std. Error	Conf. Interval
-3	-0.0333	0.023	[-0.0783 , 0.0117]
-2	-0.0286	0.0202	[-0.0681 , 0.011]
-1	0	NA	[NA , NA]
0	0.0159	0.0106	[-0.0049 , 0.0367]
1	0.0323	0.0254	[-0.0174 , 0.082]
2	0.0742**	0.0299	[0.0156 , 0.1329]
3	0.0986*	0.0523	[-0.0038 , 0.2011]
4	0.1087*	0.0625	[-0.0138 , 0.2313]
N	4907		

Note: This table shows the average treatment effects by length of exposure and the respective standard errors and confidence intervals for girls aged 12 to 16 in 1997. N is the number of observations. Event-Time refers to the period relative to the treatment year. Asterisks denote pointwise statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A28: Effect of Progresa/Oportunidades on the Probability of Marriage by Length of Exposure: Experiment (T1998 VS T2000)

Event-Time	ATT(t)	Std. Error	Conf. Interval
-1	0	NA	[NA , NA]
0	-4e-04	0.002	[-0.0044 , 0.0036]
1	6e-04	0.0031	[-0.0055 , 0.0067]
2	0.0022	0.0041	[-0.0059 , 0.0102]
3	-0.0036	0.0052	[-0.0138 , 0.0066]
4	-0.0026	0.0056	[-0.0136 , 0.0083]
5	-0.0027	0.0068	[-0.0159 , 0.0106]
N	20018		

Note: This table shows the average treatment effects by length of exposure and the respective standard errors and confidence intervals. N is the number of observations. Event-Time refers to the period relative to the treatment year. Asterisks denote pointwise statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A29: Effect of Progresa/Oportunidades on the Probability of Marriage by Length of Exposure: Experiment (T1998 VS T2000) - Girls

Event-Time	ATT(t)	Std. Error	Conf. Interval
-1	0	NA	[NA , NA]
0	1e-04	0.0033	[-0.0065 , 0.0066]
1	0.0017	0.0048	[-0.0077 , 0.0111]
2	0.0011	0.0064	[-0.0114 , 0.0136]
3	-0.006	0.0084	[-0.0226 , 0.0105]
4	-0.0034	0.0098	[-0.0226 , 0.0159]
5	-9e-04	0.0107	[-0.0219 , 0.0201]
N	9649		

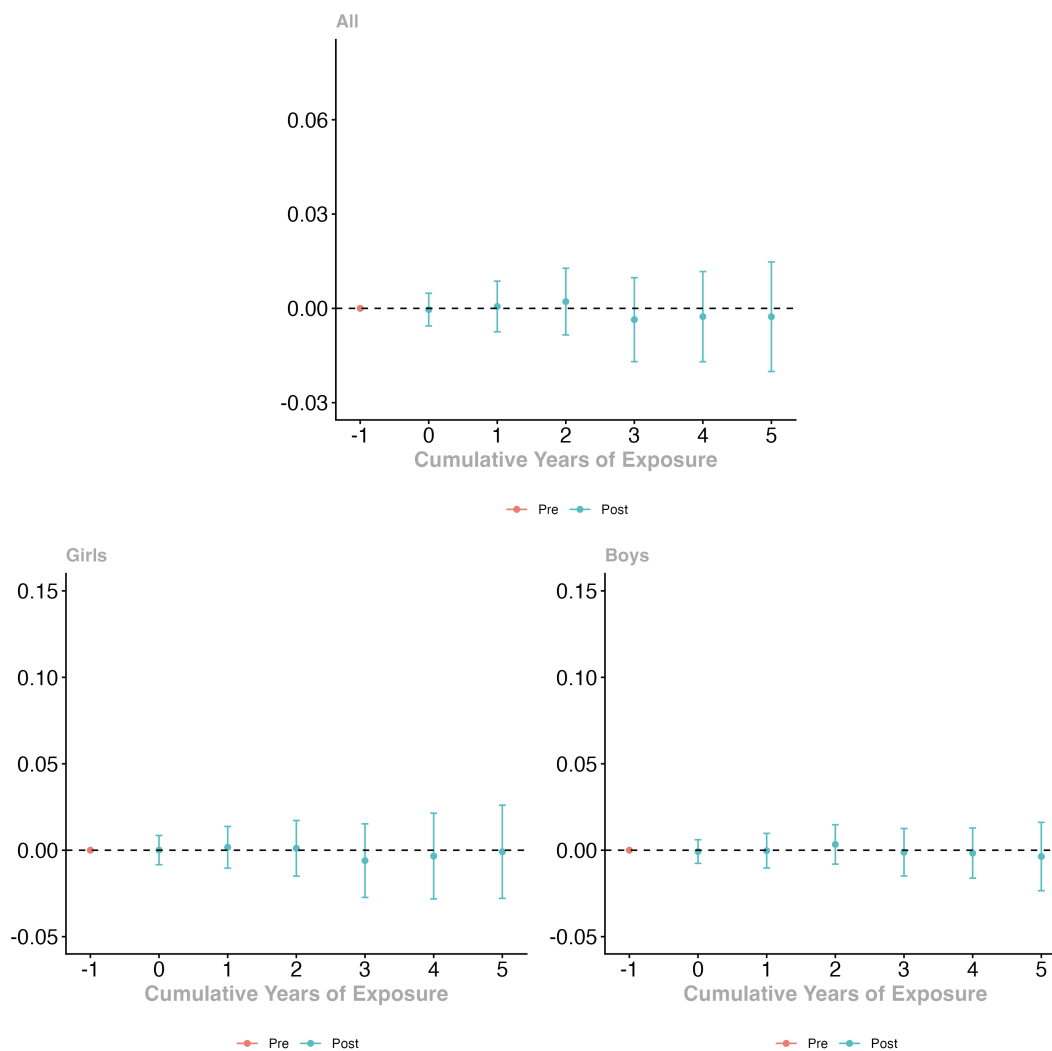
Note: This table shows the average treatment effects by length of exposure and the respective standard errors and confidence intervals. N is the number of observations. Event-Time refers to the period relative to the treatment year. Asterisks denote pointwise statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A30: Effect of Progresa/Oportunidades on the Probability of Marriage by Length of Exposure: Experiment (T1998 VS T2000) - Boys

Event-Time	ATT(t)	Std. Error	Conf. Interval
-1	0	NA	[NA , NA]
0	-8e-04	0.0028	[-0.0062 , 0.0047]
1	-3e-04	0.0041	[-0.0083 , 0.0077]
2	0.0033	0.0046	[-0.0057 , 0.0124]
3	-0.0012	0.0056	[-0.0121 , 0.0097]
4	-0.0017	0.0059	[-0.0132 , 0.0099]
5	-0.0036	0.008	[-0.0194 , 0.0121]
N	10369		

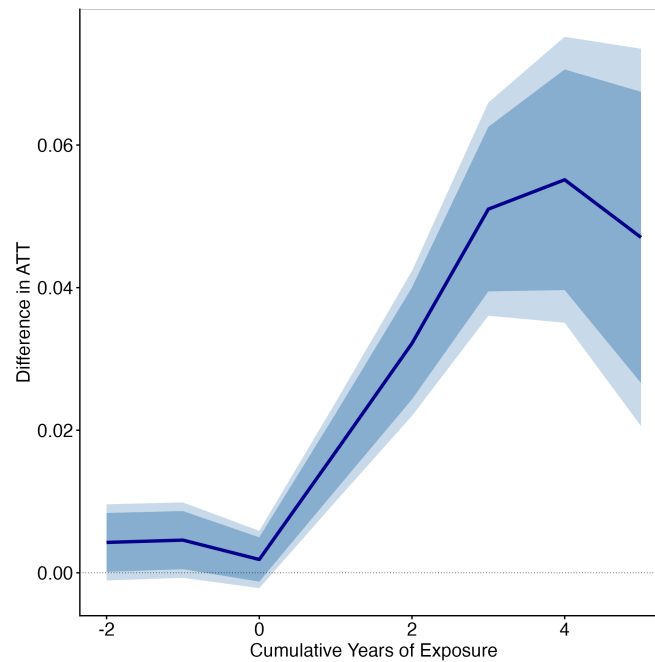
Note: This table shows the average treatment effects by length of exposure and the respective standard errors and confidence intervals. N is the number of observations. Event-Time refers to the period relative to the treatment year. Asterisks denote pointwise statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A14: Effect of Progresa/Oportunidades on the Probability of Marriage by Group: Experiment (T1998 VS T2000)



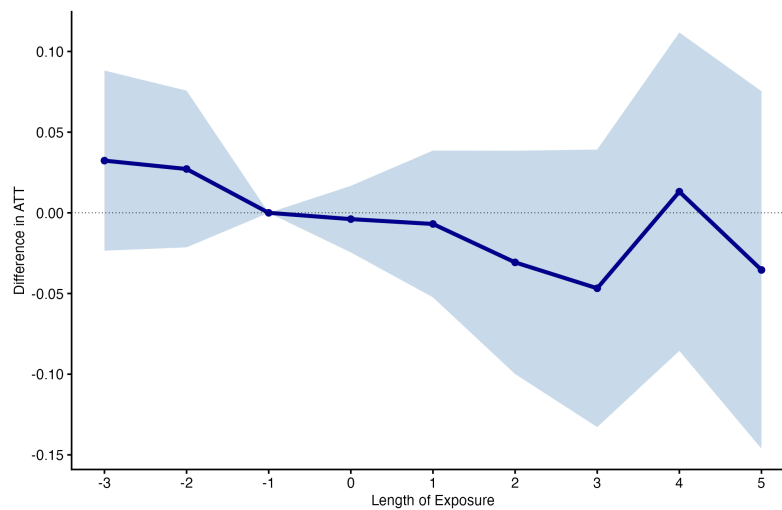
Note: This figure presents the average treatment effect on the treated, comparing T1998 and T2000. In red are the estimates before treatment started, and in blue after treatment started. Standard errors are clustered at the randomization level (locality). Confidence intervals shown are 95% uniform (simultaneous) confidence bands from the multiplier bootstrap of Callaway and Sant'Anna (2021); they hold jointly across all event times and are therefore wider than the pointwise 95% intervals reported in the tables.

Figure A15: Difference Between the ATT for the Cohort of Girls 13 y.o or older in 2001 and the Cohort of Girls 12 y.o or younger in 2001



Note: This figure presents the difference in the program's average treatment effect for girls 13 or older in 2001 and girls 12 or younger in 2001. Standard errors are clustered at the randomization level (locality). The darker inner band is the pointwise 95% confidence interval; the lighter outer band is the 95% uniform (simultaneous) confidence band.

Figure A16: Difference Between the ATT for localities with Returns to Education Above and Below the Median



Note: This figure presents the difference in the program's average treatment effect for localities with returns to education above the median and localities with returns below the median. Standard errors are clustered at the randomization level (locality). The shaded area is the pointwise 95% confidence interval.

Table A31: Predicted dropout probability and marriage rate by decile, all girls

Decile	N Individuals	N Localities	Mean dropout	SD dropout	Marriage Rate
1	1060	383	0.0057	0.0085	0.0098
2	1059	415	0.1073	0.0469	0.0222
3	1059	419	0.2743	0.0484	0.0273
4	1059	434	0.4308	0.0423	0.0347
5	1059	438	0.5756	0.0408	0.0475
6	1059	447	0.7088	0.0355	0.0530
7	1059	433	0.8289	0.0343	0.0569
8	1060	424	0.9328	0.0245	0.0721
9	1058	422	0.9910	0.0084	0.0835
10	1059	429	1.0000	0.0000	0.1430
NA	1766	425	NaN	NA	0.0179

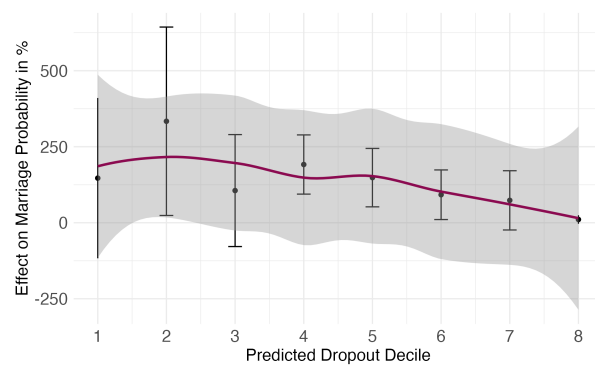
Note: In this table, each row corresponds to a predicted dropout decile, where decile 1 contains girls with the lowest predicted probability of dropping out and decile 10 the highest. Predicted dropout probabilities are estimated using a probit model run separately for each age group on 1997 baseline characteristics, and then pooled into a single global ranking. Marriage rates are averages over the full observation period. The sample is restricted to all girls in the analysis.

Table A32: Predicted dropout probability and marriage rate by decile, girls in the control group

Decile	N Individuals	N Localities	Mean dropout	SD dropout	Marriage Rate
1	257	76	0.0067	0.0092	0.0035
2	215	79	0.1078	0.0474	0.0027
3	217	83	0.2782	0.0442	0.0118
4	235	88	0.4291	0.0431	0.0233
5	204	84	0.5719	0.0391	0.0248
6	242	95	0.7060	0.0348	0.0301
7	296	93	0.8269	0.0337	0.0454
8	247	85	0.9292	0.0242	0.0635
9	226	77	0.9910	0.0084	0.0860
10	232	90	1.0000	0.0000	0.2058
NA	337	89	NaN	NA	0.0090

Note: In this table, each row corresponds to a predicted dropout decile, where decile 1 contains girls with the lowest predicted probability of dropping out and decile 10 the highest. Predicted dropout probabilities are estimated using a probit model run separately for each age group on 1997 baseline characteristics, and then pooled into a single global ranking. Marriage rates are averages over the full observation period. The sample is restricted to girls in the control group.

Figure A17: Effect of Progresa/Oportunidades on the Probability of Marriage by Dropout Probability



Note: This figure presents the average treatment effect on the treated on marriage by predicted dropout probability (in 8 quantiles). Only marriages under 18 are considered.

B Appendix: Data Limitations

B.1 Marriage Reporting

The outcome of interest, marriage status, may be subject to misreporting. Three concerns arise regarding marriage reporting. First, the control group's marriage status and age at first union were asked in 2003, whereas the treatment group was asked about marriage status in every wave. Second, existing evidence suggests that adolescents under-report marriage (Neal and Hosegood, 2015), which may be due to social desirability bias. Third, misreporting might differ by treatment status. I focus on girls in this analysis since child marriage rates for boys are very small and the program had no impact on them.

To address these concerns, I conduct a series of empirical exercises to assess the likelihood of misreporting.

First, I compare the rates of child marriage in the Census 2000 data with those in the ENCEL 2000 data. I use the year 2000 because it is the only year in which both Census and ENCEL data were collected simultaneously. Restricting the census data to localities covered by ENCEL and aggregating the ENCEL data to the locality level to match the coarser level of the Census data, I find similar proportions of girls married in 2000 in both datasets (6.14% in ENCEL and 6.24% in the Census). Since Census and ENCEL data are both self-reported, we can also compare the Census data with administrative data and if they are close, then it is likely that ENCEL data is also close to administrative data.

To do this, I compare the proportion of individuals legally married in the Census and the Mexican administrative marriage data (La Estadística de Matrimonios) in 1995 for the localities covered by the Census.²⁸ I find very close proportions in the census data (1.77 marriages per 1000 inhabitants) and the administrative marriage records (1.42 marriages per 1000 inhabitants) for both children (under 18) and adults (6.2 in the Census and 5.9 in the registry data). This

²⁸In the administrative data, we have the universe of legal marriages that occur in a given municipality and year. To have the same data in the Census, I consider those individuals who report an age at marriage equal to their current age. This way, I ensure that I am only considering marriages that occur in the same year and are comparable to the marriage records. This is the reason why I use 1995 and not 2000. In 2000 I do not have information on age at marriage in the census, thus the comparison would be incorrect.

result is reassuring, as it suggests that self-reported data is close to administrative records, thus mitigating the concern of misreporting due to social desirability bias.

Furthermore, I examine all girls in the ENCASEH and ENCEL data, between 12 and 17 years old, who reported being single in 1997. I first create a marriage variable based solely on information from 2003 (age at first union and marriage status) and compare it with the one used in the paper, which is the contemporaneous reporting not allowing for separations. The two variables are consistent if they are the same, for a given individual, for a given year. First, consistency rates are very high across the treatment groups and years, all above 97%. At baseline, in 1997, there was virtually no inconsistency, less than 1% of the girls in both T1998 and T2000 misreported their marriage. Second, I do not find statistically significant differences in consistency between T1998 and T2000 across the years (see Table B1). As C2000 was interviewed only in 2003, the two inconsistencies the pure control group can exhibit are: (i) they reported to be single in 1997, but the age at first union implies they were married by then; and (ii) they report to be single in 2003, but the age at first union is not consistent. There is virtually no inconsistency in the control group. Comparing the treated and control groups, the treated groups had between 0.4 and 1.8 p.p. more inconsistencies than the control group in 1997 and 2003, respectively.²⁹

The low rates of inconsistency within years and groups and the small difference across groups mitigate the concern of misremembering marriage status, providing evidence of the reliability of marriage reporting in the data used in the analysis.

²⁹Note that marriage in 2001 and 2002 was already created using information from 2003, since there was no survey in those years. Therefore, I do not include them in the table, as it would be redundant.

Table B1: Share with consistent civil status (contemporaneous vs retroactive) by year and group

Year	Control	T1998	T2000	β (T1998 vs T2000)	p	β (Treated vs Control)	p
1997	1.000 (N=5620)	0.994 (N=12796)	0.992 (N=7089)	0.0018	0.155	-0.0071	0.000
1998	—	0.985 (N=12796)	0.983 (N=7089)	0.0026	0.182	—	—
1999	—	0.975 (N=12796)	0.979 (N=7089)	-0.0032	0.194	—	—
2000	—	0.968 (N=12796)	0.967 (N=7089)	0.0007	0.826	—	—
2003	0.992 (N=5620)	0.973 (N=12796)	0.975 (N=7089)	-0.0021	0.427	-0.0180	0.000

Note: This table shows, for each treatment and control group, and each year, the share of individuals whose contemporaneous civil status (declared each year) matches the civil status reported in 2003. It also shows the statistical difference in the consistency rates across treatment groups and treatment and control groups, and the respective p-value. N represents the number of observations in each group and year for whom we have information each year, and in 2003. Standard errors are clustered at the locality level.

Finally, I test whether treatment leads to misreporting. Naturally, I cannot test this using the paper's data (ENCEL), given that C2000 was only asked about marriage in 2003. Therefore, I resort to census data, which is available for both treated and control municipalities in 2000. I first calculate the rate of child marriage, per municipality, in 2000 using both the paper's data (ENCEL) and census data, restricting the Census data to the municipalities covered in the ENCEL. I define misreporting as the difference between the child marriage rates in the ENCEL and the Census data, by municipality. I find that the average misreporting rates are extremely low: the difference between ENCEL and the Census's child marriage rates is -0.001. Since the program was allocated at the locality level, and I only have access to the 2000 Census at the municipality level, I create a treatment intensity variable at the municipality level. This is defined as the proportion of Progresas's beneficiaries in each municipality. I then test whether there is a correlation between treatment intensity and misreporting. I do not find a statistically significant correlation between treatment intensity and the probability of over-reporting in ENCEL (p.e. = 0.52 and CI = [-1.17, 2.20]) or the extent of misreporting (p.e. = -0.15 and CI = [-0.51, 0.22]).

B.2 Pre-program Data

To further address potential concerns about recall data and to provide additional evidence of no pre-trends in marriage, I use the 1990 Mexican Census to test whether, in 1990, localities that were going to be treated in 1998 already had different marriage rates than those that were not going to receive the program.

First, I investigate how problematic recall bias can be. The control group, C2000, answered about their 1997 characteristics in 2003. One way of testing whether the bias is substantial is to compare these individuals with a similar population in 1990, where these characteristics were asked contemporaneously. So, I look at the household characteristics of the children who were between 0 and 9 in 1990. These are the ones that will be between 6 and 16 in 1997 (the sample of the main analysis). Table B2 shows the unconditional, unweighted control-group means from the paper (C2000, recalled) and from the 1990 Census (the 0–9 cohort in the same control localities), for every covariate that both sources measure. The comparison is not exact, and there are important considerations to take when analyzing this table: (i) the paper’s data is restricted to Progresa-eligible (poorer) households, whereas the Census covers the full population of the same localities; and (ii) household composition changes naturally in 7 years. The two sources agree overall, with none of the variables registering massive disparities that cannot be explained by the two considerations above. Importantly, the variables most susceptible to recall bias, such as ownership of assets, have no counterpart in the Census and cannot be checked here. I address this limitation by running the main analysis of the paper without those variables (see Tables A11, A12, and A13). Taken together, the contemporaneous Census data agree with the recalled control-group data wherever the two measure the same thing comparably, and the discrepancies that remain are attributable to cohort and sample composition.

Table B2: Recall check: control-group baseline characteristics, paper (recalled C2000) vs. 1990 Census (0–9 cohort)

Covariate	All		Girls		Boys	
	Paper C2000	Census 0–9	Paper C2000	Census 0–9	Paper C2000	Census 0–9
<i>Housing</i>						
Dirt floor	0.69	0.41	0.68	0.41	0.70	0.40
Inferior quality wall	0.61	0.56	0.61	0.56	0.61	0.56
Inferior quality roof	0.26	0.31	0.26	0.32	0.26	0.31
No. of bedrooms	1.67	1.69	1.67	1.69	1.68	1.70
Piped water	0.43	0.37	0.44	0.38	0.41	0.37
Electricity	0.70	0.68	0.72	0.68	0.68	0.68
<i>Household head & structure</i>						
HH head/spouse went to school	0.64	0.85	0.65	0.84	0.63	0.86
HH head/spouse worked	0.93	0.80	0.93	0.79	0.93	0.80
Number in household	7.51	7.29	7.53	7.28	7.49	7.29
Age of household head	42.05	39.32	42.15	39.36	41.96	39.27
Female household head	0.12	0.08	0.12	0.09	0.12	0.08
Indigenous language in HH	0.28	0.23	0.28	0.24	0.28	0.22
<i>Household composition</i>						
Child 0–5	0.67	0.91	0.68	0.91	0.67	0.91
Teen 16–19	0.38	0.31	0.40	0.31	0.37	0.31
Woman 20–39	0.69	0.83	0.69	0.82	0.69	0.83
Woman 40–59	0.38	0.23	0.38	0.23	0.39	0.23
Woman 60+	0.13	0.09	0.14	0.09	0.13	0.09
Man 20–39	0.53	0.69	0.53	0.69	0.53	0.69
Man 40–59	0.48	0.31	0.48	0.31	0.48	0.31
Man 60+	0.14	0.09	0.15	0.09	0.13	0.09

Note: ‘Paper’ columns give the unconditional, unweighted mean for the C2000 control group in the paper’s data (children aged 6–16 in 1997, with pre-program covariates recalled in 2003); ‘Census’ columns give the corresponding unweighted mean among control-group children aged 0–9 in the 1990 Census, in the same control localities. Asset variables and child-level outcomes (marriage, own schooling, and age) are omitted, as they have no comparable 1990 Census measure for this cohort.

Then I conduct a series of exercises to show that the treatment and control localities did not differ in their marriage behavior in pre-treatment years. Table B3 shows the balance between treated and control children aged 6 to 16 in 1990 for covariates from the Census data that coincide with the variables used in the paper’s main analysis. Column 3 shows the raw summary statistics, and Column 4 shows the re-weighted statistics, using the estimated propensity score

as the weight.³⁰ Note that, even if the propensity score only targeted core poverty measures, it guarantees balance in most of the other variables that are used in the analysis.

Once we guarantee that the IPW are making groups comparable on observables, we can test whether marriage rates were different in comparable control and treated localities. Table B4 shows no difference across localities. Around 7% of girls aged 12 to 17 in the treated localities are married, compared with 5% in unweighted control localities. Once these are re-weighted, the marriage rate increases to 7% as well. Rates for boys are substantially lower, at around 1%, also similar for control and treated localities.

Importantly, the previous exercise is on a different cohort from the paper. The individuals in the analysis of this paper were between 6 and 16 years old in 1997, which means that they were roughly between 0 and 9 in 1990. For this population, we do not observe their marriage outcomes - they are too young, and the census only asks this question for children 12 years old or older. However, we can still compare how these individuals compare to older individuals regarding their observable characteristics. Table B5 shows that on the housing and poverty dimensions that drove Progresas's locality selection (dirt floor, walls, roof, piped water, electricity, indigenous status), the two cohorts are essentially identical (e.g. dirt floor 0.61 vs. 0.63, electricity 0.50 vs. 0.48, indigenous 0.26 vs. 0.27). The variables where the differences are significant are the ones somehow related to the age brackets: age, schooling, and literacy, and the mechanical life-cycle shift in household age structure (younger parents, more young children). The same pre-program covariates are also balanced across treatment and control localities, using the same propensity score specification described above.

In short, on every treatment-relevant dimension, the cohort the paper follows is as comparable across treated and control localities as the cohorts whose marriage we can observe. This can be interpreted as suggestive evidence that marriage would not diverge across treatment groups for this cohort until 1997.

³⁰The propensity score specification used in this analysis was selected to maximize covariate balance and effective sample size. It does not coincide with the specification used in the paper because that led to poor common support in the census data. Adding more variables led to a drastic reduction of the effective sample size, with no significant improvement (and sometimes worsening) on the covariate balance. Results are robust to other propensity score specifications and can be shared upon request. We also observe the same patterns if we restrict the sample to girls between 12 and 17.

Table B3: Pre-program covariate balance after reweighting : children 6–16, T1998 vs. C2000

Variable	Treated	Control (unwtd.)	Control (rewt.)	<i>p</i> -value
<i>Panel A. Housing and poverty</i>				
Dirt floor [†]	0.661	0.390	0.708	0.284
Inferior walls	0.736	0.560	0.772	0.378
Inferior roof	0.424	0.294	0.513	0.093
Number of bedrooms [†]	1.62	1.89	1.57	0.275
Piped water on lot [†]	0.247	0.395	0.207	0.423
Electricity [†]	0.446	0.702	0.406	0.530
Indigenous [†]	0.304	0.131	0.348	0.584
<i>Panel B. Child characteristics</i>				
Education (years)	3.76	3.82	3.50	0.013
Literate	0.803	0.820	0.760	0.077
Currently in school	0.710	0.719	0.674	0.148
Worked	0.167	0.180	0.206	0.098
Age	10.64	10.76	10.72	0.204
<i>Panel C. Household head and structure</i>				
Female household head	0.109	0.096	0.092	0.124
Head age	44.46	44.04	43.41	0.020
Household size	7.58	7.84	7.40	0.239
Head/spouse education (yrs)	3.21	3.50	3.30	0.469
Head/spouse ever schooled	0.711	0.818	0.755	0.117
Head/spouse worked	0.783	0.771	0.781	0.960
Any indigenous member	0.409	0.231	0.415	0.944
<i>Panel D. Household composition</i>				
Child aged 0–5	0.670	0.670	0.695	0.189
Teenager aged 16–19	0.443	0.494	0.457	0.448
Woman aged 20–39	0.674	0.694	0.683	0.553
Woman aged 40–59	0.404	0.392	0.366	0.030
Woman aged 60+	0.101	0.098	0.084	0.093
Man aged 20–39	0.541	0.555	0.538	0.882
Man aged 40–59	0.477	0.480	0.480	0.893
Man aged 60+	0.115	0.114	0.106	0.409

Note: Pre-program (1990 Census) means for treated (T1998) localities, control (C2000) localities before weighting, and control localities after inverse-probability weighting under the `poverty_core` specification. Sample: children aged 6–16 in 1990, both sexes ($N_T = 24,683$ treated, $N_C = 9,338$ control). The *p*-value tests equality of the treated and reweighted-control means, with standard errors clustered at the locality level. [†] marks the five covariates included in the `poverty_core` propensity score. The T2000-vs-C2000 comparison is analogous.

Table B4: Pre-program marriage rates, ages 12–17 (poverty_core)

	Comparison	Treated	Control (unwtd.)	Control (rewt.)	<i>p</i> -value	N_T/N_C
All	T1998 vs. C2000	0.0447	0.0289	0.0459	0.887	11,689 / 4,668
	T2000 vs. C2000	0.0432	0.0289	0.0429	0.965	7,794 / 4,668
	T1998 vs. T2000	0.0447	—	0.0432	0.754	11,689 / 7,794
Girls	T1998 vs. C2000	0.0768	0.0502	0.0797	0.841	5,718 / 2,251
	T2000 vs. C2000	0.0717	0.0502	0.0743	0.843	3,808 / 2,251
	T1998 vs. T2000	0.0768	—	0.0717	0.493	5,718 / 3,808
Boys	T1998 vs. C2000	0.0141	0.0091	0.0145	0.934	5,971 / 2,417
	T2000 vs. C2000	0.0161	0.0091	0.0133	0.558	3,986 / 2,417
	T1998 vs. T2000	0.0141	—	0.0161	0.570	5,971 / 3,986

Note: Pre-program (1990 Census) share ever married among those aged 12-17 (the census records marital status from age 12), by group. ‘Treated’ and ‘Control (rewt.)’ are the treated mean and the inverse-probability-weighted control mean under the `poverty_core` propensity-score specification; ‘Control (unwtd.)’ is the raw control mean. The *p*-value tests equality of the treated and reweighted-control means, with standard errors clustered at the locality level. The T1998-vs-T2000 rows compare the two treated cohorts and are unweighted by construction (no control reweighting), so the unweighted column is omitted, and the ‘Control (rewt.)’ column reports the T2000 mean.

Table B5: Cross-cohort summary statistics: children 6–16 vs. 0–9 in 1990

Variable	Ages 6–16	Ages 0–9	Difference
<i>Panel A. Housing and poverty</i>			
Dirt floor	0.613	0.634	-0.021
Inferior walls	0.701	0.711	-0.010
Inferior roof	0.411	0.434	-0.023
Number of bedrooms	1.67	1.55	0.13
Piped water on lot	0.265	0.255	0.010
Electricity	0.499	0.482	0.018
Indigenous	0.263	0.271	-0.008
<i>Panel B. Child characteristics</i>			
Age	10.69	4.62	6.07
Education (years)	3.79	1.68	2.12
Literate	0.810	0.541	0.269
Currently in school	0.711	0.766	-0.054
Worked	0.172	—	—
<i>Panel C. Household head and structure</i>			
Female household head	0.110	0.094	0.015
Head age	44.44	40.20	4.24
Household size	7.68	7.27	0.41
Head/spouse education (yrs)	3.26	3.62	-0.36
Head/spouse ever schooled	0.737	0.775	-0.039
Head/spouse worked	0.775	0.796	-0.021
Any indigenous member	0.356	0.367	-0.011
<i>Panel D. Household composition</i>			
Child aged 0–5	0.665	0.902	-0.237
Teenager aged 16–19	0.458	0.303	0.156
Woman aged 20–39	0.680	0.812	-0.132
Woman aged 40–59	0.398	0.255	0.143
Woman aged 60+	0.100	0.097	0.003
Man aged 20–39	0.534	0.672	-0.138
Man aged 40–59	0.481	0.336	0.145
Man aged 60+	0.113	0.096	0.018

Note: Pooled 1990 Census means across all study localities (T1998, T2000 and C2000), for children aged 6–16 (bracket A, $N = 49,445$) and 0–9 (bracket C, $N = 47,490$). Housing and poverty characteristics are nearly identical across the two cohorts; the differences are confined to the child's own age-linked attributes (age, schooling, literacy) and the mechanical life-cycle shift in household age structure (younger parents, more young children). "Worked" is not recorded for children aged 0–9.

I do a similar exercise with the 1995 Census (*Encuesta*/Survey). This dataset, despite being closer in time to the baseline of the data used in this paper, has significant drawbacks. The program was allocated at the locality level, yet the smallest geographical identifier in this dataset is the municipality. This results in virtually all municipalities having at least one treated locality. Additionally, individual-level data only exists for a fraction of the population, representative at the state level. Note that, using the ENCEL and ENCASEH data, we have information on 643 localities within 253 municipalities. Only 75 of those municipalities exist in the 1995 *Encuesta*. To circumvent these two issues and test whether marriage rates differed across treatment and control groups before the program was implemented, I do the following. First, I use two measures of treatment intensity: the share of Progresa beneficiaries in the municipality relative to the 1995 population (T1), and the share of localities in the municipality that received any Progresa transfers, as recorded in the program’s administrative register (T2). Second, I report the results for two samples — municipalities that are in the paper’s dataset (paper sample, the 75 municipalities mentioned above), and all municipalities for which I have data on program take-up and for which individual-level microdata are available in the public Census files (all-Mexico).³¹ Additionally, I run the analysis for two age groups of girls: 12-16 (recall that marriage information only exists for those aged 12 or older) and for 12-15, who are the children who have not yet turned 18 in 1997.

Unsurprisingly, given Progresa’s targeting policy, municipalities with higher program coverage are more likely to have poorer and less educated individuals/households. However, controlling for these characteristics, the program’s intensity is not predictive of child marriage rates. The proportion of girls aged 12 to 16 in 1995 who are married does not change as the program’s coverage increases. This is the case for municipalities in the paper’s data, for all Mexican municipalities in the census, and for girls between 12 and 15. These patterns do not change if I do not use the population weights. Table B6 shows these results. Each outcome, ever being married and formally married, is regressed separately on T1 and T2, controlling for state fixed effects, housing quality indicators, and age (the variables available in the dataset).

³¹The data on the program’s administrative registers and take-up was obtained from the replication package of Parker and Vogl (2023).

Standard errors are clustered at the municipality level. Girls between 12 and 16 are almost 5 p.p. less likely to be formally married in municipalities with more treated localities, but this does not result in fewer ever-married girls, nor is it robust to the other specifications.

With this set of exercises, I show that localities in pre-treatment periods had similar child marriage rates across treatment groups, thus providing additional evidence on the plausibility of the parallel trends assumption.

Table B6: Pre-program balance (outcomes): 1995 Census

Variable	Mean	T1: Beneficiary share	T2: Locality share	N
<i>Paper sample, ages 12–16</i>				
Ever married	0.033	0.017 (0.162)	-0.002 (0.042)	2,304
Formally married	0.012	-0.018 (0.100)	-0.041* (0.023)	2,304
<i>Paper sample, ages 12–15</i>				
Ever married	0.019	-0.019 (0.112)	0.004 (0.023)	1,896
Formally married	0.009	-0.004 (0.100)	-0.028 (0.021)	1,896
<i>All Mexico, ages 12–16</i>				
Ever married	0.026	-0.026 (0.068)	-0.012 (0.009)	19,569
Formally married	0.010	0.014 (0.035)	-0.005 (0.005)	19,569
<i>All Mexico, ages 12–15</i>				
Ever married	0.013	-0.014 (0.050)	-0.004 (0.004)	15,892
Formally married	0.005	0.019 (0.030)	-0.005 (0.003)	15,892

Note: The table shows the correlation between treatment intensity and each outcome, controlled by the covariates, with population weights and standard errors clustered at the municipality level (in parentheses). Each panel refers to a different sample: **Paper sample, ages 12-16** includes girls aged 12-16 in municipalities included in the main sample of the paper; **Paper sample, ages 12-15** includes girls aged 12-15 in municipalities included in the main sample of the paper. They are younger than 18 in 1997; **All Mexico, ages 12-16** includes girls aged 12-16 in all Mexican municipalities for which we have information on Progresas's coverage.; **All Mexico, ages 12-15** includes girls aged 12-15 in all Mexican municipalities for which we have information on Progresas's coverage. All regressions include state fixed effects. The column 'T1: Beneficiary share' uses as treatment intensity the proportion of Progresas/Oportunidades beneficiaries between 1997 and 2003, divided by municipal population in 1995. The column 'T2: Locality share' uses as treatment intensity the number of localities with any Progresas beneficiaries divided by the total localities in the municipality. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.3 Attrition and Missing Data

B.3.1 Attrition

In this analysis, there are three key groups of localities: T1998, which first received the program in 1998; T2000, which received it in 2000; and C2000, which did not receive the program until 2003. C2000 was included in the sample in 2003 and asked retrospective questions covering 1997 and the period from 2000 to 2003. By construction, therefore, sample attrition concerns only the two treated groups. Table B7 presents attrition rates, measured as the proportion of individuals with missing individual identifiers, across baseline and follow-up surveys. Attrition rises over time and is higher (and statistically significant) for T1998 than for T2000 from November 1999 onward. The program positively affected migration, which may partly account for attrition. Importantly, however, some individuals absent from a given survey reappear in subsequent ones. For example, roughly half of those missing in October 1998 reappeared in March 1999. Overall, between 50% and 70% of individuals missing in a given survey reappear in the next one, which allows marriage information to be recovered for most individuals in most years.

Table B7: Attrition - Missing ID

	Means	
	T1998	T2000
Individual ID lost from 1997 to 1998 (march)	0	0
Individual ID lost from 1997 to 1998 (october)	.033	.045
Individual ID lost from 1997 to 1999 (march)	.09	.11
Individual ID lost from 1997 to 1999 (november)	.1	.082
Individual ID lost from 1997 to 2000 (march)	.12	.093
Individual ID lost from 1997 to 2000 (november)	.11	.091
Individual ID lost from 1997 to 2003	.13	.11
Individual ID lost from 1997 to 2007	.27	.23

Note: This table presents the proportion of individuals, by treatment group, surveyed in 1997 and missing in subsequent surveys.

Since attrition and marriage may be correlated, I conduct a more detailed attrition analysis to

assess whether attrition is driven by marriage and whether this poses a threat to identification. Given that the program affected girls' marriage decisions but not boys', and that boys' child marriage rates are very low, the following exercises focus exclusively on girls.

Restricting the analysis to girls who satisfy all sample criteria (eligible, within the baseline age limits, and free of gender and age inconsistencies), Table B8 shows that through 2000, whenever a girl disappears from the sample, her entire household also disappears. Attrition is therefore driven primarily by the failure to locate or interview entire households in a given year, rather than by individual-specific attrition. This is consistent with the data collection protocol: interviewers were instructed to list, in all waves, all individuals residing in the household in 1997, regardless of whether they were still residents at the time of the interview. Each household respondent, preferably the spouse of the household head or the household head if female, then answered general questions about each listed individual, including their marital status.

In 2003, however, 12.7% of girls missing from the sample belong to households that are otherwise present that year. Since interviewers should still have records of these individuals, this points to a data collection or processing issue. Whether these issues are correlated with girls having left the household cannot be verified.

Table B8: Individual vs Household Attrition: Girls

Wave	Lost (N)	HH found (%)	HH gone (%)
1998m	0	0.0	100.0
1998o	445	0.0	100.0
1999m	1127	0.0	100.0
1999n	1125	0.0	100.0
2000m	1277	0.8	99.2
2000n	1209	0.0	100.0
2003	1496	12.7	87.3

Note: This table shows, for every data collection wave, the number of girls who were lost in 1997, the proportion of households that are in the survey if the girl is missing, and the proportion of households that are also missing.

Table B9 further shows that the share of missing girls in present households does not differ across the two treatment groups. Attrition rates rise over time, with nearly 13% of girls from

the 1997 baseline absent in 2003, but remain broadly similar across T1998 and T2000. Notably, between 43% and 75% of girls who disappear in a given wave reappear in a subsequent one, and most importantly for the outcome of interest, between 43% and 61% reappear by 2003. Of the roughly 12,000 girls sampled at baseline, only 1.3% are never observed in any subsequent wave.

Table B9: Attrition Patterns: Treated Girls (compared to 1997)

Wave	Lost (N)	Attrition (%)		Difference	p-value	Among those lost (%)		
		T1998	T2000			Lost forever	Returns later	In 2003
1998m	0	0.0	0.0	0.0	.	.	.	.
1998o	445	3.4	4.4	-1.1	0.096	36.9	63.1	44.0
1999m	1127	9.0	10.4	-1.3	0.453	24.6	75.4	61.0
1999n	1125	10.3	8.1	2.1	0.158	32.9	67.1	55.6
2000m	1277	11.7	9.2	2.5	0.054	37.9	62.1	53.1
2000n	1209	10.9	8.9	2.1	0.107	56.7	43.3	43.3
2003	1496	12.9	12.2	0.7	0.603	100.0	0.0	—

Note: This table shows attrition patterns by data collection wave. The first 5 columns show attrition rates by treatment group, in percentage, and the statistical difference and respective p-value. The last 3 columns show the proportion of girls who are missing in a given wave and who do not return to the survey in subsequent waves, the proportion of those who return later, and the proportion of those who return in 2003. Standard errors are clustered at the locality level.

It is nonetheless possible to examine which baseline characteristics predict attrition, and whether these overlap with the predictors of marriage. Testing this separately for T1998 and T2000 using univariate regressions (Tables B10 and B11), I find that out of 41 variables, only 9 predict both marriage and attrition, and 5 of these do so only for T1998. Crucially, none of these characteristics differ across treatment groups in their predictive power for either outcome. When multivariate regressions are estimated using all variables simultaneously as predictors of each outcome, there is no overlap between the two sets of predictors. I therefore conclude that individual characteristics, asset ownership, housing conditions, parental characteristics, and household composition do not systematically predict both attrition and marriage.

Table B10: Predictors of Attrition: Univariate Regressions, Girls

Variable	T1998		T2000		Difference	
	β	p	β	p	β	p
Age in 97	-0.002	0.049	0.000	0.945	-0.003	0.231
Education level	-0.005	0.002	-0.004	0.088	-0.001	0.739
Anyone in the HH speaks an indigenous language	-0.020	0.129	-0.018	0.450	-0.002	0.928
Dirt Floor	-0.009	0.468	0.013	0.516	-0.022	0.348
Inferior quality wall	0.003	0.804	-0.004	0.805	0.008	0.728
Inferior quality roof	0.055	0.004	0.019	0.380	0.036	0.212
No. of bedrooms	-0.011	0.043	-0.012	0.211	0.001	0.924
Piped water	-0.006	0.629	0.001	0.942	-0.007	0.735
Electricity	-0.037	0.018	-0.031	0.271	-0.006	0.847
Animals	-0.028	0.015	-0.044	0.002	0.016	0.391
Land	-0.036	0.004	-0.028	0.095	-0.008	0.683
Blender	-0.028	0.036	-0.004	0.816	-0.024	0.311
Refrigerator	-0.025	0.227	0.015	0.695	-0.039	0.354
Gas Stove	0.009	0.585	0.034	0.142	-0.024	0.393
Gas heater	-0.060	0.009	0.027	0.672	-0.087	0.193
Radio	-0.011	0.342	-0.038	0.057	0.028	0.230
TV	-0.018	0.128	-0.017	0.374	-0.001	0.948
Video player	-0.003	0.943	-0.067	0.052	0.064	0.225
Dish Washer	-0.022	0.453	0.034	0.629	-0.055	0.461
Car	-0.012	0.838	0.145	0.403	-0.158	0.390
Truck	0.021	0.604	0.017	0.725	0.003	0.960
HH Chief or Spouse have gone to school	-0.030	0.024	-0.029	0.235	-0.001	0.977
HH Chief or Spouse worked the week before	-0.068	0.004	-0.053	0.064	-0.015	0.679
Number of individuals in the HH	-0.009	0.000	-0.011	0.002	0.002	0.702
HH head age	-0.000	0.428	-0.000	0.692	-0.000	0.837
HH head is female	0.091	0.000	0.096	0.006	-0.005	0.904
At least one child between 0 and 5 y.o	0.011	0.354	-0.003	0.872	0.014	0.508
At least one teen between 16 and 19 y.o	-0.018	0.065	-0.021	0.143	0.003	0.875
At least one woman between 20 and 39 y.o	-0.014	0.239	-0.033	0.048	0.019	0.356
At least one woman between 40 and 59 y.o	-0.013	0.267	-0.006	0.702	-0.007	0.712
At least one woman over 60 y.o	-0.014	0.367	0.002	0.921	-0.016	0.565
At least one man between 20 and 39 y.o	-0.012	0.275	-0.020	0.186	0.008	0.679
At least one man between 40 and 59 y.o	-0.024	0.012	-0.023	0.107	-0.002	0.919
At least one man over 60 y.o	-0.009	0.584	0.007	0.794	-0.016	0.604
Guerrero	0.046	0.055	0.099	0.281	-0.053	0.577
Hidalgo	-0.009	0.564	0.043	0.143	-0.052	0.118
Michoacan	0.019	0.302	0.006	0.748	0.013	0.636
Puebla	0.030	0.212	0.003	0.887	0.027	0.385
Queretaro	0.033	0.304	-0.051	0.255	0.085	0.127
San Luis Potosi	0.008	0.707	0.033	0.220	-0.025	0.464

Note: This table shows (i) the coefficient of univariate regressions with attrition as an outcome variable (girl is in the sample or not) and the respective baseline characteristic as the explanatory variable, and (ii) the p-value for this coefficient. The first two columns show this information for the T1998 group, the middle two for the T2000 group, and the last show the estimate of the difference between T1998 and T2000 with the respective p-value. Standard errors are clustered at the locality level.

Table B11: Predictors of Marriage: Univariate Regressions, Girls

Variable	T1998		T2000		Difference	
	β	p	β	p	β	p
Age in 97	0.062	0.000	0.064	0.000	-0.003	0.359
Education level	0.060	0.000	0.063	0.000	-0.003	0.493
Anyone in the HH speaks an indigenous language	0.023	0.112	0.008	0.625	0.014	0.508
Dirt Floor	0.031	0.012	0.011	0.431	0.020	0.273
Inferior quality wall	0.035	0.002	0.012	0.494	0.023	0.265
Inferior quality roof	0.009	0.532	0.021	0.252	-0.011	0.625
No. of bedrooms	-0.008	0.188	-0.009	0.182	0.002	0.834
Piped water	-0.044	0.002	-0.016	0.433	-0.028	0.266
Electricity	-0.024	0.109	-0.059	0.001	0.035	0.117
Animals	0.014	0.243	0.024	0.163	-0.010	0.637
Land	-0.012	0.326	0.027	0.103	-0.039	0.058
Blender	-0.037	0.007	-0.024	0.138	-0.013	0.540
Refrigerator	-0.025	0.226	-0.034	0.301	0.008	0.832
Gas Stove	-0.014	0.324	-0.013	0.530	-0.001	0.956
Gas heater	-0.043	0.198	0.009	0.883	-0.052	0.448
Radio	-0.026	0.039	-0.026	0.108	0.000	0.997
TV	-0.032	0.006	-0.023	0.177	-0.009	0.660
Video player	-0.019	0.624	-0.122	0.005	0.102	0.080
Dish Washer	-0.055	0.065	0.041	0.472	-0.096	0.133
Car	-0.032	0.678	-0.227	0.000	0.196	0.011
Truck	-0.034	0.333	-0.023	0.642	-0.011	0.854
HH Chief or Spouse have gone to school	-0.061	0.000	-0.060	0.000	-0.001	0.971
HH Chief or Spouse worked the week before	-0.032	0.172	0.017	0.454	-0.049	0.134
Number of individuals in the HH	-0.001	0.601	0.003	0.249	-0.005	0.228
HH head age	0.002	0.001	0.001	0.035	0.000	0.592
HH head is female	0.058	0.011	-0.005	0.855	0.063	0.088
At least one child between 0 and 5 y.o	-0.053	0.000	-0.035	0.034	-0.019	0.365
At least one teen between 16 and 19 y.o	0.092	0.000	0.086	0.000	0.006	0.758
At least one woman between 20 and 39 y.o	-0.082	0.000	-0.058	0.001	-0.024	0.262
At least one woman between 40 and 59 y.o	0.032	0.012	0.022	0.165	0.010	0.612
At least one woman over 60 y.o	-0.007	0.718	0.010	0.690	-0.017	0.591
At least one man between 20 and 39 y.o	-0.047	0.000	-0.041	0.005	-0.006	0.733
At least one man between 40 and 59 y.o	0.033	0.002	0.042	0.003	-0.009	0.601
At least one man over 60 y.o	0.017	0.361	0.001	0.954	0.016	0.584
Guerrero	0.043	0.161	0.035	0.317	0.008	0.865
Hidalgo	-0.010	0.536	0.006	0.834	-0.015	0.620
Michoacan	0.010	0.656	-0.006	0.760	0.016	0.592
Puebla	0.012	0.503	0.028	0.269	-0.017	0.591
Queretaro	-0.013	0.518	-0.065	0.003	0.052	0.076
San Luis Potosi	0.012	0.599	0.011	0.663	0.001	0.979

Note: This table shows (i) the coefficient of univariate regressions with marriage in 2003 as an outcome variable (whether the girl was married by 2003) and the respective baseline characteristic as the explanatory variable, and (ii) the p-value for this coefficient. The first two columns show this information for the T1998 group, the middle two for the T2000 group, and the last show the estimate of the difference between T1998 and T2000 with the respective p-value. Standard errors are clustered at the locality level.

Despite the evidence suggesting that attrition is driven primarily by household-level rather than individual-level factors, I also estimate the relationship between a girl being absent in a given wave and having been married in the previous wave. If married girls are more likely to exit the sample, then the main estimates represent a lower bound, since married girls would be underrepresented in later waves. This concern, however, only threatens identification if the marriage attrition relationship differs across treatment groups. I find that this correlation is positive and statistically significant in some waves for both T1998 and T2000 (see Table B12). Except for 2003, the marriage attrition relationship does not differ between the two groups, either in pre- or post-treatment periods. In 2003, married girls in T2000 were 3 p.p. more likely to be absent from the sample than married girls in T1998, significant at 10%. This indicates that while marriage is associated with a higher likelihood of subsequent sample exit, this relationship does not vary across treatment arms for most years, and when it does it is to a small extent, and therefore it does not undermine the identification strategy.

Table B12: Attrition and Marriage

Wave	Married	T1998		T2000		β married (lagged)		
		Rate	N	Rate	N	T1998	T2000	Difference
1998m	No	0.000	7346	0.000	4103	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
	Yes	0.000	102	0.000	56	[.]	[.]	[.]
1998o	No	0.024	7346	0.033	4103	0.045 (0.024)	0.092 (0.046)	-0.047 (0.051)
	Yes	0.069	102	0.125	56	[0.059]	[0.045]	[0.356]
1999m	No	0.071	6963	0.085	3879	0.038 (0.022)	0.071 (0.033)	-0.033 (0.040)
	Yes	0.109	202	0.155	116	[0.093]	[0.034]	[0.409]
1999n	No	0.083	6963	0.061	3879	0.041 (0.022)	0.051 (0.030)	-0.010 (0.037)
	Yes	0.124	202	0.112	116	[0.068]	[0.087]	[0.786]
2000m	No	0.081	6756	0.058	3814	0.036 (0.019)	0.036 (0.023)	0.000 (0.030)
	Yes	0.117	333	0.094	191	[0.059]	[0.129]	[0.998]
2000n	No	0.070	6756	0.057	3814	0.050 (0.021)	0.017 (0.019)	0.033 (0.028)
	Yes	0.120	333	0.073	191	[0.016]	[0.369]	[0.233]
2003	No	0.063	6439	0.056	3613	-0.001 (0.011)	0.032 (0.017)	-0.033 (0.020)
	Yes	0.062	561	0.088	328	[0.940]	[0.056]	[0.099]

Note: The table reports attrition rates and the relationship between pre-wave marriage and attrition, for each survey wave and treatment group. "Married (lagged)" indicates whether the girl was observed as married in the calendar year prior to the wave. Attrition is defined as being lost from the baseline sample by the given wave. The columns β T1998 and β T2000 reports the OLS coefficient on lagged marriage status from a linear probability model estimated separately within each treatment group; a positive coefficient indicates that girls who were already married are more likely to attrit by that wave. The column β Difference reports the coefficient on married (lagged) $\times$ T1998 from a pooled regression on the two treatment groups, testing whether the marriage-attrition relationship differs between T1998 and T2000. A significant coefficient would suggest that attrition is differentially driven by marriage across treatment arms, which could bias the comparison. Standard errors in parentheses, p -values in brackets, clustered at the locality level.

B.3.2 Missing Data and Final Sample

I start by providing a step-by-step description of the restrictions and exclusions I implemented. The initial dataset has 166,960 observations. The final dataset is constructed through the following sequential steps: (i) I exclude individuals that are not eligible for Progresa/Oportunidades (non-poor) since they are out of scope of the project; (ii) I drop those observations whose age in 1997 is missing; (iii) I restrict the sample to the population of interest - those aged 6 to 16 in 1997; (iv) I drop those observations whose age in 2003 is larger than 24, since it implies an inconsistency in age that directly contradicts condition (iii); (v) I drop those whose gender is missing; (vi) I drop those whose gender is inconsistent across waves; (vii) I drop observations whose age

is decreasing across waves; and (viii) I drop observations who have duplicated entries in the dataset. After these restrictions, there are 33,165 unique observations in the database, as shown in Table B13.

Table B13: Number of Kept Observations, per criteria and treatment group

Restriction step	Total	Treated 1998	Treated 2000	Control
After dropping non-poor	119055	52667	27779	35396
After dropping missing age (1997)	113244	52524	27704	33016
After restricting to ages 6–16	36657	18025	9797	8835
After dropping age in 2003 > 24	36459	17963	9768	8728
After dropping missing gender	36412	17963	9768	8681
After dropping gender inconsistencies	34773	16886	9206	8681
After dropping decreasing age	33165	15854	8630	8681
After dropping duplicates	33165	15854	8630	8681

Note: This table shows the number of observations retained after applying each criterion.

As one can see, there are inconsistencies across surveys (around 3400 observations). Namely, individuals' ages do not progress as expected, or their gender changes from female to male or vice versa, which might indicate a mismatch in the IDs or misreporting of gender or age (see Tables B14 and B15). These inconsistencies are not statistically different across T1998 and T2000. For the main analysis, I exclude all those observations. If, instead, I forced missing values on marriage in those years in which the observation has an inconsistency, the estimator would only consider those observations that have information on two consecutive years.

Then, excluding observations for which the 1997 education level is missing, 29,606 remain. Of these, after excluding those for whom the 1997 marriage status was missing, 29,111 remain. Extending this exercise for all outcome and control variables, I have a final sample of 25,645 observations - as shown in Tables B16 and B17.

First, since individuals in the treatment groups are more likely to have missing information regarding the outcome of interest (see Table B18), I perform a robustness check using Lee bounds with inverse probability weights and tight bounds. Then, treating the data as if it were a repeated cross-section, I estimate a lower bound for the aggregate effect for girls of 2 p.p., statistically different from zero at 1%, CI = [0.016, 0.025].

Table B14: Attrition — Age inconsistency

	Means	
	T1998	T2000
Age in 1998 (march) not consistent with age in 1997	.036	.033
Age in 1998 (october) not consistent with age in 1997	.037	.039
Age in 1999 (march) not consistent with age in 1997	.16	.16
Age in 1999 (november) not consistent with age in 1997	.054	.054
Age in 2000 (march) not consistent with age in 1997	.16	.17
Age in 2000 (november) not consistent with age in 1997	.11	.099
Age in 2003 not consistent with age in 1997	.081	.074
Age is inconsistent in at least one year	.32	.33
Age is decreasing in at least one year	.064	.065
Age is inconsistent with 1997 in 2000 and 2003	.22	.21

Note: This table presents the proportion of individuals for whom age is not consistent across surveys, by treatment group.

Table B15: Attrition — Gender inconsistency

	Means	
	T1998	T2000
Gender changes from 1997 to 1999 (march)	0	0
Gender changes from 1997 to 1999 (november)	0	0
Gender changes from 1997 to 2000	.036	.034
Gender changes from 1997 to 2003	.023	.017
Gender changes from 1997 to 2007	.036	.034
Gender changes from 1997 in at least one year	.06	.058

Note: This table presents the proportion of individuals for whom gender is not consistent across surveys, by treatment group.

The data also lack information for some individuals regarding baseline characteristics used to estimate the propensity score and the outcome regression. Missing rates are extremely low for both treatment groups, but between 5 and 7% of the control group did not have information on asset holdings and household head in 1997 (see Tables B19 and B20). I do not resort to imputation of these missing values: since I estimate the probability of treatment with these variables, imputation would introduce bias in the estimates due to the non-zero covariance across the predictors. Therefore, I opt to exclude those observations from the sample.

Importantly, 34% of individuals in the control group lacked education information at baseline. Since the literature suggests that education is a very important predictor of marriage decisions, I exclude observations with missing education, and include the variable in the outcome and propensity score regression.

Table B16: Number of Kept Observations, per variable and treatment group (Part 1)

Variable	Total	Treated 1998	Treated 2000	Control
Education Level	29606	15486	8398	5722
Married in 1997	29111	15171	8231	5709
Married in 1998	28222	14619	7894	5709
Married in 1999	27597	14166	7722	5709
Married in 2000	27081	13813	7559	5709
Married in 2001	27081	13813	7559	5709
Married in 2002	27081	13813	7559	5709
Married in 2003	26086	13126	7251	5709
Dirt Floor	26027	13093	7232	5702
Inferior Quality Wall	25981	13056	7226	5699
Inferior Quality Roof	25963	13048	7216	5699
No. of Bedrooms	25919	13024	7207	5688
Piped Water	25901	13020	7198	5683
Electricity	25895	13020	7198	5677
Animals	25868	13013	7195	5660
Land	25835	12986	7192	5657
Blender	25831	12982	7192	5657
Refrigerator	25829	12982	7190	5657
Gas Stove	25826	12979	7190	5657
Gas Heater	25775	12940	7178	5657
Radio	25759	12924	7178	5657
TV	25741	12914	7170	5657
Video Player	25738	12914	7170	5654
Dish Washer	25733	12910	7169	5654
Car	25726	12903	7169	5654
Truck	25726	12903	7169	5654

Note: This table shows the number of kept observations after excluding those that have missing data for each of the variables that are included in the outcome regression, the propensity score specification, and the outcomes of interest.

Table B17: Number of Kept Observations, per variable and treatment group (Part 2)

Variable	Total	Treated 1998	Treated 2000	Control
Anyone in the HH Speaks an Indigenous Language	25726	12903	7169	5654
HH Chief or Spouse Have Gone to School	25664	12876	7144	5644
HH Chief or Spouse Worked the Week Before	25661	12874	7144	5643
Number of Individuals in the HH	25661	12874	7144	5643
Age of Household Head	25645	12874	7144	5627
Female Household Head	25645	12874	7144	5627
At least one child between 0 and 5 y.o	25645	12874	7144	5627
At least one teen between 16 and 19 y.o	25645	12874	7144	5627
At least one woman between 20 and 39 y.o	25645	12874	7144	5627
At least one woman between 40 and 59 y.o	25645	12874	7144	5627
At least one woman over 60 y.o	25645	12874	7144	5627
At least one man between 20 and 39 y.o	25645	12874	7144	5627
At least one man between 40 and 59 y.o	25645	12874	7144	5627
At least one man over 60 y.o	25645	12874	7144	5627
Guerrero	25645	12874	7144	5627
Hidalgo	25645	12874	7144	5627
Michoacan	25645	12874	7144	5627
Puebla	25645	12874	7144	5627
Queretaro	25645	12874	7144	5627
San Luis Potosi	25645	12874	7144	5627
Age in 1997	25645	12874	7144	5627

Note: This table shows the number of kept observations after excluding those that have missing data for each of the variables that are included in the outcome regression, the propensity score specification, and the outcomes of interest.

Table B18: Missing in Outcome

	Means		
	T1998	T2000	C2000
Missing marriage status in 1997	.023	.023	.0081
Missing marriage status in 1998	.057	.06	.0081
Missing marriage status in 1999	.067	.059	.0081
Missing marriage status in 2000	.081	.072	.0084
Missing marriage status in 2003	.15	.14	.0068
Missing Age at Marriage	.037	.043	0

Note: This table presents the proportion of individuals, by treatment group, with missing outcome information in each survey.

Table B19: Missing in main controls (Part 1)

	Means		
	T1998	T2000	C2000
Missing education in 1997	.026	.029	.34
Missing age in 1997	0	0	0
Missing indigenous background information	0	0	0
Missing if head or spouse went to school	.0021	.003	.069
Missing if head or spouse worked recently	.00011	0	.07
Missing gender of household head	0	0	.0014

Note: This table presents the proportion of individuals for whom the main control variables are missing at baseline, by treatment group.

Table B20: Missing in main controls (2)

	Means		
	T1998	T2000	C2000
Missing floor quality information	.0034	.0026	.059
Missing wall quality information	.003	.0015	.059
Missing roof quality information	.0014	.0014	.058
Missing no. bedrooms information	.0023	.0014	.066
Missing water provision information	.0014	.002	.059
Missing electricity provision information	.00078	.00061	.06
Missing animals ownership information	.0012	.00072	.062
Missing land ownership information	.003	.00041	.059
Missing blender ownership information	.0011	.00031	.059
Missing refrigerator ownership information	.0013	.00051	.059
Missing stove ownership information	.001	0	.059
Missing heater ownership information	.0036	.0021	.059
Missing radio ownership information	.0017	0	.059
Missing TV ownership information	.0013	.00092	.059
Missing video player ownership information	.0011	.0002	.059
Missing washing machine ownership information	.0013	.0002	.059
Missing car ownership information	.0017	.00031	.059
Missing truck ownership information	.0015	0	.059

Note: This table presents the proportion of individuals for whom the main control variables are missing at baseline, by treatment group.